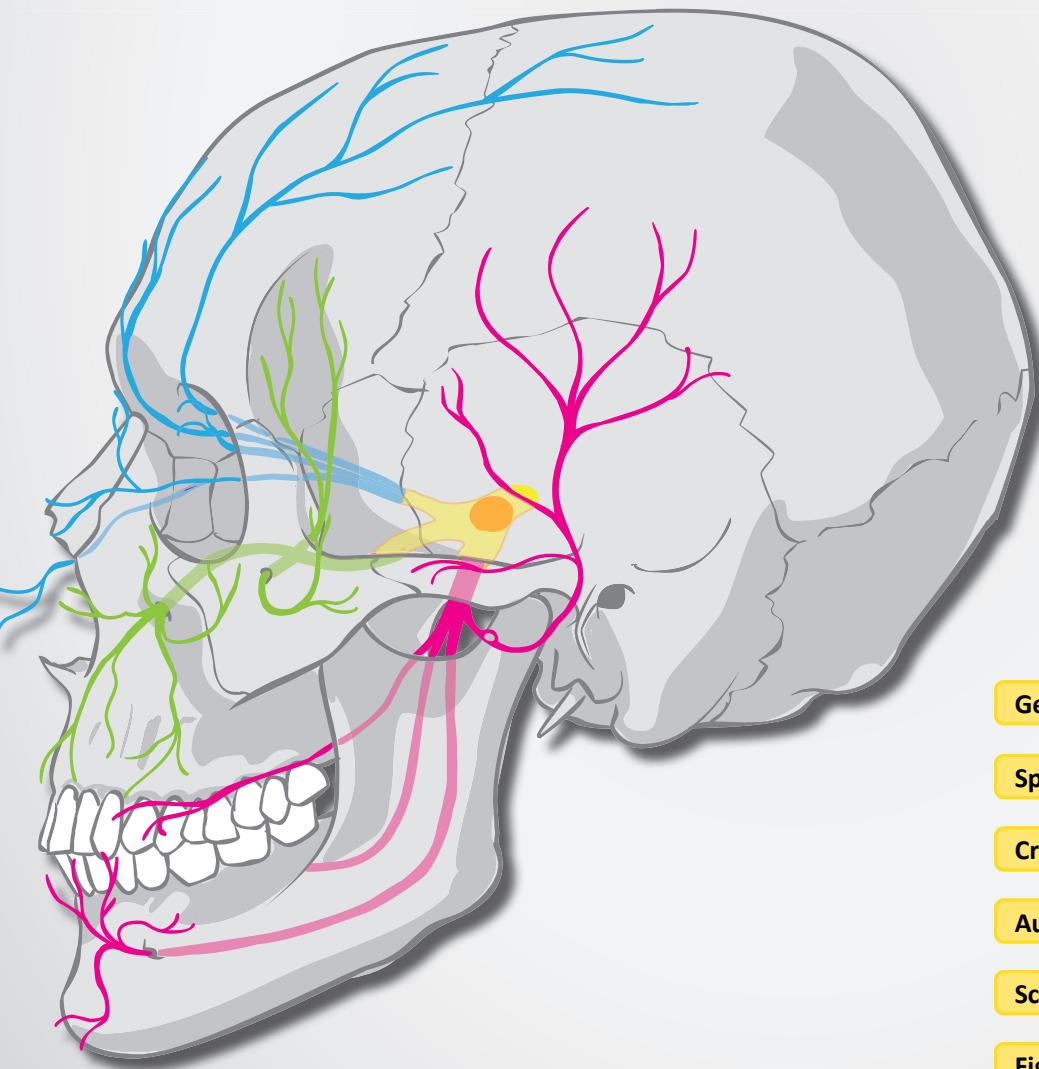


11

Peripheral nervous system

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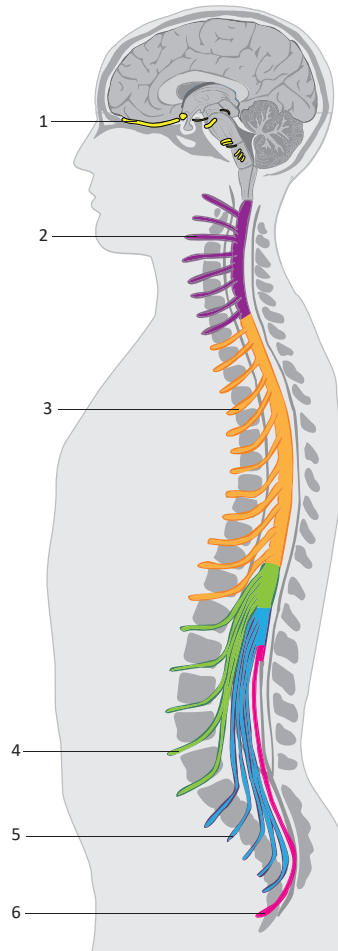
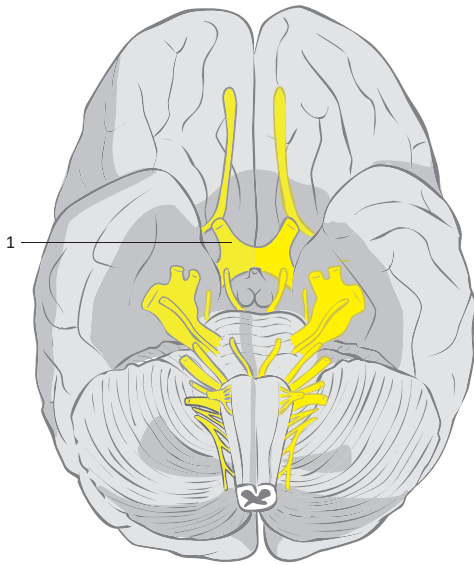
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The **peripheral nervous system (PNS)** consists of **nerves** and neural **ganglia**. Nerves enable **bidirectional communication** between the central nervous system and the rest of the body (referred to as the periphery). There are **31 pairs of spinal nerves** originating from the spinal cord and **12 pairs of cranial nerves** from the brainstem. The **autonomic nervous system (ANS)** is a division of the PNS. It is subdivided into the **sympathetic, parasympathetic and enteric nervous systems**. Ganglia are groups of cell bodies of nerves, located outside of the CNS. **Spinal ganglia** contain the cell bodies of sensory nerves. **Autonomic ganglia** contain synapsing visceromotor neurones; they contain a preganglionic axon synapsing on the dendrites of the postganglionic neurone. The cell body of the postganglionic neurone is located in the ganglion. Nerve fibres are axons and are either **myelinated** or **non-myelinated**. They are grouped together into nerve fascicles, which are further grouped together to form a peripheral nerve.

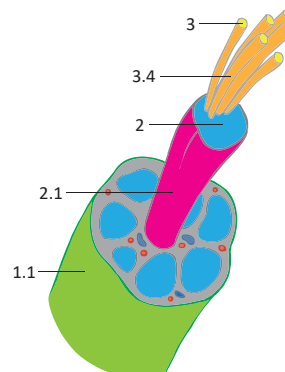
Classification

- 1 **Cranial nerves** (*nervi craniales*) – 12 pairs
- 2 **Cervical nerves** (*nervi cervicales*) – 8 pairs
- 3 **Thoracic nerves** (*nervi thoracici*) – 12 pairs
- 4 **Lumbar nerves** (*nervi lumbales*) – 5 pairs
- 5 **Sacral nerves** (*nervi sacrales*) – 5 pairs
- 6 **Coccygeal nerve** (*nervus coccygeus*) – 1 pair



Peripheral nerve

- 1 **Nerve** (*nervus*)
 - the peripheral nerve is a bundle of axons surrounded by fibrous tissue and supplied by small vessels called the vasa nervorum
 - 1.1 **Epineurium** – a sheath of fibrous tissue covering a peripheral nerve
 - emits septa into the nerve, dividing the fascicles
- 2 **Fascicle** (*fasciculus*) – a bundle of axons (nerve fibres)
 - 2.1 **Perineurium** – a fibrous sheath covering one fascicle
- 3 **Types of nerve fibres**
 - 3.1 **Axons of alpha motor neurones**
 - 3.1.1 **Schwann cells**
 - responsible for myelin production
 - 3.2 **Peripheral branches of pseudounipolar neurones**
 - 3.3 **Axons of autonomic neurones**
 - 3.4 **Endoneurium** – a fibrous sheath surrounding each myelinated nerve fibre



CNS is an abbreviation for the central nervous system.

PNS is an abbreviation for the peripheral nervous system.

A **peripheral nerve** does not contain any neuronal bodies (*perikarya*).

The **nerve fibres** travelling in a peripheral nerve are arranged in a wavy pattern. This prevents them from being harmed when they are stretched during limb movements.

Clinical notes

From the clinical point of view, there are two types of motor neurones. The upper motor neurone (UMN) is located in the cerebral cortex and the lower motor neurone (LMN) is located in the anterior horns of the spinal cord and in the nuclei of the cranial nerves.

Flaccid paralysis is caused by damage to the lower motor neurone. Muscle tone is decreased (hypotonia), partial or complete impairment of voluntary movement (paresis or plegia, respectively) and tendon reflexes are weakened or completely absent (hyporeflexia or areflexia, respectively). Viral poliomyelitis is an example of lower motor neuron disease. Spinal muscular atrophy (SMA) is a group of genetically determined diseases of the lower motor neurones.

Spastic paralysis is caused by damage to the upper motor neurone. It occurs as a result of lesions in the cerebral motor cortex or from damage to the axons of UMN that descend from the cortex to the spinal cord. This may arise after ischaemic insults such as occur in strokes and in periventricular perinatal hypoxia that can occur from complicated labour. Spastic paralysis manifests as partial or complete impairment of voluntary movements (paresis or plegia), increased muscle tone (hypertonia) and signs of spasticity (hyperreflexia and irritative pyramidal signs including the Babinski sign). Cerebral palsy is a classic example of an upper motor neurone disorder.

To **suture nerves** a special microsurgical technique is used to apply stitches to the perineurium of individual fascicles.

The **spinal nerve** is formed from an anterior and a posterior spinal root, which fuse and leave the vertebral canal through the intervertebral foramen. The **anterior root** consists of **efferent (motor) fibres** and the **posterior root** consists of **afferent (sensory) fibres**. The posterior root contains the **spinal ganglion** (*ganglion spinale*), containing the cell bodies of the afferent nerves entering the spinal cord. Outside the vertebral canal, the spinal nerve divides into a **posterior branch** (*ramus posterior*) and an **anterior branch** (*ramus anterior*). These branches contain **both afferent and efferent fibres**.

1 Grey matter of the spinal segment

- 1.1 **Anterior horn** (*cornu anterius*) – the cell bodies (*perikarya*) of the somatomotor neurones
- 1.2 **Lateral horn** (*cornu laterale*) – the cell bodies of the visceromotor neurones
- 1.3 **Intermediate zone** (*zona intermedia*) – the cell bodies of the viscerosensory neurones
- 1.4 **Posterior horn** (*cornu posterius*) – the cell bodies of the somatosensory neurones

2 Spinal nerve

2.1 Rootlets (*fila radicularia*)

- 2.1.1 **Anterior root** (*radix anterior/ventralis*) – emerges from the anterolateral groove (*sulcus anterolateralis*)
- 2.1.2 **Posterior root** (*radix posterior/dorsalis*) – emerges from the posterolateral groove (*sulcus posterolateralis*)
- 2.1.3 **Spinal ganglion** (*ganglion sensorium nervi spinalis*) – a sensory ganglion within the posterior root

2.2 Trunk of spinal nerve (*truncus nervi spinalis*)

- 2.2.1 **Anterior branch** (*ramus anterior*) – innervates the anterior part of the trunk and limbs
- 2.2.2 **Posterior branch** (*ramus posterior*) – innervates the back
- 2.2.3 **Meningeal branch** (*ramus meningeus*) – innervates the spinal dura mater
- 2.2.4 **White ramus communicans** (*ramus communicans albus*)
 - a branch of the spinal nerve going to the sympathetic ganglion
 - contains preganglionic and afferent fibres, which either transmit signals to other neurones and leave the ganglion as the grey ramus communicans (*ramus communicans griseus*),
 - interganglionic branches or pass through the ganglion and continue as the splanchnic nerves (*nn. splanchnici*)

3 Ganglion of sympathetic trunk (*ganglion trunci sympathici*)

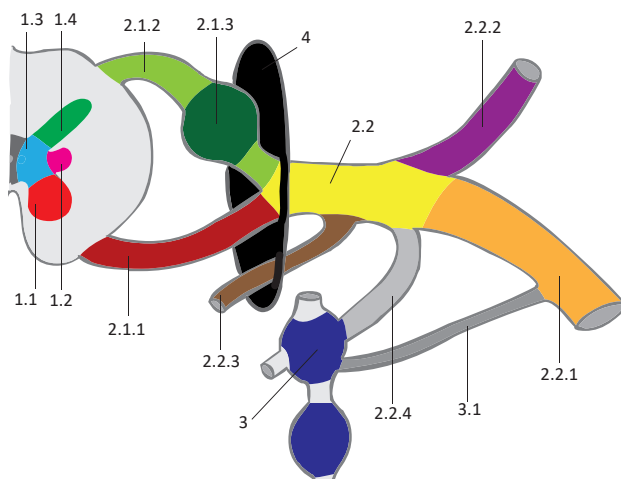
- the synapse between pre and postganglionic neurones in the sympathetic ganglia
- in contrast to parasympathetic ganglia, a significant amount of signal divergence occurs (as one preganglionic neuron connects to several postganglionic neurones)

- 3.1 **Grey ramus communicans** – originates from the sympathetic ganglion and merges into the spinal nerve containing postganglionic fibers

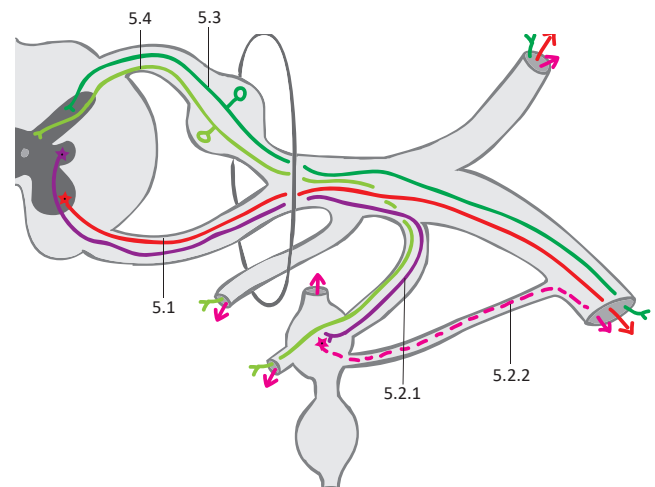
4 Intervertebral foramen (*foramen intervertebrale*) – the exit for the spinal nerve

5 Fibres according to their function in the peripheral nerve

- 5.1 **Somatomotor fibres** – regulate movements and tension in skeletal muscles
- 5.2 **Visceromotor fibres** – participate in the regulation of activity of smooth muscle cells, myocardial cells and glands
 - 5.2.1 **Preganglionic fibers** (C8–L3, S2–S4) – join the ganglia of the sympathetic trunk via the white rami communicantes
 - 5.2.2 **Postganglionic sympathetic fibers** (C8–L3) – emerge from the ganglia of the sympathetic trunk
 - rejoin the spinal nerves via the grey rami communicantes
 - other fibres give rise to the sympathetic trunk or leave the ganglia independently
 - 5.2.3 **Preganglionic parasympathetic fibers** (S2–S4)
 - do not synapse in the ganglia of the sympathetic trunk
- 5.3 **Somatosensory fibers** – transfer information from mechanoreceptors, exteroceptors, proprioceptors and thermoreceptors on the body surface
- 5.4 **Viscerosensory fibers** – transfer information from baroreceptors, chemoreceptors and receptors inside the internal organs (wall tension, special receptors for pain and chemical stimuli)



Scheme of the spinal nerve



Scheme of the spinal nerve

The nervous system **constantly monitors changes of external and internal environments**. A **reflex is a physiological phenomenon** that occurs in response to a change in the **internal or the external environment**. The reflex arc provides the anatomical basis for a reflex. It consists of a peripheral receptor, which transmits a signal to the CNS, where the signal is processed and a command is given to an effector organ. Reflex arcs may contain a **various number of synapses**. The simplest reflex arc is the monosynaptic reflex, which consists of two neurones: an afferent sensory neurone and an efferent motor neurone.

Receptors

- react to changes in the external and internal environments
- transduce these changes to action potentials, which pass information to the CNS

Division according to the location of the stimulus:

- 1 **Exteroreceptors** – receptors that react to stimuli in the external environment
- 2 **Interoreceptors** – receptors that react to stimuli in the internal environment
 - 2.1 **Proprioceptors** – receptors located in muscles, tendons and joint capsules that provide information about the location of different parts of the body
 - 2.2 **Visceroreceptors** – receptors located in the internal organs and vessels

Division according to the nature of the stimulus:

- 1 **Mechanoreceptors** – react to mechanical stimuli (touch, pressure, vibration, stretch)
- 2 **Chemoreceptors** – react to chemical stimuli
- 3 **Thermoreceptors** – react to temperature
- 4 **Photoreceptors** – react to light

Afferent (centripetal) pathways

- conduct signals from receptors to the CNS
 - are formed by axons of sensory neurons
 - their cell bodies are located outside the CNS in the dorsal root ganglia
- 1 **Somatosensory** (somatic sensory) – transfer signals from receptors in the skin and structures of the locomotor system
 - 2 **Viscerosensory** (visceral sensory) – transfer signals from visceroreceptors in the internal organs
 - 3 **Special sensory** – carry signals from the special sense organs (vision, hearing, motion and balance, smell and taste)

Efferent (centrifugal) pathways

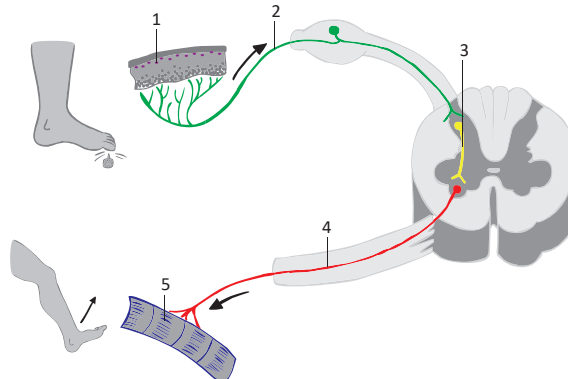
- transmit impulses from the CNS to effector organs in the periphery through efferent neurones
 - the efferent neurone is the (somatomotor or visceromotor) lower motor neuron
 - the cell body of the efferent neuron is located in the spinal cord, its axon leaves the CNS to form the motor part of the peripheral nerve
- 1 **Somatomotor tracts** – innervate striated muscle
 - 2 **Visceromotor tracts** – innervate smooth muscles and glandular cells

Effectors

- an organ or a tissue executing the commands of the CNS
- 1 **Muscle cells** – contraction of skeletal muscles (somatomotor / somatic motor) myocardial and smooth muscles (visceromotor / visceral motor)
 - 2 **Glands** – secretion (sweat, saliva, tears and glands of hollow organs)

The reflex arc

- 1 **Receptor**
- 2 **Afferent branch**
- 3 **Neurons within the CNS**
- 4 **Efferent branch**
- 5 **Effector**



A **mixed nerve** (*nervus mixtus*) contains several different types of nerve fibres. Most nerves are mixed nerves and contain somatomotor, somatosensory, visceromotor and viscerosensory fibres.

Examples of purely sensory cutaneous nerves: saphenous nerve, sural nerve and lateral cutaneous nerve of the forearm.

The **stretch (myotatic) reflex** is a monosynaptic reflex. The afferent limb originates in the muscle spindle, travels in the spinal nerve and posterior root, enters the spinal cord and synapses on an alpha motor neurone in the anterior horn. The efferent limb (the alpha motor neurone) travels to the neuromuscular junction and synapses on a muscle. Transection of a peripheral nerve will abolish its corresponding reflex, resulting in areflexia. Damage to the spinal cord above a reflex arc results in destruction of inhibiting impulses from higher neuronal centres, resulting in hyper-reflexia.

Polysynaptic reflex arcs contain one or more interneurons between the afferent and efferent limbs.

Muscles originate from the segmentally organised somites. Each somite gives rise to muscle tissue (myotome), part of the skeleton (sclerotome) and an area of skin and subcutaneous tissue (dermatome). Each somite is innervated by one segment of the spinal cord. The structures that the somite differentiates into retain this innervation as they mature and migrate to form different parts of the body. The myotome divides into an epaxial and a hypaxial division. The epaxial muscles are located posterior to the vertebral column and are innervated by the posterior branches of the spinal nerves. The hypaxial muscles are located ventral to the vertebral column and are innervated by the anterior branches of the spinal nerves. The musculature of the limbs originates from the hypaxial division; thus, all plexuses are derived from the anterior branches of the spinal nerves.

Sensory radicular area (*area radicularis sensoria*) is the area of the skin (dermatome) innervated by one posterior spinal root.

Motor radicular area (*area radicularis motoria*) is the muscles (myotome) that are innervated by one anterior spinal root.

Dermatome (*area nervina*) is the area (peripheral nerve field) innervated by one peripheral nerve.

There are **31 pairs of spinal nerves**. After leaving the intervertebral foramen, the spinal nerve divides into **two main branches: the posterior branch (*ramus posterior*) and the anterior branch (*ramus anterior*)**. Both branches contain afferent and efferent fibres. The **posterior branches** retain segmental organisation and provide motor innervation of the epaxial muscles of the back and the sensory innervation of the nape, back and buttocks. The **anterior branches form plexuses (*plexus nervosi*)** for the limbs and retain **segmental organisation** only in the **thoracic nerves (*nervi thoracici*)**.

Spinal nerves

- 1 **Cervical nerves (*nervi cervicales*)** C1 to C8
 - 8 pairs of cervical nerves, which innervate the skin and muscles of the upper limbs, head and neck
- 2 **Thoracic nerves (*nervi thoracici*)** T1 to T12
 - 12 pairs of thoracic nerves, which innervate the intercostal muscles, skin and muscles of the back and thorax
- 3 **Lumbar nerves (*nervi lumbales*)** L1 to L5
 - 5 pairs of lumbar nerves, which innervate the muscles and skin of the abdomen, thighs and skin of the genital organs
- 4 **Sacral nerves (*nervi sacrales*)** S1 to S5
 - 5 pairs of sacral nerves, which innervate the skin and muscles of the lower extremities
- 5 **Coccygeal nerve (*nervus coccygeus*)** Co
 - 1 pair, rudimentary in humans

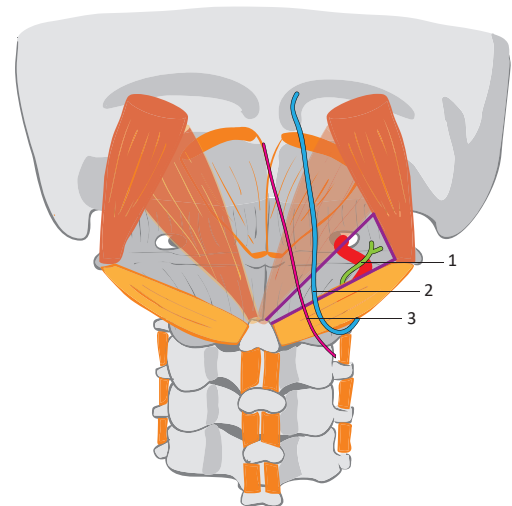
Posterior branches of the spinal nerves

- are thinner and shorter than the anterior branches
 - innervate the posterior (epaxial) part of the trunk
 - provide motor innervation for the epaxial muscles of the back
 - provide sensory innervation for the skin of the nape, back and buttocks
 - retain segmental organisation
- 1 **Suboccipital nerve (*nervus suboccipitalis*)** – the posterior branch of C1
 - a purely motor nerve for the suboccipital muscles and the semispinalis capitis
 - 2 **Greater occipital nerve (*nervus occipitalis major*)**
 - the posterior branch of C2
 - a purely somatosensory nerve for the skin posterior to the interauricular line (a line joining the auricles in the frontal plane)
 - 3 **Third occipital nerve (*nervus occipitalis tertius*)**
 - the posterior branch of C3
 - a purely somatosensory nerve innervating a narrow area of the skin medial to the greater occipital nerve
 - 4 **Superior clunial nerves (*nervi clunium superiores*)**
 - the posterior branches of L1 to L3
 - provides somatosensory innervation of the superior gluteal region
 - 5 **Middle clunial nerves (*nervi clunium medii*)**
 - the posterior branches of S1 to S3
 - provides somatosensory innervation of part of the gluteal region

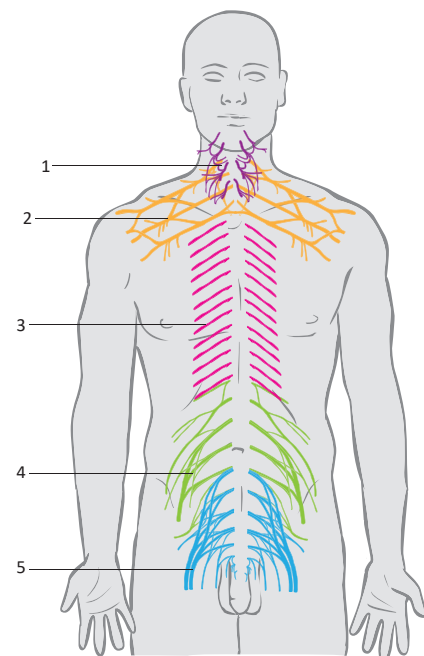
Anterior branches of the spinal nerves

- are longer and thicker branches than the posterior branches
- provide both motor and sensory innervation of the limbs and the hypaxial part of the trunk
- provide motor innervation of the muscles of the anterior abdomen, thorax and neck (except for the suprahyoid muscles)
- provide sensory innervation of the skin on the anterior aspect of the abdomen and thorax

- 1 **Cervical plexus (*plexus cervicalis*)** – C1 to C4
- 2 **Brachial plexus (*plexus brachialis*)** – C4 to T1
- 3 **Thoracic nerves (*nervi thoracici*)** – T1 to T12
- 4 **Lumbar plexus (*plexus lumbalis*)** – T12 to L4
- 5 **Sacral plexus (*plexus sacralis*)** – L4 and L5 and S1 to S4
- 6 **Coccygeal plexus (*plexus coccygeus*)** – S5 and Co



Suboccipital triangle and occipital nerves, posterior view

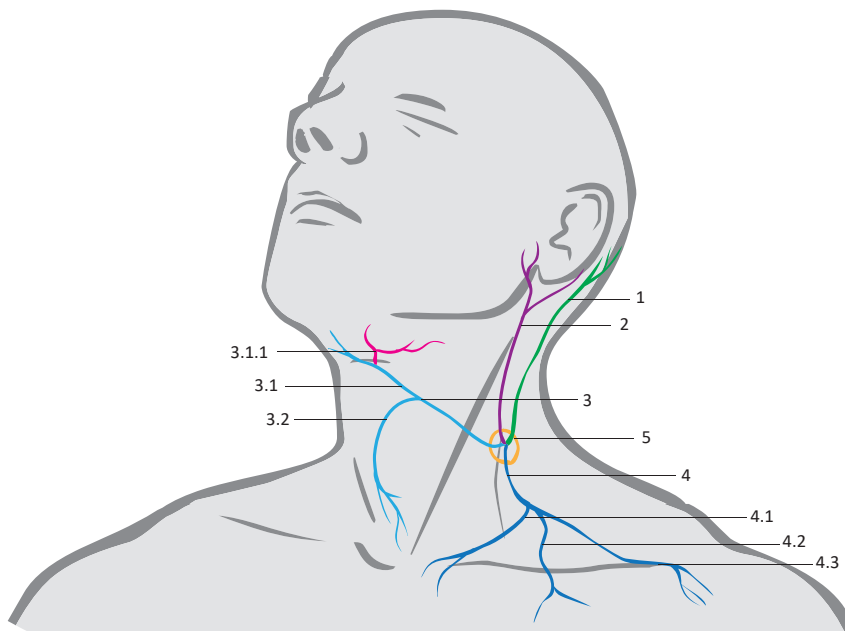


Anterior branches of the spinal nerves

The **cervical plexus** consists of spinal nerves **C1 to C4** and is located in the lateral cervical region. It produces both sensory and motor nerves. The sensory nerves reach the skin through the **the nerve point** (*punctum nervosum*), a gap in the superficial cervical fascia located approximately half-way along the posterior margin of the sternocleidomastoid. The **motor nerves** innervate the **anterior group** of cervical muscles. The **phrenic nerve** passes through the mediastinum and innervates the diaphragm.

Somatosensory branches

- 1 **Lesser occipital nerve** (*nervus occipitalis minor*) – C2 to C3
 - a somatosensory nerve, which travels along the posterior margin of the sternocleidomastoid in a cranial direction
 - innervates the lateral part of the nape region
- 2 **Great auricular nerve** (*nervus auricularis magnus*) – C2 to C3
 - a somatosensory nerve, which travels along the sternocleidomastoid muscle in a cranial direction
 - 2.1 **Anterior branch** (*ramus anterior*) – innervates the skin over the anterior part of the auricle of the inferior part in the parotidomasseteric region
 - 2.2 **Posterior branch** (*ramus posterior*) – innervates the skin of the posterior part of the auricle and the mastoid region
- 3 **Transverse cervical nerve** (*nervus transversus colli*) – C3
 - runs ventrally across the sternocleidomastoid muscle
 - 3.1 **Superior branch** (*ramus superior*) – innervates the skin of the suprahyoid region
 - 3.1.1 **Superficial ansa cervicalis** (*ansa cervicalis superficialis*)
 - a loop forming a connection with the cervical branch of the facial nerve (*ramus colli nervi facialis*), which carries motor fibres for the platysma
 - 3.2 **Inferior branch** (*ramus inferior*) – innervates the skin of the infrahyoid region
- 4 **Supraclavicular nerves** (*nervi supraclaviculares*) – C3 to C4
 - three groups of sensory nerves
 - pass superficially over the clavicle giving off branches in a caudal direction
 - 4.1 **Medial supraclavicular nerves** (*nervi supraclaviculares mediales*)
 - innervate the skin over the manubrium
 - 4.2 **Intermediate supraclavicular nerves** (*nervi supraclaviculares intermedii*)
 - innervate the skin of the infraclavicular region (*regio infraclavicularis*)
 - 4.3 **Lateral supraclavicular nerves** (*nervi supraclaviculares laterales*)
 - innervate the skin of the acromial region (*regio acromialis*)
- 5 **Nerve point** (*punctum nervosum*) – the point where the sensory cervical nerves pass through the superficial layer of the cervical fascia, emerging dorsally to the sternocleidomastoid muscle



Somatosensory branches of the cervical plexus

The **hypoglossal nerve** joins and travels with the superior root of the deep ansa cervicalis (*radix superior anse cervicalis profundae*). However, it does not participate in the motor innervation of the infrahyoid muscles.

Clinical notes

Surgical procedures in the area of the carotid bifurcation require very careful precision not to injure the motor branches of the cervical plexus.

The **phrenic nerve** must not be harmed when opening the pericardium during heart surgery.

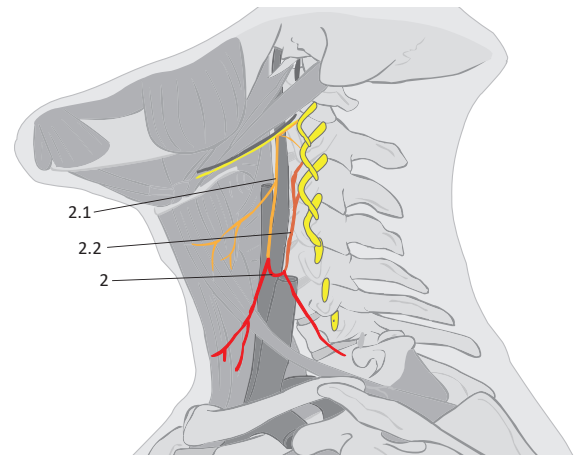
Phrenic nerve irritation results in twitches of the diaphragm causing hiccups (singultus).

Unilateral damage to the phrenic nerve leads to ipsilateral paresis of the diaphragm and unilateral respiratory impairment.

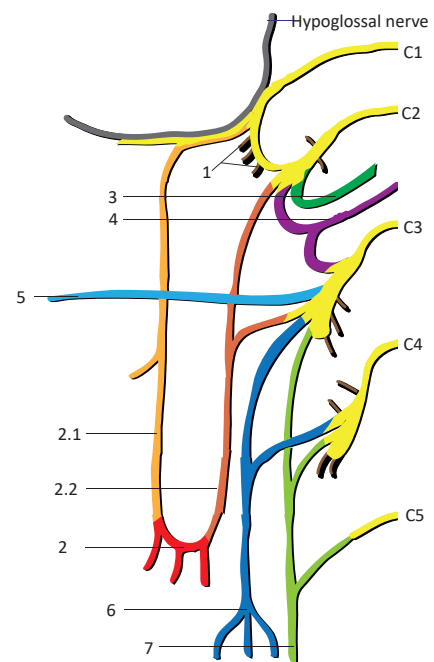
Transverse lesions of the spinal cord above C4 often lead to respiratory arrest. This is common in vertebral column trauma.

Plexus and branches

- **1 Muscular branches** (*rami musculares*) – C1 to C4
 - innervate the rectus capitis lateralis, rectus capitis anterior, longus capitis, longus colli muscles, scalenus anterior and scalenus medius
- **2 Deep cervical loop** (*ansa cervicalis profunda*)
 - innervates the infrahyoid muscles,
 - is formed by the superior and inferior roots
 - **2.1 Superior root** (*radix superior*) – C1 and C2
 - motor branches of the cervical plexus
 - travels with the hypoglossal nerve, but does not exchange of fibres
 - leaves the hypoglossal nerve as it passes the internal carotid artery
 - then descends to join the inferior root
 - **2.2 Inferior root** (*radix inferior*) – C2 to C3
 - motor branches of the cervical plexus
 - joins the superior root
- **3 Lesser occipital nerve** (*nervus occipitalis minor*) – C2 to C3
- **4 Great auricular nerve** (*nervus auricularis magnus*) – C2 to C3
- **5 Transverse cervical nerve** (*nervus transversus colli*) – C3
- **6 Supraclavicular nerves** (*nervi supraclaviculares*) – C3 to C4
- **7 Phrenic nerve** (*nervus phrenicus*) – C3 to C5



Deep cervical loop, lateral view



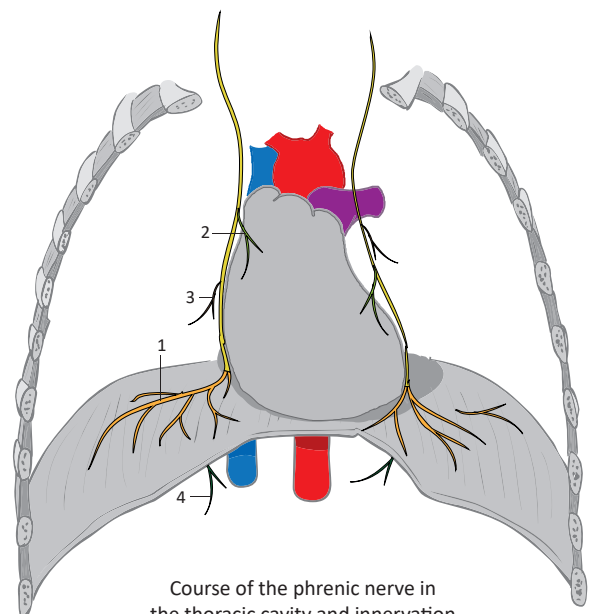
Scheme of the cervical plexus

Mixed branch

- **Phrenic nerve** (*nervus phrenicus*)
 - from C4, with accessory fibres from C3 and C5
 - descends bilaterally on the surface of the anterior scalenus muscle and travels through the superior thoracic aperture to reach the thoracic cavity
- **Right phrenic nerve** (*nervus phrenicus dexter*)
 - passes between the right subclavian artery and vein
 - descends on the right side from the right brachiocephalic vein and right superior vena cava
 - runs on the surface of the pericardium and enters the diaphragm near the caval opening
- **Left phrenic nerve** (*nervus phrenicus sinister*)
 - passes between the left subclavian artery and vein
 - descends behind the left brachiocephalic vein ventrally to the aortic arch and pulmonary hilum
 - runs on the surface of the pericardium and enters the diaphragm at or ventral to the central tendon

Branches of the phrenic nerve

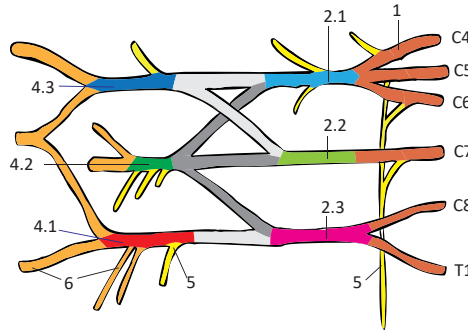
- **1 Phrenic branches** (*rami phrenici*)
 - provide motor innervation of the diaphragm
- **2 Pericardial branches** (*rami pericardiaci*)
 - provide somatosensory innervation of the pericardium
- **3 Pleural branches** (*rami pleurales*)
 - provide somatosensory innervation of the mediastinal and diaphragmatic pleura
- **4 Phrenicoabdominal branches** (*rami phrenicoabdominales*)
 - provide somatosensory innervation of the parietal peritoneum under the diaphragm (down to the gallbladder and pancreas)
- **5 Sensory branches for the thymus**



Course of the phrenic nerve in the thoracic cavity and innervation of the diaphragm, anterior view

The initial part of the brachial plexus consists of the anterior branches of spinal nerves **C5–C8** and connections from **spinal segments C4 and T1**. These roots converge to form **three parallel trunks: the superior (C4–C6), middle (C7), and inferior (C8–T1) trunks**. The trunks pass through the **scalenic fissure** into the omoclavicular triangle and then divide to form the cords. The clavicle divides the plexus topographically into a **supraclavicular and an infraclavicular part**.

- 1 Anterior branches of spinal nerves (*rami anteriores nervorum spinalium*)
- 2 Trunks (*trunci*)
 - 2.1 Superior trunk (*truncus superior*)
 - 2.2 Middle trunk (*truncus medius*)
 - 2.3 Inferior trunk (*truncus inferior*)
- 3 Divisions
- 4 Cords (*fasciculi*)
 - 4.1 Medial cord (*fasciculus medialis*)
 - 4.2 Posterior cord (*fasciculus posterior*)
 - 4.3 Lateral cord (*fasciculus lateralis*)
- 5 Supraclavicular part (*pars supraclavicularis*)
- 6 Infraclavicular part (*pars infraclavicularis*)



The **trunks of the brachial plexus divide to form the cords**. Three posterior divisions form the posterior cord, two anterior divisions form the lateral cord and one anterior division forms the medial cord.

The **cords** of the brachial plexus are named according to their position to the axillary artery.

Erb's point (*punctum supraclaviculare*) is located 2–3 cm above a point between the middle and lateral thirds of the clavicle, anterior to the transverse projection of vertebra C6 and lateral to the sternocleidomastoid. Hypersensitivity in this area indicates damage to the brachial plexus.

The **nerves** innervating the muscles of the anterior side of the arm, forearm and hand originate from the anterior branches of the brachial plexus (from the medial and lateral cords). The posterior groups of the arm and forearm are innervated by the radial nerve, which is a branch of the posterior cord.

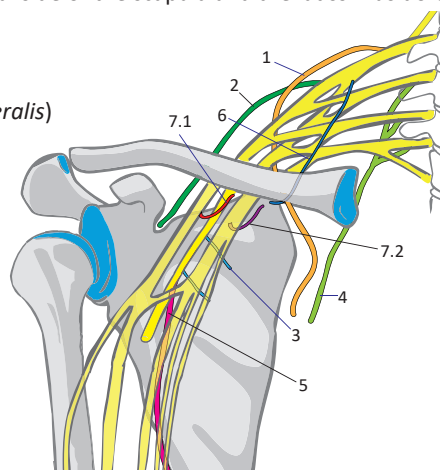
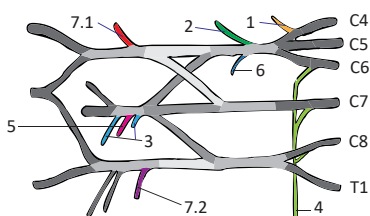
The **musculocutaneous nerve** is the only nerve of the arm that penetrates a muscle (the coracobrachialis).

2.2.1

Supraclavicular part of the brachial plexus

The **supraclavicular part of the brachial plexus** (*pars supraclavicularis plexus brachialis*) gives off nerves in the omoclavicular triangle, which innervate the **spinothoracic, thoracohumeral and scapular muscles**. Somatomotor fibres predominate in these nerves.

- 1 Dorsal scapular nerve (*nervus dorsalis scapulae*) – C5
 - penetrates the scalenus medius and follows the deep branch of the transverse cervical artery
 - innervates the levator scapulae and rhomboid major and minor
- 2 Suprascapular nerve (*nervus suprascapularis*) – C4 to C5
 - follows the suprascapular artery and veins towards the suprascapular notch
 - passes underneath the superior transverse scapular ligament to reach the supraspinous fossa
 - passes through the spinoglenoid notch to reach the infraspinous fossa
 - innervates the supraspinatus and infraspinatus, and provides sensory innervation for part of the capsule of the shoulder joint
- 3 Subscapular nerves (*nervi subscapulares*) – C5 to C7
 - two branches innervating the subscapularis and teres major
- 4 Long thoracic nerve (*nervus thoracicus longus*) – C6 to C7
 - provides motor innervation for the serratus anterior
 - travels with the lateral thoracic artery and vein on the lateral side of the thorax
- 5 Thoracodorsal nerve (*nervus thoracodorsalis*) – C6 to C8
 - innervates the latissimus dorsi
 - travels with the thoracodorsal vessels on the lateral side of the scapula and the latissimus dorsi
- 6 Subclavian nerve (*nervus subclavius*) – C5 to C6
 - a small nerve for the subclavius
- 7 Pectoral nerves (*nervi pectorales*) – C5–T1
 - 7.1 Lateral pectoral nerve (*nervus pectoralis lateralis*)
 - 7.2 Medial pectoral nerve (*nervus pectoralis medialis*)
 - emerges from the clavipectoral triangle
 - innervates the pectoralis major and minor



Clinical notes

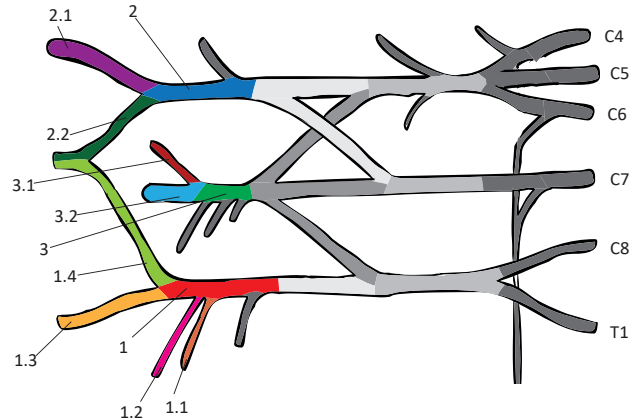
Paresis of the long thoracic nerve impairs the function of the serratus anterior. The scapula no longer adheres strongly to the thorax and so protrudes from the patient's back as what is called a winged scapula (*scapula alata*). Abduction of the arm above the horizontal plane is also impaired.

The **median nerve** is most commonly damaged in the carpal tunnel, which results in carpal tunnel syndrome. Shoulder dislocation and stab and bullet wounds can damage the nerve in the axilla. Colles' fracture and slash wounds on the wrist may damage the nerve in the forearm. Damage to the median nerve results in impaired pronation of the forearm (pronator teres and quadratus), impaired opposition and flexion of the thumb (opponens pollicis and the deep head of the flexor pollicis brevis) and impaired flexion of the second and third finger. Flexion of the fourth and fifth finger is preserved, due to the part of the flexor digitorum profundus innervated by the ulnar nerve. The clinical sign is called the "benedict hand (preacher's hand)".

The nerves of the **infraclavicular part of the brachial plexus** (*pars infraclavicularis plexus brachialis*) emerge from the **cords**. The cords are located around the **axillary artery** and give both mixed and purely sensory branches, which innervate the entire upper extremity. The cords eventually split into **7 terminal branches**. The **axillary nerve** and the **radial nerve** arise from the posterior cord, the **musculocutaneous nerve** from the lateral cord, and the **ulnar nerve** from the medial cord. The medial and the lateral cords give off the medial and lateral roots of the median nerve, which fuse to form the median nerve.

Infraclavicular part

- 1 **Medial cord** (*fasciculus medialis*)
 - 1.1 **Medial cutaneous nerve of arm** (*nervus cutaneus brachii medialis*)
 - 1.2 **Medial cutaneous nerve of forearm** (*nervus cutaneus antebrachii medialis*)
 - 1.3 **Ulnar nerve** (*nervus ulnaris*)
 - 1.4 **Medial root of median nerve** (*radix medialis nervi mediani*)
- 2 **Lateral cord** (*fasciculus lateralis*)
 - 2.1 **Musculocutaneous nerve** (*nervus musculocutaneus*)
 - 2.2 **Lateral root of median nerve** (*radix lateralis nervi mediani*)
- 3 **Posterior cord** (*fasciculus posterior*)
 - 3.1 **Axillary nerve** (*nervus axillaris*)
 - 3.2 **Radial nerve** (*nervus radialis*)



Lateral cord

- 1 **Musculocutaneous nerve** (C5–C7)

Course

 1. penetrates the coracobrachialis
 2. passes between the biceps brachii and the brachialis
 3. its terminal branch runs on the lateral aspect of the forearm

Branches and structures innervated

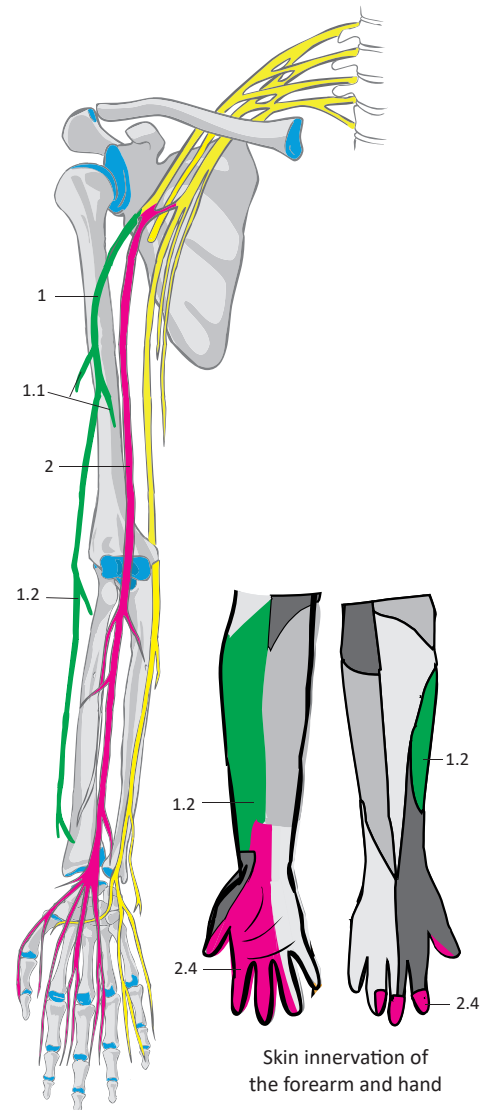
 - 1.1 **Motor branches for the anterior muscle group of the arm:**
coracobrachialis, brachialis, biceps brachii
 - 1.2 **Lateral cutaneous nerve of forearm / lateral antebrachial cutaneous nerve** (*nervus cutaneus antebrachii lateralis*)
– terminal sensory branch for the lateral forearm
- 2 **Median nerve** (C5–T1) – arises from the lateral and medial roots

Course

 1. follows the brachial artery and veins within the medial brachial intermuscular septum (*septum intermusculare brachii mediale*)
 2. enters the cubital fossa (*fossa cubitalis*)
 3. runs within the pronator canal (*canalis pronatorius*) between the two heads of the pronator teres and then passes between the two heads of the flexor digitorum superficialis
 4. runs between the flexor digitorum superficialis and profundus in the forearm
 5. runs more superficially in the distal part of the forearm, between the palmaris longus and the flexor carpi radialis
 6. enters the palm through the carpal tunnel (*canalis carpi*)

Branches and structures innervated:

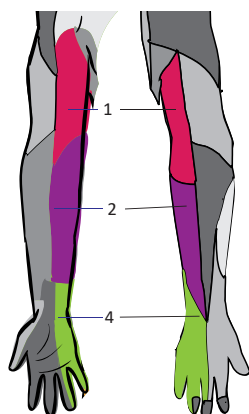
 - 2.1 **Articular branches** (*rami articulares*) – sensory innervation for the elbow joint
 - 2.2 **Muscular branches and anterior interosseous nerve** (*rami musculares et nervus interosseus antebrachii anterior*)
– motor innervation for almost all of the anterior antebrachial muscles:
flexor carpi radialis, palmaris longus, pronator teres, flexor digitorum superficialis, flexor pollicis longus, flexor digitorum profundus (for the 2nd and 3rd finger), pronator quadratus
 - 2.3 **Palmar branch** (*ramus palmaris*)
– provide sensory innervation for the skin of the thenar eminence
 - 2.4 **Common palmar digital nerves** (*nervi digitales palmares communes*) – I, II and III
– motor branches for the muscles of the thenar eminence: abductor pollicis brevis, opponens pollicis and the superficial head of the flexor pollicis brevis,
– motor branches for the 1st and 2nd lumbricals
– split into the proper palmar digital nerves (*nervi digitales palmares proprii*) which innervate the skin of digits 1–3 and the radial side of the 4th digit and reach the distal phalanges on both the palmar and the dorsal sides



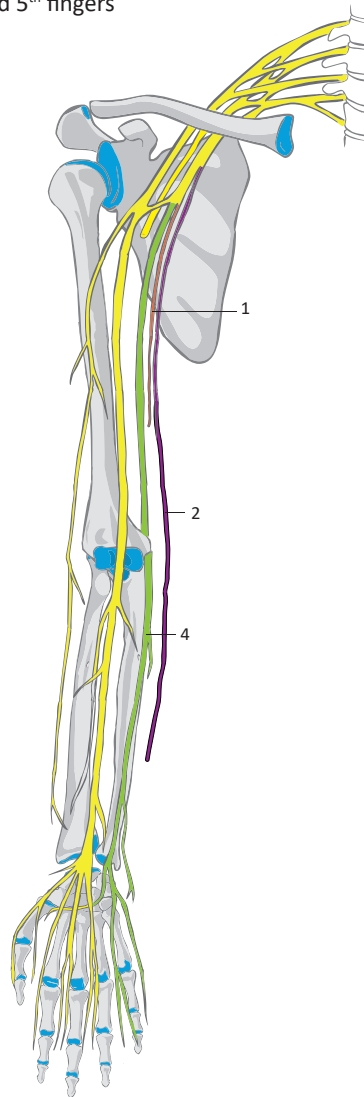
Skin innervation of the forearm and hand

Medial cord

- **1 Medial cutaneous nerve of arm / medial brachial cutaneous nerve**
(*nervus cutaneus brachii medialis*) – C8–T1
– sensory innervation of the medial aspect of the arm
- **2 Medial cutaneous nerve of forearm / medial antebrachial cutaneous nerve**
(*nervus cutaneus antebrachii medialis*) – C8–T1
– sensory innervation of the medial part of the forearm
– follows the basilic vein through the hiatus basilicus
- 3 Median nerve – C5–T1**
– arises from the lateral and medial roots
- **4 Ulnar nerve (*nervus ulnaris*) – C7–T1**
Course
 1. travels with the superior collateral ulnar artery and veins in the medial brachial intermuscular septum
 2. passes through the groove for the ulnar nerve behind the medial epicondyle of the humerus, here it is covered by fascia and skin only
 3. passes through the cubital canal between the two heads of the flexor carpi ulnaris
 4. runs between and innervates the flexor carpi ulnaris and the flexor digitorum profundus
 5. enters the palm via the ulnar canal**Branches and structures innervated:**
 - 4.1 Articular branches (*rami articulares*)**
– sensory innervation of the elbow joint
 - 4.2 Muscular branches (*rami musculares*) in the forearm**
– provide motor innervation for the flexor carpi ulnaris and the part of the flexor digitorum profundus that acts on the 4th and 5th fingers
 - 4.3 Dorsal branch of ulnar nerve**
(*ramus dorsalis nervi ulnaris*)
 - 4.3.1 Common and proper digital dorsal nerves**
(*nervi digitales dorsales communes et proprii*)
– provide sensory innervation of the 4th and 5th fingers and the ulnar side of the 3rd finger
 - 4.4 Superficial branch of ulnar nerve**
(*ramus superficialis nervi ulnaris*)
– travels in the palm of the hand
 - 4.4.1 Muscular branch (*ramus muscularis*)**
– innervates the palmaris brevis
 - 4.4.2 Common and proper palmar digital nerves**
(*nervi digitales palmares communes et proprii*)
– provide sensory innervation for the 5th finger and the ulnar side of the 4th finger
 - 4.5 Deep branch of ulnar nerve**
(*ramus profundus nervi ulnaris*)
– innervates all hypothenar muscles, palmar and dorsal interossei, lumbricals III and IV, adductor pollicis and deep head of the flexor pollicis brevis



Skin innervation of the arm, forearm and hand



Right upper limb, forearm in supination

The lateral branches of the 2nd and 3rd intercostal nerves form the intercostobrachial nerve (*n. intercostobrachialis*), which runs through the axillary lymph nodes and joins the medial cutaneous nerve of the arm.

The ulnar nerve is particularly vulnerable as it travels behind the medial epicondyle. This point is colloquially known as the “funny bone”.

The most commonly assessed reflexes of nerves originating in the brachial plexus are:

- the bicipital reflex (C5)
- the brachioradial reflex (C5–C6)
- the tricipital reflex (C7)
- the finger flexors (C8)

Clinical notes

The ulnar nerve is most commonly harmed in narrow anatomical spaces such as the cubital and ulnar canals. Fractures of the elbow and forearm often damage both the ulnar and median nerves. Injury to the ulnar nerve results in an impairment in motor function giving rise to the clinical appearance of a “claw hand”.

Axillary nerve palsy can occur due to a fracture of the surgical neck of the humerus. It manifests as impaired abduction of the arm and predisposes to subluxations of the shoulder due to instability of the head of the humerus.

Radial nerve injuries are fairly common. They are usually caused by fractures of the distal part of the humerus affecting the groove for the radial nerve (Holstein-Lewis fractures). Fractures of the forearm may also affect the radial nerve. Radial nerve injuries manifest as an absent tricipital reflex and weakness in elbow extension. Another prominent sign of radial nerve palsy is an inability to extend the wrist causing a condition called wrist drop.

Selective surgical transection of the sensory branches of the posterior interosseous nerve of the forearm can relieve otherwise uncontrollable pain of the wrist (such that may occur in osteoarthritis or rheumatoid arthritis).

Posterior cord

● 1 Axillary nerve (*nervus axillaris*) – C5–C6

Course

- passes with the posterior circumflex humeral artery and veins through the humerotricipital foramen to the posterior part of the surgical neck of the humerus

Branches and structures innervated:

- 1.1 **Muscular branches** (*rami musculares*)
 - innervate the deltoid and teres minor muscles
- 1.2 **Articular branches** (*rami articulares*)
 - sensory innervation for a part of the shoulder joint capsule
- 1.3 **Superior lateral cutaneous nerve of arm / superior lateral brachial cutaneous nerve** (*nervus cutaneus brachii lateralis superior*)
 - sensory innervation of the skin of the deltoid region

● 2 Radial nerve (*nervus radialis*)

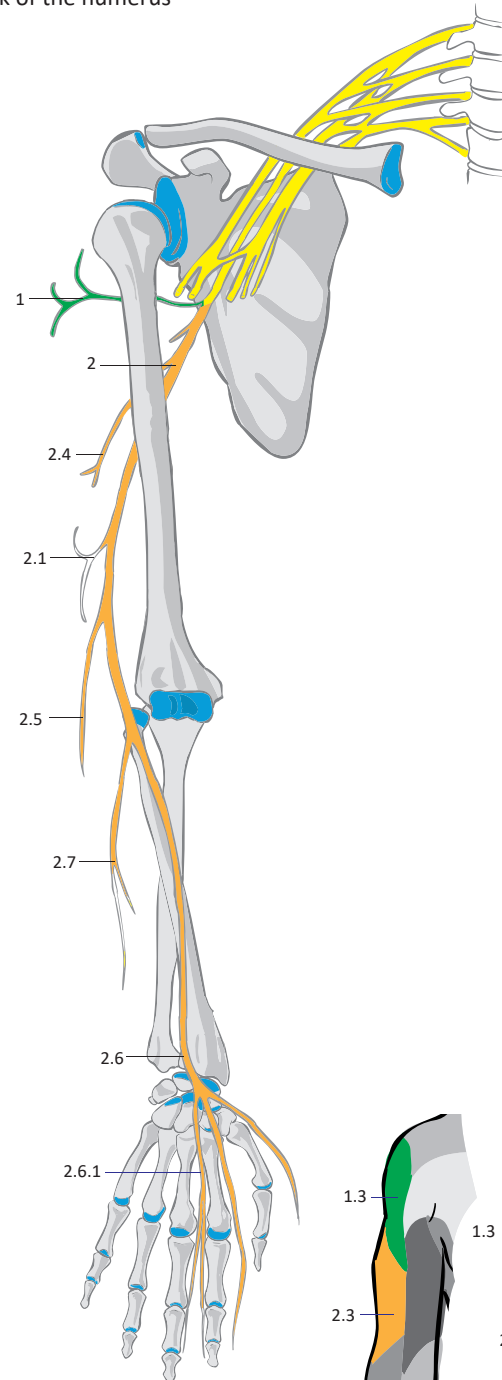
- C5–C8 and an additional branch from T1

Course

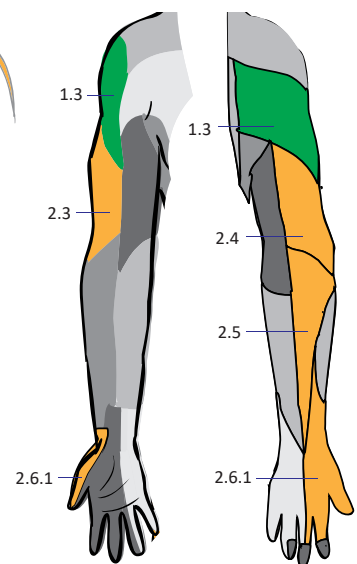
1. spirals down the radial canal (a space bound by the groove for the radial nerve and the medial and lateral heads of the triceps brachii on the posterior surface of the humerus) with the deep brachial artery and veins
2. perforates the lateral intermuscular septum to enter the anterior compartment of the arm, where it travels between brachialis and the brachioradialis
3. runs through the cubital fossa, where it divides into the superficial and deep branches
4. the deep branch enters the dorsal forearm through the supinator canal (a space between two heads of the supinator)
5. the superficial branch runs with the radial artery and veins to the radial fossa, where it divides into somatosensory branches for the dorsum of the hand

Branches and structures innervated:

- 2.1 **Muscular branches in the arm** (*rami musculares*)
 - innervate the triceps brachii and anconeus
- 2.2 **Muscular branches in the forearm** (*rami musculares*)
 - innervate all posterior and lateral muscles of the forearm
- 2.3 **Inferior lateral cutaneous nerve of arm / inferior lateral brachial cutaneous nerve** (*nervus cutaneus brachii lateralis inferior*)
 - sensory innervation of the distal part of the deltoid region and lateral part of the arm
- 2.4 **Posterior cutaneous nerve of arm / posterior brachial cutaneous nerve** (*nervus cutaneus brachii posterior*)
 - sensory innervation of the posterior side of the arm
- 2.5 **Posterior cutaneous nerve of forearm / posterior antebrachial cutaneous nerve** (*nervus cutaneus antebrachii posterior*)
 - sensory innervation of the posterior side of the forearm
- 2.6 **Superficial branch** (*ramus superficialis*)
 - branches from the radial nerve in the cubital fossa and travels with the radial vessels
- 2.6.1 **Dorsal digital nerves** (*nervi digitales dorsales*)
 - sensory innervation of the thumb and forefinger and the radial side of the 3rd finger (proximal and middle phalanges only)
- 2.7 **Deep branch** (*ramus profundus*) – runs through the supinator canal to get to the posterior compartment of the forearm, where it passes between the deep and superficial layers, giving off muscular branches
- 2.7.1 **Posterior interosseous nerve** (*nervus interosseus antebrachii posterior*)
 - runs on the posterior side of the interosseous membrane of the forearm with the posterior interosseous vessels
 - innervates the periosteum of both the radius and ulna, the interosseous membrane and both radio-ulnar joints



Right upper limb, forearm in pronation



Skin innervation of arm, forearm and hand

The **intercostal nerves** are the **anterior branches of the thoracic nerves**. They retain **segmental organisation** and do not form plexuses. They run dorsally on the inner surface of the external intercostal muscles towards the costal angle. They follow the ribs ventrally between the internal and innermost intercostal muscles. Each intercostal space contains a **neurovascular bundle** consisting of an intercostal vein, artery and nerve. The nerve is the most inferior structure of the neurovascular bundle. The **subcostal nerve** is the anterior branch of the **last thoracic nerve (T12)** and runs inferiorly to the last rib.

1 Muscular branches (*rami musculares*)

– innervates the external, internal and innermost intercostal muscles, subcostales, transversus thoracis, rectus abdominis, external oblique, internal oblique, transversus abdominis, quadratus lumborum and pyramidalis

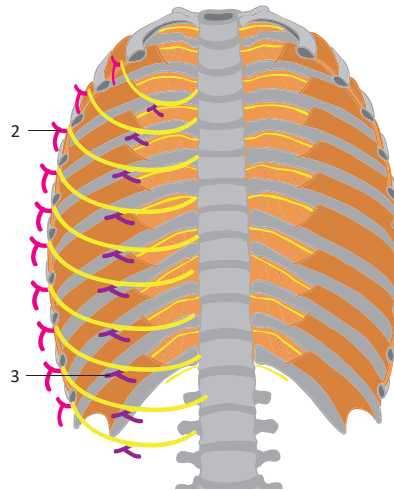
2 Lateral cutaneous branches

(*rami cutanei laterales*) – somatosensory innervation of the lateral part of the mammary and inframammary regions and abdomen

3 Anterior cutaneous branches

(*rami cutanei anteriores*) – somatosensory innervation of the medial part of the mammary and inframammary region and abdomen

4 Branches for the parietal pleura and peritoneum



VAN is a mnemonic for the structures travelling in the intercostal space. The Vein is the most cranial structure, followed by the Artery and then the Nerve.

The most commonly assessed reflexes in the abdomen:

- the epigastric region is stimulated for assessing the T7–8 segment
- the mesogastric region is stimulated for assessing the T9–10 segment
- the hypogastric region is stimulated for assessing the T11–12 segment

The cremasteric reflex (L2) is provoked by stimulating the skin of the medial thigh with a sharp object. This induces contraction the cremaster muscle, causing the ipsilateral testis to ascend towards the body.

The **lumbar plexus** is formed from the **anterior branches of spinal nerves L1–L4** and a contributing branch from **T12**. The plexus is located within the **psoas major**, lateral to the lumbar vertebrae.

Lumbar plexus nerves

1 Iliohypogastric nerve (*nervus iliohypogastricus*)

2 Ilioinguinal nerve (*nervus ilioinguinalis*)

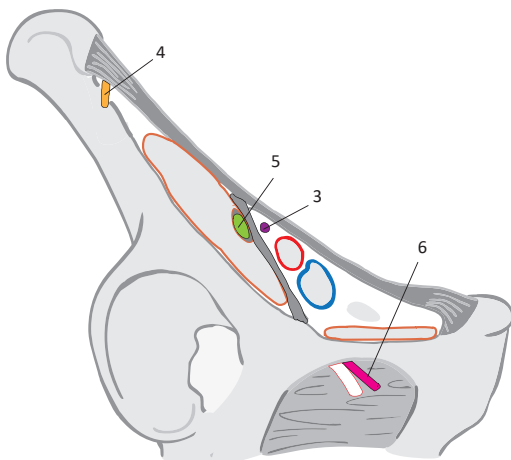
3 Genitofemoral nerve (*nervus genitofemoralis*)

4 Lateral cutaneous nerve of thigh

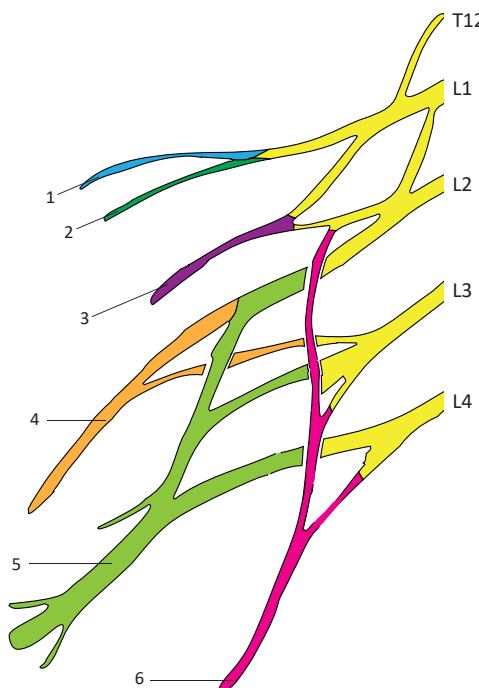
(*nervus cutaneus femoris lateralis*)

5 Femoral nerve (*nervus femoralis*)

6 Obturator nerve (*nervus obturatorius*)



Vascular and muscular spaces and the obturator canal



Scheme of the lumbar plexus

Clinical notes

Herpes zoster occurs after reactivation of the varicella-zoster virus lying latent inside a spinal ganglion. A painful vesicular rash appears on the dermatome supplied by the spinal nerve in which the virus is residing.

Pelvic fractures, hip dislocations, surgeries and incorrectly applied intramuscular injections can harm the femoral nerve. Induration of the inguinal lymphatic nodes and aneurysms of the femoral artery are the most common non-traumatic causes of femoral nerve injury. The motor symptoms of femoral nerve injury occur as are a result of impaired function of the quadriceps femoris. Walking without support and walking upstairs become difficult. It is impossible for the patient to raise his/her lower limb when lying supine. Standing and squatting become unstable. In certain cases, hyperextension of the knee (*genu recurvatum*, back knee) may occur when standing.

Meralgia paresthetica results from compression of the lateral cutaneous nerve of the thigh in the muscular space. Its symptomatology consists of acute intermittent pain and loss of sensation on the lateral thigh above the knee. It is common in obese and diabetic patients.

Branches

- **1 Iliohypogastric nerve** (*nervus iliohypogastricus*) – T12–L1
 - emerges laterally from the psoas major
 - passes between the internal oblique and the transversus abdominis
 - runs above the iliac crest travelling ventrocaudally
 - terminates in the hypogastric and inguinal regions
 - 1.1 **Muscular branches** (*rami musculares*)
 - motor branches for the internal oblique and transversus abdominis
 - 1.2 **Lateral cutaneous branch** (*ramus cutaneus lateralis*)
 - somatosensory innervation of the skin of the lateral abdomen
 - 1.3 **Anterior cutaneous branch** (*ramus cutaneus anterior*)
 - innervates the skin around the superficial inguinal ring
- **2 Ilioinguinal nerve** (*nervus ilioinguinalis*) – L1
 - emerges laterally from the psoas major and continues below the iliohypogastric nerve
 - travels through the inguinal canal (*canalis inguinalis*)
 - innervates the skin of the anterior part of the scrotum (in men) and the skin on the anterior part of the labia majora (in women)
- **3 Genitofemoral nerve** (*nervus genitofemoralis*) – L2–L3
 - penetrates the psoas major and leaves it on its ventral surface
 - 3.1 **Genital branch** (*ramus genitalis*) – passes through the inguinal canal
 - innervates the cremaster, dartos fascia and a small area on the medial side of the thigh (in men)
 - innervates part of the labia majora and a small area on the medial side of the thigh (in women)
 - 3.2 **Femoral branch** (*ramus femoralis*) – passes through the vascular space
 - innervates the skin of the medial side of the thigh
- **4 Lateral cutaneous nerve of thigh / lateral femoral cutaneous nerve** (*nervus cutaneus femoris lateralis*) – L2–L3
 - emerges laterally from the psoas major
 - runs laterocaudally on the iliacus towards the anterior superior iliac spine
 - travels through the muscular space beneath the lateral attachment of the inguinal ligament to reach the lateral thigh
 - provides somatosensory innervation for the lateral thigh
- **5 Femoral nerve** (*nervus femoralis*) – L2–L4

Course: emerges laterally from the psoas major, travels above the iliopsoas, passes medially through the muscular space to reach the anterior thigh and femoral triangle, then splits into muscular and cutaneous branches

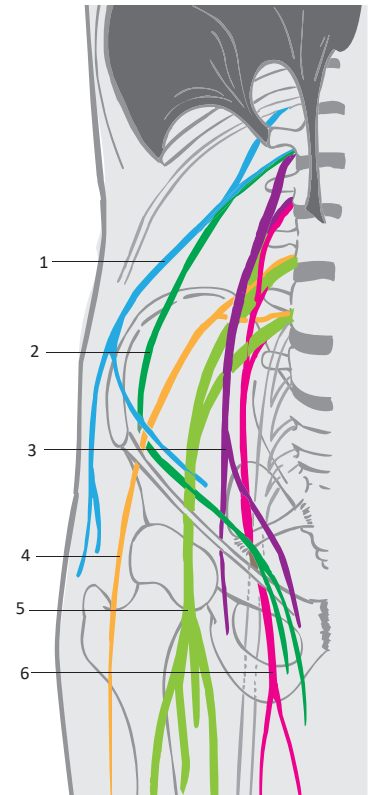
Branches and structures innervated:

 - 5.1 **Muscular branches** (*rami musculares*)
 - innervate the iliopsoas, quadriceps femoris, sartorius and pectineus
 - 5.2 **Anterior cutaneous branches** (*rami cutanei anteriores*)
 - provide somatosensory innervation for the anterior side of the thigh
 - 5.3 **Saphenous nerve** (*nervus saphenus*) – the terminal somatosensory branch, which follows the great saphenous vein distally on the leg
 - runs in the adductor canal and leaves it ventrally through the vastoadductor lamina
 - is accompanied by the descending genicular artery under the skin of the knee
 - travels with the great saphenous vein under the skin of the leg
 - 5.3.1 **Infrapatellar branch** (*ramus infrapatellaris*)
 - provides somatosensory innervation of the skin inferior to the patella
 - 5.3.2 **Medial cutaneous nerve of leg / medial crural cutaneous nerve** (*rami cutanei cruris mediales*) – provides somatosensory innervation for the skin of the medial part of the leg
- **6 Obturator nerve** (*nervus obturatorius*) (L2–L4)

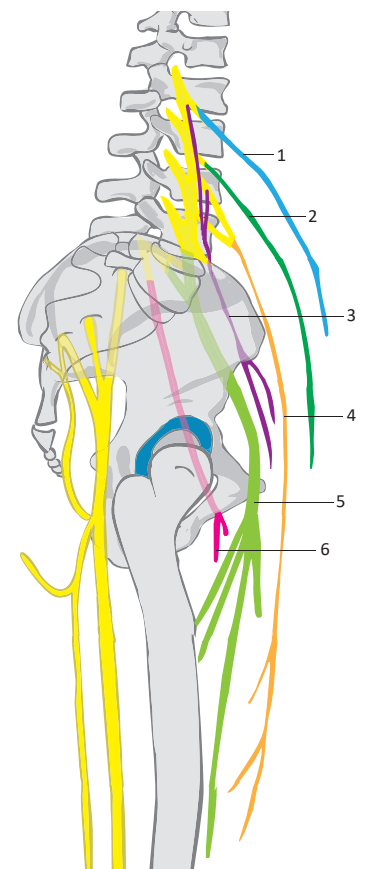
Course: emerges medially from the psoas major and passes through the obturator canal

Branches and structures innervated

 - 6.1 **Muscular branches** (*rami musculares*) – innervates the obturator externus
 - 6.2 **Anterior branch** (*ramus anterior*) – innervates the gracilis, adductor longus, adductor brevis, pectineus (a diploneural muscle also innervated by the femoral nerve) and provides somatosensory innervation for the medial thigh
 - 6.3 **Posterior branch** (*ramus posterior*) – innervates the adductor magnus (a diploneural muscle, also innervated by the sciatic nerve)
 - 6.3.1 **Articular branches** (*rami articulares*)
 - participate in the somatosensory innervation of the hip joint



Anterior view of the right lumbar plexus

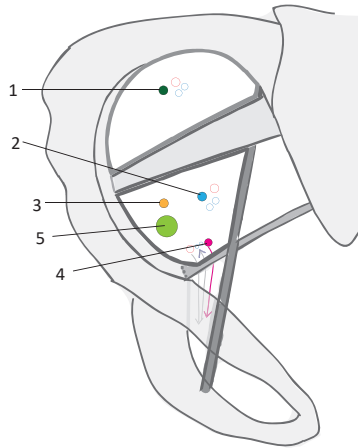


Lateral view of the right lumbar plexus

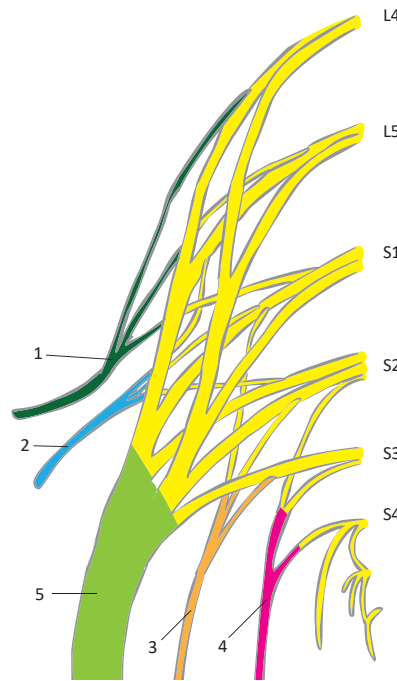
The **sacral plexus** arises from the spinal nerves of **S1 to S4** and receives contributing branches from **L4 and L5** via the lumbosacral trunk (*truncus lumbosacralis*). It is located on the **pelvic surface of the sacral bone**. Along with the lumbar plexus it provides somatomotor and somatosensory innervation for the lower limb.

Nerves of the sacral plexus

- 1 **Superior gluteal nerve** (*nervus gluteus superior*)
- 2 **Inferior gluteal nerve** (*nervus gluteus inferior*)
- 3 **Posterior cutaneous nerve of thigh** (*nervus cutaneus femoris posterior*)
- 4 **Pudendal nerve** (*nervus pudendus*)
- 5 **Sciatic nerve** (*nervus ischiadicus*)



Posterior view of the left suprapiriform and infrapiriform foramina



Nerves of the sacral plexus

Branches

1 Superior gluteal nerve (*nervus gluteus superior*) – L4–S1

- runs through the suprapiriform foramen with superior gluteal artery and veins
- innervates the gluteus medius, gluteus minimus and tensor fasciae latae

2 Inferior gluteal nerve (*nervus gluteus inferior*) – L5–S2

- runs through the infrapiriform foramen with the inferior gluteal artery and veins
- innervates the gluteus maximus

3 Posterior cutaneous nerve of thigh / posterior femoral cutaneous nerve

(*nervus cutaneus femoris posterior*) – S1–S3

- runs through the infrapiriform foramen and passes over the inferior border of the gluteus maximus
- provides somatosensory innervation to the dorsal thigh,
- distally travels with the cranial extension of the small saphenous vein

3.1 Inferior clunial nerve (*nervi clunium inferiores*)

- provides somatosensory innervation to the distal part of the gluteal region

3.2 Perineal branches (*rami perineales*) – provide somatosensory innervation to the perineal skin

4 Pudendal nerve (*nervus pudendus*) – S2–S4

Course: runs through the infrapiriform foramen with the internal pudendal artery and veins, turns around the ischial spine into the lesser sciatic foramen, enters the ischioanal fossa and passes inside the pudendal canal to reach the external genital organs

Branches and structures innervated:

4.1 Muscular branches (*rami musculares*) – innervate the levator ani and ischiococcygeus

4.2 Inferior rectal branches (*nervi rectales inferiores*) – innervate the external anal sphincter and provide somatosensory innervation to the skin around the intergluteal crest

4.3 Perineal nerves and muscular branches (*nervi perineales et rami musculares*)

- motor branches for the transversus perinei superficialis, transversus perinei profundus, bulbospongiosus and ischiocavernosus

4.3.1 Posterior scrotal nerves (*nervi scrotales posteriores*)

- provide somatosensory innervation to the skin on the dorsal part of scrotum

4.3.2 Posterior labial nerves (*nervi labiales posteriores*)

- provide somatosensory innervation to the dorsal part of the labia

4.4 Dorsal nerve of penis (*nervus dorsalis penis*)

- provides somatosensory innervation to the penis and urethral mucosa

4.4 Dorsal nerve of clitoris (*nervus dorsalis clitoridis*) – somatosensory innervation of the clitoris

The **sciatic nerve** is the longest and thickest nerve in the human body. At its thickest point, it may be thicker than the spinal cord

The **peroneal nerve** (*nervus peroneus*) is a synonym for the fibular nerve.

Clinical notes

Paresis of the inferior gluteal nerve results in impaired function of the gluteus maximus, limiting extension of the hip joint. This causes difficulties in walking upstairs.

Disorders of the superior gluteal nerve lead to an impairment in abduction of the hip joint. Walking and standing on one foot become difficult and patients develop a duck-like gait. The Trendelenburg sign is positive. It is demonstrated by asking the patient to stand on the affected leg; the unsupported side of the pelvis will drop lower than the side the patient is standing on.

Pelvic fractures, hip dislocations and incorrectly applied injections can harm the sciatic nerve. Mild forms of sciatic nerve injury hinder the function of the extensors of the thigh and the flexors of the leg.

Damage to the tibial nerve usually occurs in the malleolar canal and can result from cut wounds, ankle fractures and incorrectly applied plasters. It presents as impaired plantar flexion (an inability to stand on tiptoes) due to impaired function of the triceps surae. The ankle jerk reflex is absent. Due to the predominance of the un-antagonised dorsal flexion of the tibialis anterior, the heel is the first part of the foot to touch the ground while walking. This results in a syndrome called “club foot” (*pes calcaneus*).

The **fibular nerve** is covered only by skin where it lies near the head of the fibula. In this location the nerve is most vulnerable to trauma and compression from incorrectly applied casts. Dislocations and sprains of the knee may also harm this nerve. The anterior muscles of the leg become impaired, resulting in a “flapping foot” and an inability to stand on the heels. The arches of the foot gradually deteriorate due to dysfunction of the fibularis longus.

5 Sciatic nerve (nervus ischiadicus) – L5–S2 with contributions from L4 and S3

Course: passes through the infrapiriform foramen between the muscles of the posterior thigh then descends distally towards the popliteal fossa

Branches and structures innervated – splits into two main branches

(usually in the distal third of the thigh): the tibial nerve and the common fibular nerve

5.1 **Muscular branches** (*rami musculares*) – innervates the biceps femoris, semitendinosus, semimembranosus and part of the adductor magnus

5.2 **Articular branches** (*rami articulares*) – partial somatosensory innervation of the hip joint

● 5.3 Common fibular nerve (*nervus fibularis communis*)

– runs medially to the tendon of the biceps femoris and passes through the popliteal fossa into the fibular canal between the fibularis longus and the soleus behind the head of the fibula

5.3.1 **Articular branches** (*rami articulares*)

– somatosensory innervation of the knee joint and tibiofibular joint

5.3.2 **Lateral sural cutaneous nerve** (*nervus cutaneus surae lateralis*)

– provides somatosensory innervation to the lateral calf

– the sural communicating branch (*ramus communicans fibularis*) emerges from it and joins the medial sural cutaneous nerve; together they form the sural nerve (*nervus suralis*)

● 5.3.3 Superficial fibular nerve (*nervus fibularis superficialis*)

– runs between the fibula and the fibularis longus

– continues between the fibular muscles, which it innervates (muscular branches),

– penetrates the fascia, reaches the surface and runs superficially over the extensor retinacula

5.3.3.1 **Medial dorsal cutaneous nerve** (*nervus cutaneus dorsalis medialis*) – splits into the dorsal digital nerves of the foot (*nn. digitales dorsales pedis*) which innervate the medial side of the 1st toe, lateral side of the 2nd toe and medial side of the 3rd toe

5.3.3.2 **Intermedial dorsal cutaneous nerve** (*nervus cutaneus dorsalis intermedius*)

– splits into the dorsal digital nerves of the foot which innervate the lateral side of the 3rd toe, the whole of the 4th toe and the medial side of the 5th toe

● 5.3.4 Deep fibular nerve (*nervus fibularis profundus*)

– runs through the fibular canal and anterior intermuscular septum of the leg into a space between the tibialis anterior and extensor hallucis longus

– follows the anterior tibial artery and veins under the extensor retinaculum and emerges on the surface in the 1st interdigital space

5.3.4.1 **Lateral dorsal digital nerve of toe** (*nervus digitalis dorsalis hallucis lateralis*)

– somatosensory innervation of the lateral side of the 1st toe

5.3.4.2 **Medial dorsal digital nerve of toe** (*nervus digitalis dorsalis digiti secundi medialis*)

– somatosensory innervation of the medial side of the 2nd toe

● 5.4 Tibial nerve (*nervus tibialis*)

– passes through the popliteal fossa with the popliteal artery and vein

– travels with the posterior tibial artery and veins under the tendinous arch of the soleus

– descends between the posterior muscles of the calf

– reaches the sole via the malleolar canal (*canalis malleolaris*) and then divides into the medial and lateral plantar nerves (*n. plantaris medialis et lateralis*)

5.4.1 **Medial sural cutaneous nerve** (*nervus cutaneus surae medialis*)

– runs with the small saphenous vein under the skin

– joins the sural communicating branch (*ramus communicans fibularis*) from the lateral sural cutaneous nerve to form the sural nerve

● 5.4.1.1 Sural nerve (*nervus suralis*) – somatosensory innervation of the calf and lateral side of the heel, follows the small saphenous vein

5.4.2 **Muscular branches** (*rami musculares*) – innervates the posterior muscles of the calf

● 5.4.3 Medial plantar nerve (*nervus plantaris medialis*) – follows the medial plantar artery and veins

– innervates the abductor hallucis, flexor digitorum brevis and the 1st and 2nd lumbrical

5.4.3.1 **Proper plantar digital nerve of toe** (*nervus digitalis plantaris proprius hallucis*)

– innervates the flexor hallucis brevis and the skin of the medial side of 1st toe

5.4.3.2 **Common plantar digital nerves** (*nervi digitales plantares communes*)

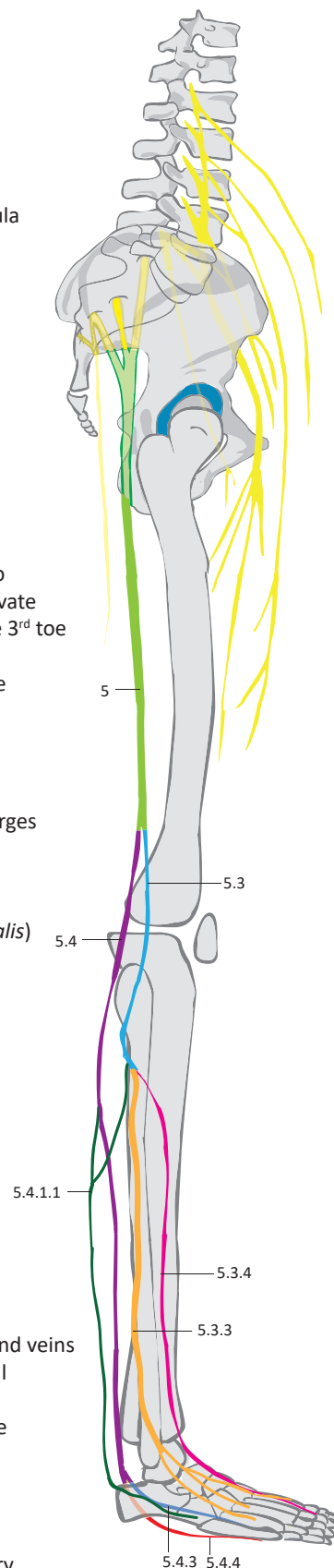
– split into the proper digital plantar nerves (*nervi digitales plantares proprii*), which innervate the lateral side of the 1st toe, both sides of the 2nd and 3rd toe and the medial side of the 4th toe

● 5.4.4 Lateral plantar nerve (*nervus plantaris lateralis*) – travels with the lateral plantar artery and veins to the 5th toe and innervates the quadratus plantae

5.4.4.1 **Superficial branch** (*ramus superficialis*) – gives off the proper plantar digital nerves of the 4th and 5th toe

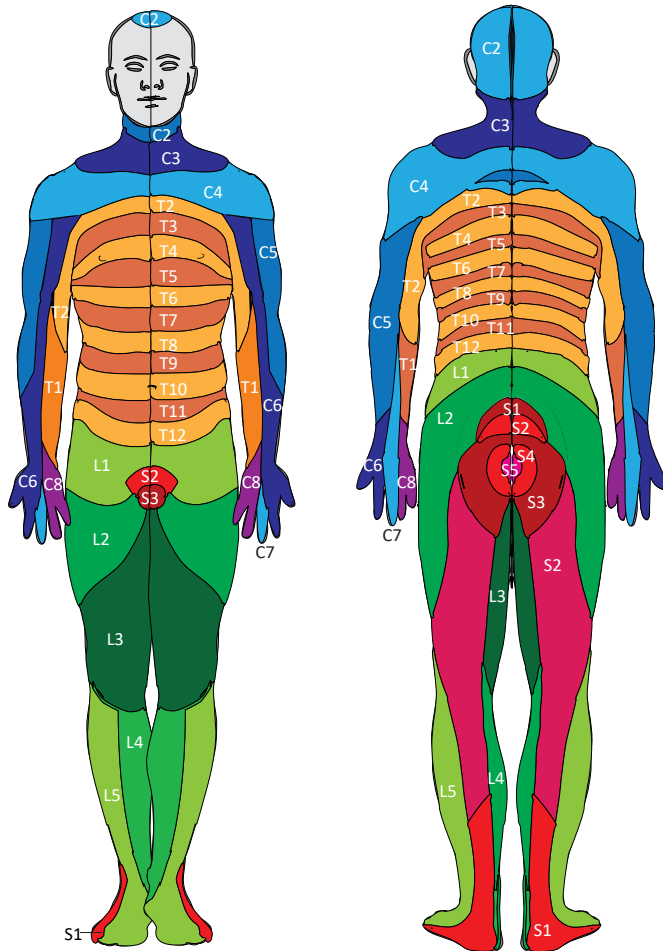
5.4.4.2 **Deep branch** (*ramus profundus*) – travels to the interossei and the oblique head of the adductor hallucis

– innervates the abductor digiti minimi, flexor digiti minimi brevis, interossei, adductor hallucis and the lumbricals of the 4th and 5th toes



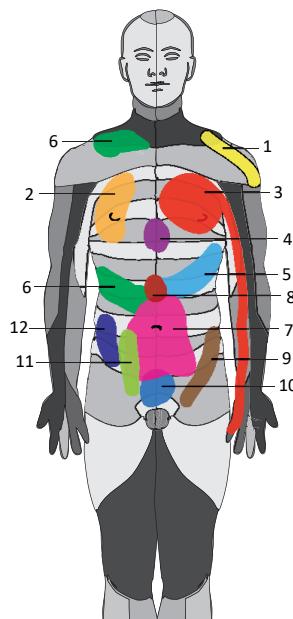
A **dermatome** (*area radicularis sensoria*) is an area of skin innervated by a posterior root of a single spinal nerve. A **myotome** (*area radicularis motorica*) is a group of muscles innervated by an anterior root of a single spinal nerve. Adjacent dermatomes may overlap, but never cross the midline of the body. Dermatomes retain segmental organisation on the thorax and abdomen and are arranged longitudinally along the long axes of the limbs. In the second intercostal space, the dermatomes skip from C4 to T2. The dermatome T4 contains the skin of the nipple; **T10** contains the **navel** (*umbilicus*).

Dermatomes



Head's zones – areas of referred pain

- 1 Diaphragm (C3–C5)
- 2 Lungs (T2–T5)
- 3 Heart (C8–T4)
- 4 Oesophagus (T4–T6) – the middle of the sternum
- 5 Stomach (T6–T9) – the epigastrium and the left part of the thorax
- 6 Liver, gall bladder (T7–T9) – the right subcostal area (Murphy's point), under the right scapula and the neck (both anterior and posterior aspects)
- 7 Intestine
 - 7.1 Small intestine (T8–L1)
 - 7.2 Large intestine to the Cannon-Böhm point (T11–L1) – aboral to the Cannon-Böhm point (L1–L2)
 - 7.3 Vermiform appendix (T11–T12)
- 8 Pancreas (T7–9) – dorsally to the inferior thoracic vertebra
- 9 Kidneys (T10–L1)
- 10 Ureters and urinary bladder (T11–L2)
- 11 Uterine tubes (T10–L1) and uterus (T12–L1)
- 12 Ovaries and testes (T10–T11)
- 13 Uterine cervix and prostate (S2–4)



A **myotome** is a group of muscles, which preserves its segmental innervation and therefore is innervated by one anterior branch of a spinal nerve.

A **dermatome** is innervated by one posterior root of a spinal nerve. This is why the pain often spreads in strips.

The muscles innervated from one spinal segment share the same embryological origin.

One spinal segment does not exclusively supply one nerve. Therefore, the radial nerve (C5–C8) is partially formed by the same segments as the axillary nerve (C5–C6).

Clinical notes

Head's zones are areas of skin innervated by the same spinal cord segments as particular internal organs. Due to rich interconnections of neurons within the spinal cord (convergence), diseases of the internal organs may cause pain or hyperesthesia of certain skin areas (referred or radiating pain). The skin of the anterior thorax and left upper limb corresponds to the heart. Cardiac pain can radiate further to the ulnar side of the left forearm and the little finger. This occurs in diseases such as angina pectoris and myocardial infarction.

The proper knowledge of dermatomes helps determine the level of spinal cord injury.

Intervertebral disc herniation is the most common cause of harm to the spinal nerves. It typically occurs at the spinal roots of L5 or S1. The pain typically radiates into the corresponding dermatomes. This differs from pain in peripheral nerve fields caused by injury to a peripheral nerve.

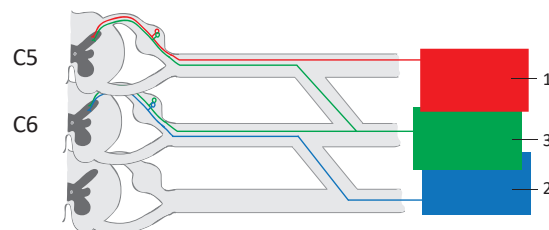
Pain in the L5 dermatome spreads on the lateral side of the thigh and leg down to the dorsal side of the foot and big toe. This segment innervates several muscles; there is weakness of plantar extension and the patient is unable to walk on his heels or extend the big toe.

Pain in the S1 segment radiates on the posterior thigh and calf towards the 5th toe. The function of the gluteus maximus is impaired and the patient has difficulties standing up from sitting and struggles to walk on tip toes.

Peripheral nerve field refers to the structures innervated by a single peripheral nerve. **The peripheral skin nerve field** (*area nervina sensoria*) is the area of skin innervated by a single peripheral nerve. **The peripheral muscle nerve field** (*area nervina motorica*) is the group of muscles innervated by a single peripheral nerve.

Area radicularis (dermatome) versus area nervina (nerve field) – examples

- 1 **Dermatome C5** – the area of skin innervated by all the nerves derived from the C5 spinal nerve root
- 2 **Dermatome C6** – the area of skin innervated by all the nerves derived from the C6 spinal nerve root
- 3 **Peripheral skin nerve field C5–C6** – the skin area innervated by a peripheral nerve, originating from spinal roots C5 and C6



Peripheral skin nerve fields (*areae nervinae sensoriae*)

1 Trigeminal nerve

- 1.1 Ophthalmic nerve – supraorbital nerve
- 1.2 Maxillary nerve – infraorbital nerve
- 1.3 Mandibular nerve – mental nerve

2 Cervical plexus

- 2.1 Lesser occipital nerve (C2–C3)
- 2.2 Great auricular nerve (C2–C3)
- 2.3 Transverse cervical nerve (C3)
- 2.4 Supraclavicular nerves (C3–C4)

3 Brachial plexus

- 3.1 Axillary nerve (C5–C6) – superior lateral cutaneous nerve of arm
- 3.2 Radial nerve (C5–T1) – inferior lateral and posterior cutaneous nerve of arm
- 3.3 Medial cutaneous nerve of arm (C8–T1)
- 3.4 Medial cutaneous nerve of forearm (C8–T1)
- 3.5 Musculocutaneous nerve (C5–C7) – lateral cutaneous nerve of forearm
- 3.6 Radial nerve (C6–C8) – posterior cutaneous nerve of forearm
- 3.7 Ulnar nerve (C7–T1)
- 3.8 Radial nerve (C6–C8) – superficial branch
- 3.9 Median nerve (C6–T1)

4 Thoracic nerves

- 4.1 Intercostal nerves – anterior cutaneous branches
- 4.2 Intercostal nerves – lateral cutaneous branches

5 Lumbar plexus

- 5.1 Iliohypogastric nerve (T12–L1)
- 5.2 Ilioinguinal nerve (L1)
- 5.3 Lateral cutaneous nerve of thigh (L2–L3)
- 5.4 Genitofemoral nerve (L1–L2)
- 5.5 Femoral nerve (L2–L4) – anterior cutaneous branches
- 5.5.1 Saphenous nerve

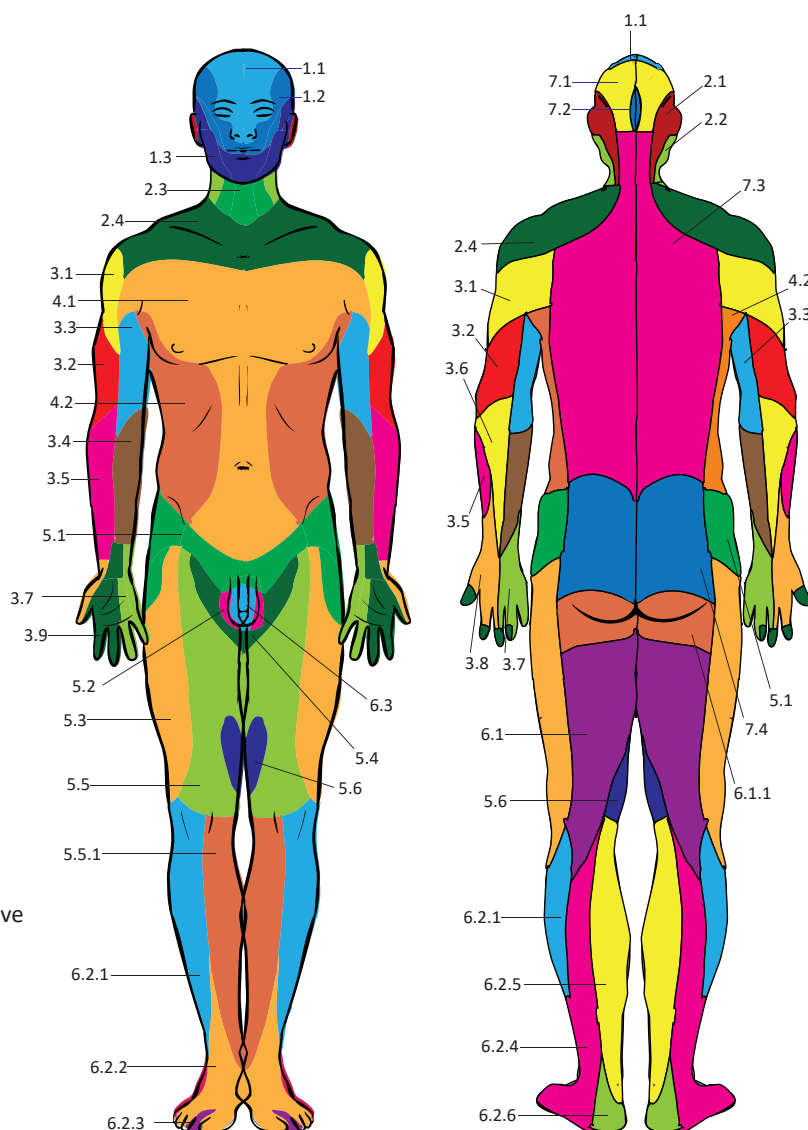
6 Sacral plexus

- 6.1 Posterior cutaneous nerve of thigh (S1–S3)
 - 6.1.1 Inferior clunial nerves
- 6.2 Sciatic nerve (L4–S3)
 - 6.2.1 Common fibular nerve – lateral sural cutaneous nerve
 - 6.2.2 Superficial fibular nerve
 - 6.2.3 Deep fibular nerve
 - 6.2.4 Sural nerve
 - 6.2.5 Tibial nerve – medial sural cutaneous nerve
 - 6.2.6 Medial plantar nerve
 - 6.2.7 Lateral plantar nerve
- 6.3 Pudendal nerve (S2–S4)

7 Posterior branches

- 7.1 Greater occipital nerve (C2)
- 7.2 Third occipital nerve (C3)
- 7.3 Posterior branches of spinal nerves
- 7.4 Superior (L1–L3) and middle (S1–S3) clunial nerves

Area radicularis (dermatome) versus area nervina (nerve field)



The twelve pairs of the cranial nerves emerge from the brain and brain stem. The first cranial nerve, the **olfactory nerve**, phylogenetically originates from the telencephalon. The second cranial nerve, the **optic nerve**, arises from the **diencephalon**. The remaining ten pairs emerge from the **brainstem**. From the developmental point of view, only these ten nerves are true cranial nerves.

Division

- N. I **Olfactory nerve** (*nervus olfactorius*)
– **afferent**: sense of smell
- N. II **Optic nerve** (*nervus opticus*)
– **afferent**: sense of vision
- N. III **Oculomotor nerve** (*nervus oculomotorius*)
– **somatomotor**: four eye muscles (the superior, inferior and medial rectus and the inferior oblique) and the levator palpebrae superioris
– **visceromotor (parasympathetic)**: the ciliary body and the sphincter pupillae
- N. IV **Trochlear nerve** (*nervus trochlearis*)
– **somatomotor**: the superior oblique
- N. V **Trigeminal nerve** (*nervus trigeminus*)
– **somatomotor**: the 4 masticatory muscles (masseter, temporalis and lateral and medial pterygoids), mylohyoid, anterior belly of the digastric, tensor veli palatini and tensor tympani
– **somatosensory**: the skin of the head ventral to the interauricular line, the dura mater and oral and nasal mucosa
- N. VI **Abducent nerve** (*nervus abducens*)
– **somatomotor**: one eye muscle (the lateral rectus)
- N. VII **Facial nerve** (*nervus facialis*)
– **somatomotor**: the stapedius, mimetic muscles, platysma, stylohyoid and posterior belly of the digastric
– **visceromotor (parasympathetic)**: sublingual, submandibular and lacrimal glands and other small glands (nasal, palatine and nasopharyngeal)
– **special sensory**: sense of taste in the anterior 2/3 of the tongue
- N. VIII **Vestibulocochlear nerve** (*nervus vestibulocochlearis*)
– **special sensory**: sense of hearing and balance
- N. IX **Glossopharyngeal nerve** (*nervus glossopharyngeus*)
– **somatomotor**: the stylopharyngeus
– **visceromotor (parasympathetic)**: innervation of the parotid gland and buccal glands
– **viscerosomatosensory**: carotid body and carotid sinus
– **somatosensory**: mucosa of the posterior part of the tongue, isthmus of the fauces, posterior part of the oral cavity, superior half of the pharynx, palatine tonsils and tympanic cavity
– **special sensory**: taste in the posterior third of the tongue
- N. X **Vagus nerve** (*nervus vagus*)
– **somatomotor**: muscles of the pharynx, larynx and soft palate
– **visceromotor (parasympathetic)**: trachea, bronchi, lungs, heart, oesophagus, liver, pancreas, small intestine, vermiform appendix, ascending colon, transverse colon, spleen, adrenals, kidneys, ovaries, testes
– **viscerosensory**: the aforementioned organs
– **somatosensory**: part of the auricle, external acoustic meatus, dura mater, larynx and pharynx
– **special sensory**: taste in the epiglottis
- N. XI **Accessory nerve** (*nervus accessorius*)
– **somatomotor**: trapezius and sternocleidomastoid
some branches join the vagus nerve to innervate the laryngeal muscles
- N. XII **Hypoglossal nerve** (*nervus hypoglossus*)
– **somatomotor**: 3 extraglossal (genioglossus, styloglossus and hyoglossus) and 4 intraglossal muscles (superior and inferior longitudinal, transverse and vertical muscles)

Classification according to the type of nerve fibres

Sensory:

- Somatosensory** – n. V, IX, X
Viscerosensory – n. IX, X
Special sensory – n. I, II, VII, VIII, IX, X

Motor:

- Somatomotor (of somite origin)** – n. III, IV, VI, XII
Branchiomotor (of branchial origin) – n. V, VII, IX, X, XI
Visceromotor (parasympathetic) – n. III, VII, IX, X

The cranial nerves are marked by Roman numerals I–XII.

A mnemonic for the origin of the cranial nerves: 1–1–2–4–4

- 1 nerve emerges from the telencephalon
1 nerve emerges from the diencephalon
2 nerves emerge from the mesencephalon
4 nerves emerge from the pons
4 nerves emerge from the medulla oblongata

The true cranial nerves originate from the brainstem nuclei.

The false cranial nerves are the olfactory nerve (n. I) and the optic nerve (n. II). The spinal root (radix spinalis) of the accessory and hypoglossal nerves arise from several spinal cord segments. They are drawn inside the skull during embryogenesis. This explains the close relation of the hypoglossal nerve to the deep cervical ansa.

The **branchial motor** (branchiomotor) neurones are somatomotor neurones for the muscles originating in the branchial (pharyngeal) arches.

The **ganglia of the cranial nerves**, just as the ganglia of the spinal nerves, contain pseudounipolar neurones. However, these ganglia have a different embryologic origin. They originate from neural crest cells and cells of the epibranchial (epipharyngeal) placodes.

The **lateral mixed system** is the name for the common origin of the glossopharyngeal, vagus and accessory nerves from the medulla oblongata and their shared pathway through the jugular foramen out of the skull. These nerves also share some nuclei in the brainstem.

The **nuclei of the cranial nerves**, see page 411.

Origin of the cranial nerves

Telencephalon

- N. I **Olfactory nerve**
– the olfactory triangle

Diencephalon

- N. II **Optic nerve**
– the lateral geniculate body

Mesencephalon

- N. III **Oculomotor nerve**
– the groove for the oculomotor nerve in the interpeduncular fossa
- N. IV **Trochlear nerve**
– caudal to the inferior colliculus

Pons

- N. V **Trigeminal nerve**
– next to the middle cerebellar peduncle
(*pedunculus cerebellaris medius*)
- N. VI **Abducens nerve**
– the medullopontine sulcus at the border with the medulla oblongata
- N. VII **Facial nerve**
– the cerebellopontine angle
- N. VIII **Vestibulocochlear nerve**
– the cerebellopontine angle

Medulla oblongata

- N. IX **Glossopharyngeal nerve** – the retro-olivary groove
- N. X **Vagus nerve** – the retro-olivary groove
- N. XI **Accessory nerve** – the retro-olivary groove
- N. XII **Hypoglossal nerve** – the pre-olivary groove

How the nerves leave the skull

Anterior cranial fossa (*fossa cranii anterior*)

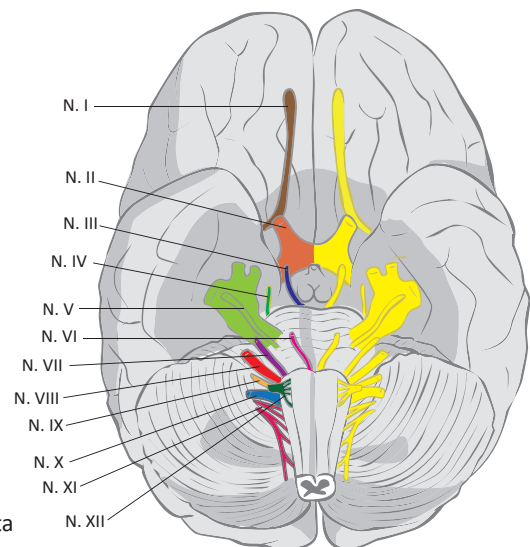
- N. I **Olfactory nerve**
– through the cribriform plate of the ethmoidal bone

Middle cranial fossa (*fossa cranii media*)

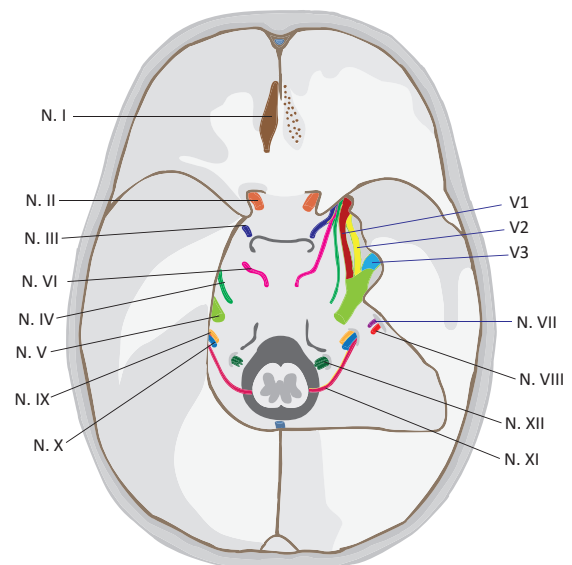
- N. II **Optic nerve**
– through the optic canal of the sphenoid bone
- N. III **Oculomotor nerve**
– through the superior orbital fissure of the sphenoid bone
- N. IV **Trochlear nerve**
– through the superior orbital fissure of the sphenoid bone
- N. V **Trigeminal nerve** – through the sphenoid bone
 - V₁ **Ophthalmic nerve** (*nervus ophthalmicus*)
– through the superior orbital fissure
 - V₂ **Maxillary nerve** (*nervus maxillaris*)
– through the foramen rotundum
 - V₃ **Mandibular nerve** (*nervus mandibularis*)
– through the foramen ovale
- N. VI **Abducent nerve**
– through the superior orbital fissure of the sphenoid bone
- N. VII **Facial nerve**
– through the internal acoustic meatus of the temporal bone
- N. VIII **Vestibulocochlear nerve**
– through the internal acoustic meatus of the temporal bone

Posterior cranial fossa (*fossa cranii posterior*)

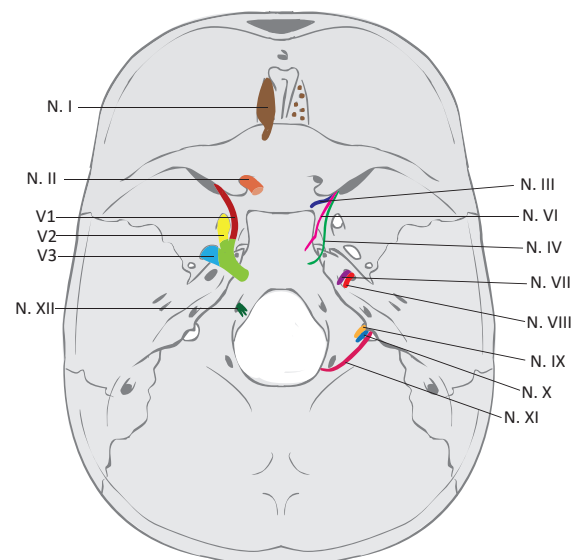
- N. IX **Glossopharyngeal nerve**
- N. X **Vagus nerve**
- N. XI **Accessory nerve**
– through the posterolateral part of the jugular foramen
(between the temporal and occipital bones)
- N. XII **Hypoglossal nerve**
– through the hypoglossal canal



Origins of the cranial nerves



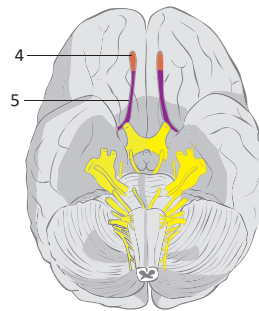
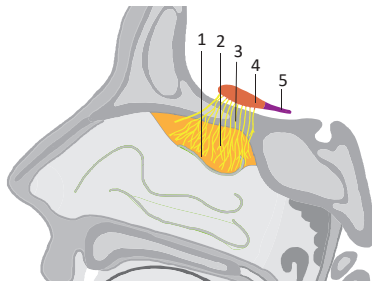
Course of the cranial nerves through the dura mater



Course of the cranial nerves through the skull

The **olfactory nerve** conducts the sensation of smell to the brain via the olfactory bulb and tract. It develops from a **pouch of the telencephalon**. It consists of the axons of the olfactory cells (**fila olfactoria**) in the **nasal mucosa**, which penetrate the cribriform plate of the ethmoid bone and enter the anterior cranial fossa, where they synapse with the **mitral cells** in the olfactory bulb. The axons of mitral cells travel in the olfactory tract to the **olfactory trigone** and the **olfactory area** of the cerebral cortex.

- 1 **Olfactory region** (*pars olfactoria cavitatis nasi*) – specialised olfactory epithelium on the roof of the nasal cavity and superior part of the nasal septum
- 2 **Olfactory cells** – bipolar neurons
- 3 **Olfactory nerve** (*fila olfactoria / nervus olfactorius*)
 - consists of bundles of axons of the olfactory cells
 - passes through the cribriform plate to synapse in the olfactory bulb
- 4 **Olfactory bulb** – consists of olfactory nerve synapsing on mitral cells
- 5 **Olfactory tract** (*tractus olfactorius*) – the axons of the mitral cells
 - travels to the cortex and subcortical structures



The **neurons of the olfactory tract** send impulses to the olfactory cortex for the conscious perception of smell. They also send impulses to some subcortical structures (the amygdala, septum verum and dorsomedial thalamus). The subcortical connections work in cooperation with the limbic system to create emotional reactions and evoke both good and bad memories in response to smell.

The **fibres of the oculomotor nerve** destined for the superior rectus decussate within the brain stem and join the contralateral nerve.

The **fibres of the trochlear nerve** decussate within the tectum of the mesencephalon.

Contraction of the ciliary muscle (*m. ciliaris*) focuses the eye on near objects. Relaxation focuses the eye on distant objects.

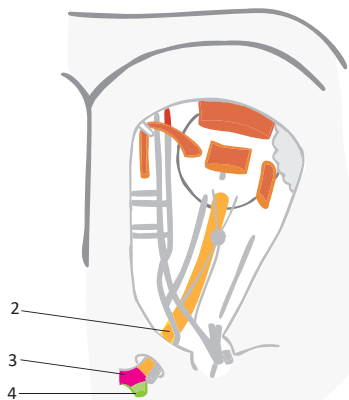
The **sphincter pupillae** (*m. sphincter pupillae*) narrows the pupil (miosis).

Visual pathway, see page 466.

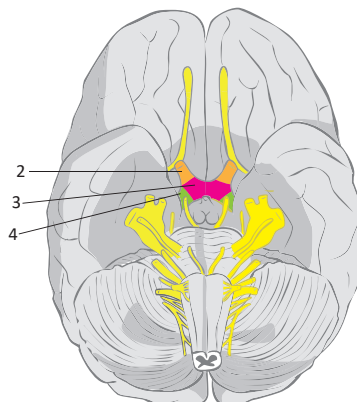
Olfactory pathway, see page 462.

The optic nerve carries **visual information** to the brain via the optic tract. It develops from a pouch of the diencephalon. It consists of the **axons of the retinal ganglion cells** (the 3rd order neurons of the optic pathway). It originates on the posterior side of the eyeball and enters the cranial cavity through the **optic canal**. It passes through the **optic chiasma**, where **about half of its fibres decussate with the contralateral nerve** and then continues as the **optic tract** to the **lateral geniculate nucleus** in the metathalamus. The meninges cover the optic nerve in its entire course from the brain to the eyeball. Dura mater forms the external sheath of the optic nerve and the leptomeninges form the internal sheath of the optic nerve. The central retinal artery enters the optic nerve in the orbit and supplies the retina.

- 1 **Retina** – contains the 1st, 2nd and 3rd order neurons of the optic pathway
- 2 **Optic nerve** – the axon of the 3rd order neurons of the optic pathway (optic ganglion)
- 3 **Optic chiasma** – the decussation of the right and the left optic nerves
- 4 **Optic tract** – the continuation of the optic nerve to the metathalamus
- 5 **Lateral geniculate nucleus** – the site of synapse with the 4th order neurons of the optic pathway



Superior view of the right orbit and the course of the optic nerve



Clinical notes

Hyposmia is a reduced ability to smell.

Anosmia is a complete loss of smell.

Hyperosmia is increased sensitivity to smell.

Parosmia is a qualitative smell defect.

Unilateral paresis of the oculomotor nerve causes divergent strabismus, dilated pupil (mydriasis), impaired accommodation (cycloplegia), ptosis and double vision (diplopia).

Unilateral paresis of the trochlear nerve causes diplopia while looking down and laterally.

There are **two nuclei** for the oculomotor nerve in the **midbrain**: the nucleus of the oculomotor nerve and the accessory posterior nucleus of the oculomotor nerve. The oculomotor nerve emerges from the midbrain in the **interpeduncular fossa**. It penetrates the dura mater to run laterally in the wall of the **cavernous sinus**. It **enters the orbit** through the superior orbital fissure. It carries **parasympathetic fibres and somatomotor fibres**. The parasympathetic fibres innervate the ciliary muscle, which focuses the eye on near objects (accommodation), and the sphincter pupillae, which constricts the pupil. The somatomotor fibres innervate most of the extraocular muscles. The **parasympathetic visceromotor** fibres synapse in the **ciliary ganglion**, which is located posterior to the eyeball in the orbit.

Nuclei

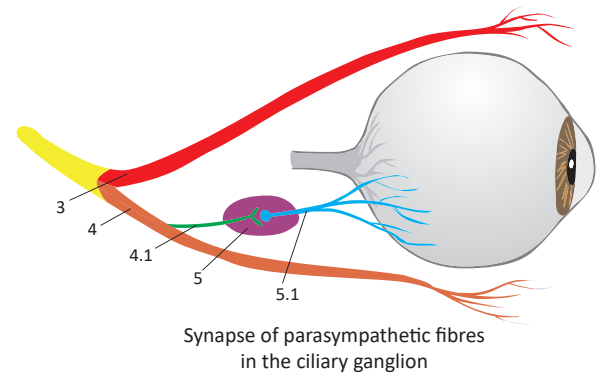
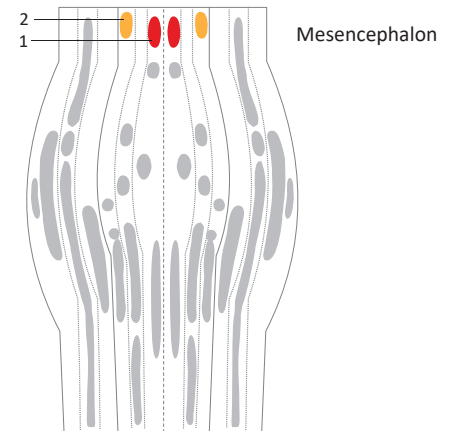
- 1 **Nucleus of oculomotor nerve** (*nucleus nervi oculomotorii*)
 - a somatomotor nucleus
 - innervates the superior, inferior and medial rectus, inferior oblique and levator palpebrae superioris
- 2 **Accessory posterior nucleus of oculomotor nerve / Edinger-Westphal nucleus** (*nucleus accessorius dorsalis nervi oculomotorii*)
 - a visceromotor parasympathetic nucleus
 - innervates the sphincter pupillae and ciliary muscles

Course

1. emerges from midbrain in the interpeduncular fossa
2. penetrates the dura mater and runs laterocranially in the cavernous sinus
3. enters the orbit via the superior orbital fissure within the common tendinous ring (*anulus tendineus communis*)

Branches and structures innervated:

- 3 **Superior branch** (*ramus superior*)
 - innervates the superior rectus and levator palpebrae superioris
- 4 **Inferior branch** (*ramus inferior*) –
 - innervates the inferior and medial rectus and inferior oblique
 - 4.1 **Branch to ciliary ganglion** (*ramus ad ganglion ciliare*)
 - parasympathetic preganglionic fibres from the Edinger-Westphal nucleus for the ciliary ganglion
- 5 **Ciliary ganglion** (*ganglion ciliare*) – a parasympathetic ganglion
 - is located within the orbit, posterior to the eyeball and lateral to the optic nerve
 - 5.1 **Short ciliary nerves** (*nervi ciliares breves*)
 - contain parasympathetic postganglionic fibres, which leave the ciliary ganglion to innervate the ciliary muscle and sphincter pupillae



3.4

N. IV Trochlear nerve – Nervus trochlearis

The trochlear nerve decussates in the midbrain before emerging under the inferior colliculus. It is the **only nerve that leaves from the posterior side of the brainstem**. It passes the lateral surface of the mesencephalon to enter and run on the lateral wall of the **cavernous sinus**. It then enters the orbit via the **superior orbital fissure**. The trochlear nerve is exclusively **somatomotor** and innervates only one muscle: **the superior oblique**.

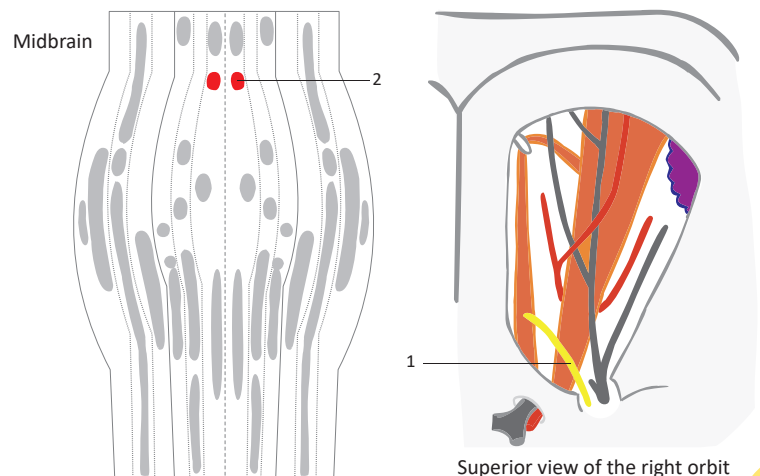
- 1 **Trochlear nerve** (*nervus trochlearis*)

Nucleus

- 2 **Nucleus of trochlear nerve** (*nucleus nervi trochlearis*)
 - a somatomotor nucleus in the mesencephalon
 - innervates the superior oblique

Course

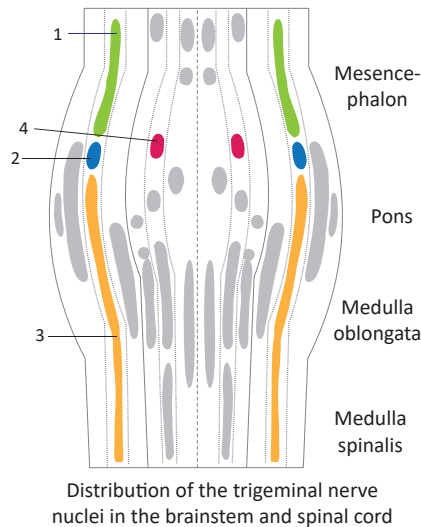
1. emerges under the inferior colliculus on the posterior side of the mesencephalon
3. passes around the cerebral peduncle within the cisterna ambiens
4. runs laterocranially in the cavernous sinus
5. enters the orbit via the superior orbital fissure, but not within the common tendinous ring



The trigeminal nerve leaves the **pons** at the level of the middle cerebellar peduncle. The nerve has two roots: a **sensory root** (*radix sensoria*) and a **motor root** (*radix motoria*). The trunk of the nerve is formed from both roots and runs towards the apex of petrous part of the temporal bone, where the somatosensory **trigeminal ganglion** is located in the **trigeminal impression** (*impressio trigeminalis*). The nerve then splits into **three terminal branches**: the ophthalmic nerve, the maxillary nerve and the mandibular nerve. The skin and organs of the face and head up to the interauricular line receive **somatosensory fibres** from branches of the trigeminal nerve. All three branches of the trigeminal nerve carry somatosensory fibres, but only the **mandibular nerve contains motor fibres**. They innervate the **masticatory muscles** and four other **muscles**, which originate from the **first pharyngeal arch**. Parasympathetic and sympathetic fibres join peripheral branches of trigeminal nerve at autonomic ganglia to innervate glands and smooth muscle of the head and neck.

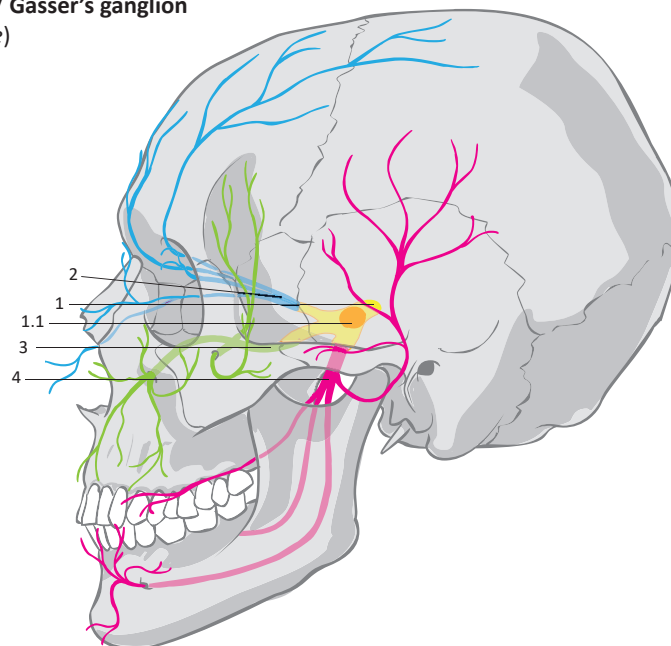
Nuclei

- **1 Mesencephalic nucleus of trigeminal nerve** (*nucleus mesencephalicus nervi trigemini*)
 - the most dorsal nucleus in the mesencephalon
 - receives proprioceptive information from the masticatory, mimetic and extra-ocular muscles
- **2 Principal nucleus of trigeminal nerve** (*nucleus principalis nervi trigemini*)
 - is located in the pons
 - receives information about fine touch, vibration and some proprioception
- **3 Spinal nucleus of trigeminal nerve** (*nucleus spinalis nervi trigemini*) – is located in the medulla oblongata and spinal cord (C1–3)
 - receives information about crude touch, pain and temperature
 - receives the same modalities from n. IX and n. X
- **4 Motor nucleus of trigeminal nerve** (*nucleus motorius nervi trigemini*) – is located in the pons
 - innervates the muscles of the first pharyngeal arch: masticatory muscles (masseter, temporalis, lateral and medial pterygoid), tensor tympani, tensor veli palatini, anterior belly of the digastric and the mylohyoid



Main branches

- **1 Trigeminal nerve** (*nervus trigeminus*)
 - **1.1 Trigeminal ganglion / Gasser's ganglion** (*ganglion trigeminale*)
- **2 Ophthalmic nerve** (*nervus ophthalmicus*)
- **3 Maxillary nerve** (*nervus maxillaris*)
- **4 Mandibular nerve** (*nervus mandibularis*)



The trigeminal nerve is the largest cranial nerve.

The **mesencephalic nucleus** of the trigeminal nerve receives proprioception information from the masticatory muscles. A feedback loop is located between it and the motor nucleus of the trigeminal nerve. This ensures optimal masticatory muscles tension and prevents damage of the teeth.

The **function of this loop may be demonstrated as follows**: if you try to bite your tongue very strongly, the feedback loop diminishes the tension of the masticatory muscles in order to prevent injury.

The **principal nucleus** of the trigeminal nerve was formerly called the pontine nucleus of the trigeminal nerve.

The **sensory part of the trigeminal nerve** was formerly called the greater portion (*portio major*) and the motor part the lesser portion (*portio minor*).

The **points at which the branches of the trigeminal nerve leave the skull** are palpable on the face:

V1 (as the supraorbital nerve) – in the eyebrows, on a line passing vertically through the pupil

V2 (as the infraorbital nerve) – inferior to the eye, on a line between the pupil and angle of the mouth

V3 (as the mental nerve) – on the mandible, on a line passing vertically downwards from the angle of the mouth.

Painful palpation at any of these points signifies pathological irritation of the trigeminal nerve.

The **superior and inferior dental and gingival branches** (*rami gingivales et dentales superiores and inferiores*) form the superior and inferior dental plexus.

Clinical notes

Trigeminal neuralgia / prosopalgia is characterised by intensive unbearable episodic facial pain in areas innervated by branches of the trigeminal nerve (the supraorbital, infraorbital and mental nerves). The most common cause is impingement of the trigeminal nerve by an aberrant loop of the superior cerebellar artery at the junction between the CNS and PNS, where oligodendrocytes and Schwann cells meet (the Obersteiner-Redlich zone). This may be treated by microvascular decompression. Non-lesional causes can be treated by gamma-knife surgery (radiosurgery) of the trigeminal ganglion.

V₁ Ophthalmic nerve (*nervus ophthalmicus*)

Fibres

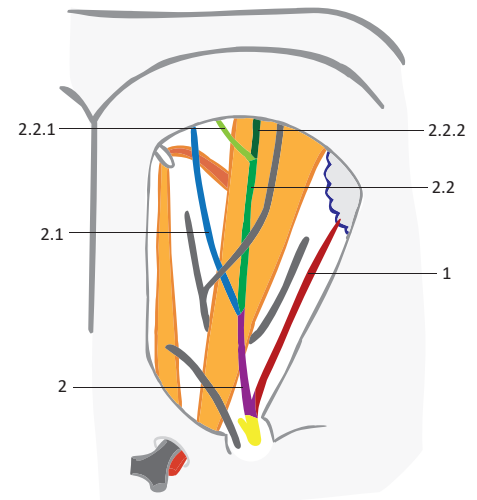
– somatosensory innervation of the face (forehead, upper eyelids, nasal dorsum), part of the dura mater, the orbit and its content, part of the mucosa in the nasal cavity and the paranasal sinuses

Course

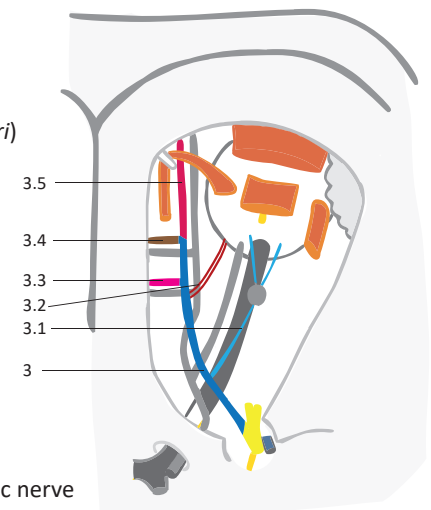
– passes through the superior orbital fissure into the orbit
– some of its branches continue to the forehead

Branches and structures innervated

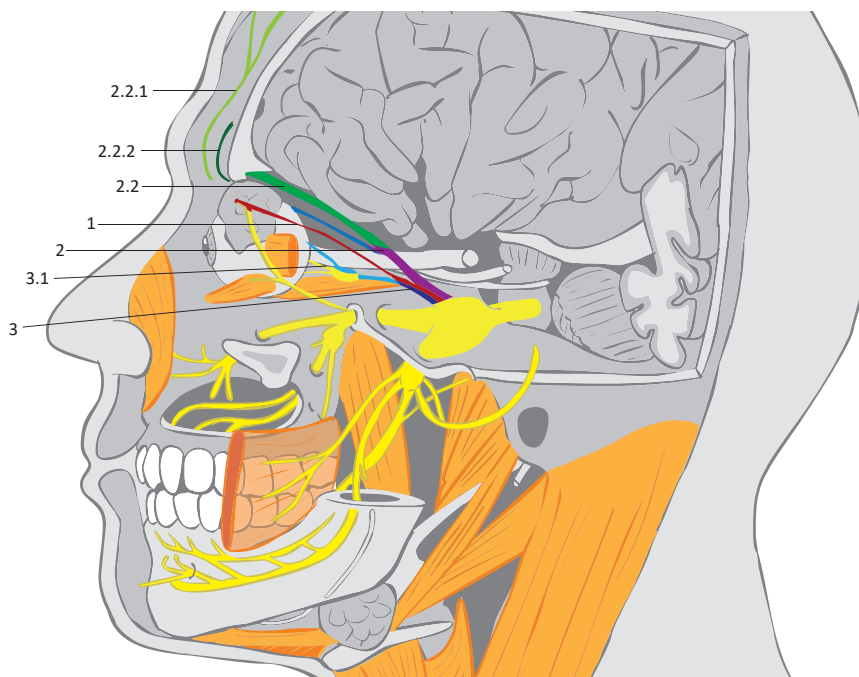
- **1 Lacrimal nerve** (*nervus lacrimalis*)
 - runs along the lateral rectus towards the lacrimal gland
 - receives a communicating branch with the zygomatic nerve (*ramus communicans nervi lacrimalis cum nervo zygomatico*), which brings parasympathetic fibres to the lacrimal gland
 - innervates the skin of the lateral part of the eyelid and the adjacent part of the conjunctiva
- **2 Frontal nerve** (*nervus frontalis*)
 - runs between the orbital periosteum and the levator palpebrae superioris
 - **2.1 Supratrochlear nerve** (*nervus supratrochlearis*) – innervates the skin and conjunctiva of the medial part of the eyelid and the skin of the forehead
 - **2.2 Supraorbital nerve** (*nervus supraorbitalis*)
 - **2.2.1 Medial branch** (*ramus medialis*) – leaves the orbit via the frontal notch – innervates the skin of the upper eyelid and forehead
 - **2.2.2 Lateral branch** (*ramus lateralis*) – leaves the orbit via the supraorbital notch – innervates the skin of the upper eyelid and forehead
- **3 Nasociliary nerve** (*nervus nasociliaris*)
 - runs from the lateral part of the orbit to its medial wall
 - **3.1 Communicating branch with ciliary ganglion** (*ramus communicans cum ganglio ciliari*) – carries sensory fibres for the eyeball, which pass through the ciliary ganglion and then continue as the short ciliary nerves (*nervi ciliares breves*) to reach the eyeball
 - **3.2 Long ciliary nerves** (*nervi ciliares longi*) – sensory innervation of the eyeball
 - **3.3 Posterior ethmoidal nerve** (*nervus ethmoidalis posterior*) – leaves the orbit via the posterior ethmoidal foramen and orbitoethmoidal canal
 - **3.4 Anterior ethmoidal nerve** (*nervus ethmoidalis anterior*) – leaves the orbit through the anterior ethmoidal foramen, runs through the orbitocranial canal and then through the cribriform plate into the nasal cavity – sensory innervation of the frontal sinus, anterior nasal mucosa and skin of the apex and dorsum of the nose
 - **3.5 Infratrochlear nerve** (*nervus infratrochlearis*) – the terminal branch of the ophthalmic nerve – sensory innervation of the medial part of both eyelids and the adjacent conjunctiva



Superior level of the orbit – right side



Middle level of the orbit – right side



V₂ Maxillary nerve (*nervus maxillaris*)**Fibers**

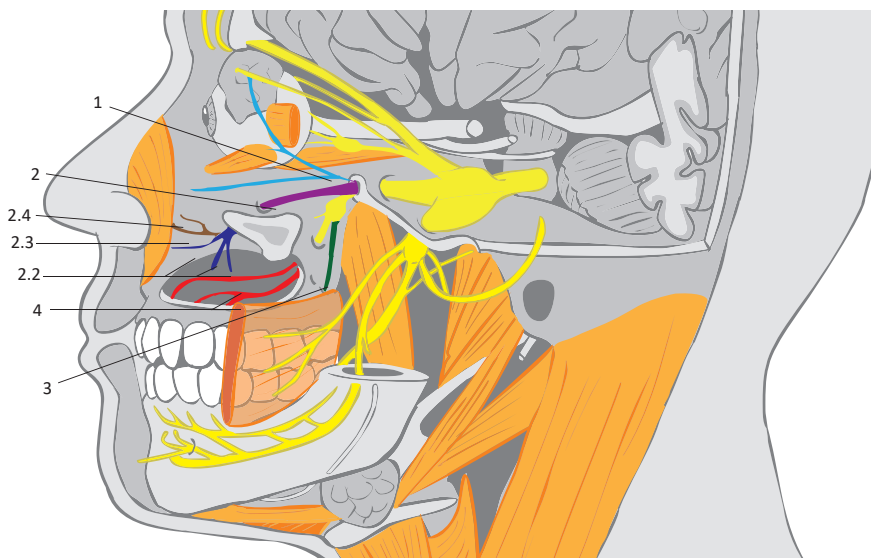
– somatosensory innervation of the middle part of the face (lower eyelid, upper lip, ala of nose and upper part of the cheek), upper teeth and part of the nasal, oral and nasopharyngeal mucosa

Course

– runs through the foramen rotundum to reach the pterygopalatine fossa
 – branches at the pterygopalatine ganglion

Branches and structures innervated

- **1 Zygomatic nerve (*nervus zygomaticus*)** – reaches the orbit via the inferior orbital fissure, runs on the inferior orbital wall
 - enters the zygomatico-orbital foramen and branches within the zygomatic bone
 - 1.1 Zygomaticofacial branch (*ramus zygomaticofacialis*)**
 - reaches the upper lateral part of the cheek via the zygomaticofacial foramen
 - innervates the skin of the zygomatic region
 - 1.2 Zygomaticotemporal branch (*nervus zygomaticotemporalis*)**
 - reaches the temporal fossa by passing through the zygomaticotemporal foramen
 - innervates the skin of the temporal region
- **2 Infraorbital nerve (*nervus infraorbitalis*)** – enters the orbit through the inferior orbital fissure, runs in the infra-orbital groove then enters the infra-orbital canal
 - leaves the infra-orbital canal through the infra-orbital foramen to reach the anterior surface of the maxilla, where it branches into:
 - 2.1 Inferior palpebral branches (*rami palpebrales inferiores*)**
 - innervate the skin and conjunctiva of the lower eyelid
 - **2.2 Superior labial branches (*rami labiales superiores*)** – innervate the skin of the upper lip and the adjacent part of the mucosa in the oral vestibule
 - **2.3 External nasal branches (*rami nasales externi*)** – innervate the skin of the ala of the nose
 - **2.4 Internal nasal branches (*rami nasales interni*)** – innervate the nasal vestibule mucosa
- **3 Posterior superior alveolar branches (*rami alveolares superiores posteriores*)**
 - travel towards the maxillary tuberosity and enter the alveolar foramina, then travel in the alveolar canals to reach the posterior upper teeth
 - 3.1 Superior dental branches (*rami dentales superiores*)**
 - innervate the teeth of the upper jaw
 - 3.2 Superior gingival branches (*rami gingivales superiores*)** – innervate the adjacent gingiva
- **4 Anterior superior alveolar branches and middle superior alveolar branch (*rami alveolares superiores anteriores et ramus alveolaris superior medius*)**
 - originate in the infra-orbital canal and groove
 - 4.1 Superior dental branches (*rami dentales superiores*)**
 - innervate the remaining teeth of the upper jaw
 - 4.2 Superior gingival branches (*rami gingivales superiores*)** – innervate the adjacent gingiva
- 5 Posterior nasal branches (*rami nasales posteriores*)**
 - innervate the posterior 2/3 of the nasal mucosa
- 6 Greater palatine nerve (*nervus palatinus major*) and lesser palatine nerves (*nervi palatini minores*)** – innervate the mucosa of the hard and soft palates
- 7 Pharyngeal nerve (*nervus pharyngeus*)** – innervate the mucosa of the nasopharynx



The three branches of the infraorbital nerve (for the eyelid, nose and upper lip) in the infra-orbital foramen are collectively known as the *pes anserinus minor*.

The function of the trigeminal nerve may be assessed by the following reflexes:

The masseter reflex is provoked by tapping the teeth of the lower jaw. Stretching tendon of the masseter evokes its contraction.

The corneal reflex is provoked by gently touching the conjunctiva. It results in the contraction of the orbicularis oculi (causing a wink). The ophthalmic nerve provides the afferent limb of the reflex arc and the facial nerve provides the efferent limb.

The trigeminal nerve innervates most of the meninges. Each of its branches gives off one meningeal branch:

V1 – tentorial branch (*ramus tentorius*) before passing through the superior orbital fissure

V2 – meningeal branch (*ramus meningeus*) before passing through the foramen rotundum

V3 – meningeal branch (*ramus meningeus*) is given off in the infratemporal fossa and enters the cranial cavity via the foramen spinosum. The ventral part of the falx cerebri is innervated by the anterior ethmoidal nerve.

A tiny part of the meninges around the jugular foramen is innervated by the vagus nerve.

Clinical notes

Regional anaesthesia is different for the upper and lower jaw. For the lower teeth it is administered at a single point posterior to the 3rd molar, next to the trunk of the inferior alveolar nerve. For the upper teeth there are 3 points (anterior, middle, posterior) to administer anaesthesia to the individual branches of the infraorbital nerve as they leave the maxilla.

Herpes zoster ophthalmicus is caused by reactivation of the varicella-zoster virus in the ophthalmic nerve. It presents with a painful rash in of the skin innervated by the ophthalmic nerve and can affect the conjunctiva and cornea.

V₃ Mandibular nerve – *nervus mandibularis*

Fibres

- somatosensory innervation of the skin of the lower part of the face (lower lip, part of the cheek and chin), teeth of the lower jaw and part of the oral mucosa
- innervation of 8 muscles: 4 masticatory muscles, anterior belly of digastric, mylohyoid, tensor veli palatini and tensor tympani

Course

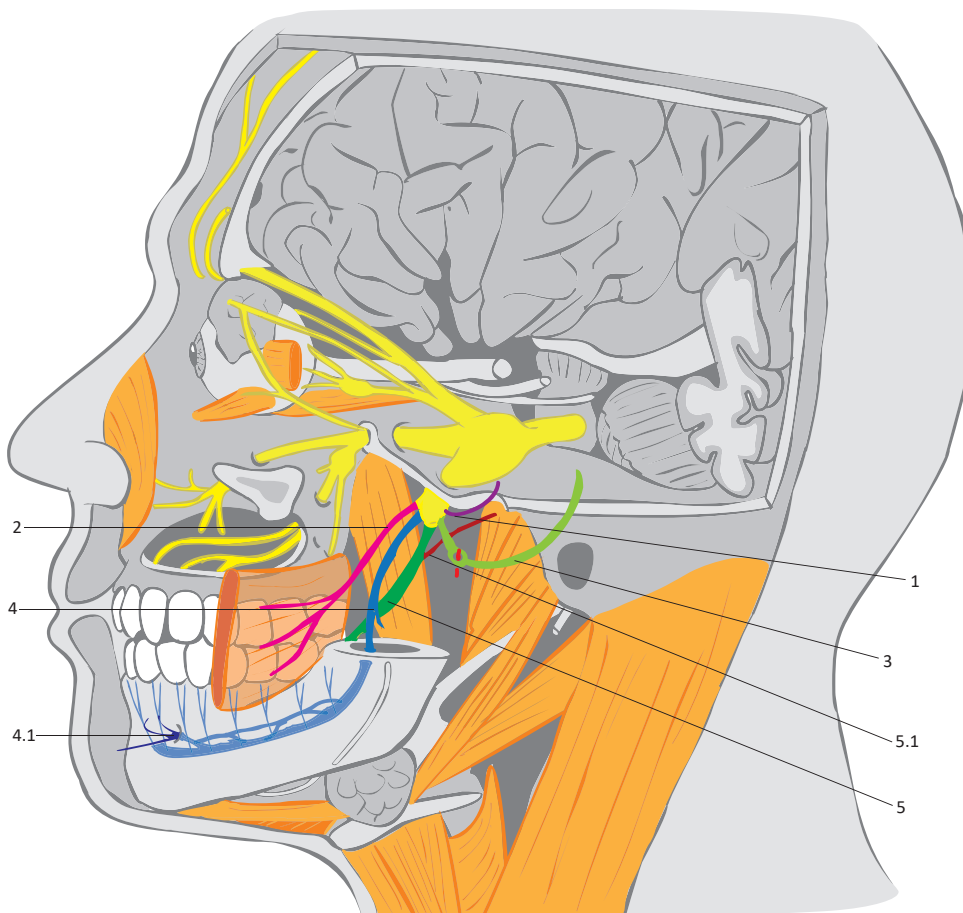
- travels through the foramen ovale to reach the infratemporal fossa, where it gives off its terminal branches
- the otic ganglion is found on the medial side of the nerve

Somatomotor branches

- 1 **Deep temporal branches** (*nervi temporales profundi*) – innervate the temporalis
- 2 **Medial and lateral pterygoid nerves** (*nervus pterygoideus medialis et lateralis*) – innervate the medial and lateral pterygoid muscles
- 3 **Masseteric nerve** (*nervus massetericus*) – innervates the masseter
- 4 **Mylohyoid muscle** (*nervus mylohyoideus*) – innervates the mylohyoid and the anterior belly of the digastric
- 5 Branches for the tensor veli palatini and tensor tympani

Somatosensory branches

- 1 **Meningeal branch** (*ramus meningeus*) – innervates the meninges of the middle cranial fossa
- 2 **Buccal nerve** (*nervus buccalis*) – innervates the skin of the face and the adjacent part of the oral mucosa
- 3 **Auriculotemporal nerve** (*nervus auriculotemporalis*) – innervates part of the auricle and the skin of the temporal region
 - forms a loop around the middle meningeal artery
- 4 **Inferior alveolar nerve** (*nervus alveolaris inferior*) – innervates the teeth, gingiva and oral mucosa
 - passes through the mandibular foramen and canal
 - 4.1 **Mental nerve** (*nervus mentalis*) – the terminal branch emerging from the mental foramen
 - innervates the skin covering the body of the mandible
- 5 **Lingual nerve** (*nervus lingualis*) – a thick nerve, somatosensory innervation of the anterior 2/3 of the tongue
 - 5.1 **Chorda tympani** – a branch of the facial nerve containing visceromotor and sensory taste fibers
 - joins the lingual nerve and continues to the submandibular ganglion and into the tongue
 - gustatory fibers carry taste information from the anterior 2/3 of the tongue (from the apex to the terminal sulcus, excluding the vallate papillae)
 - 5.2 **Submandibular ganglion** (*ganglion submandibulare*) – a parasympathetic ganglion for fibres from the facial nerve



The abducent nerve originates at the border between the pons and the medulla oblongata, the **medullopontine sulcus** (*sulcus bulbopontinus*). It travels ventrolaterally on the internal surface of the base of the skull, penetrates the meninges and runs medially within the **cavernous sinus**. It reaches the **orbit** through the **superior orbital fissure** and common tendinous ring and innervates the **lateral rectus**, which causes the eyeball to abduct.

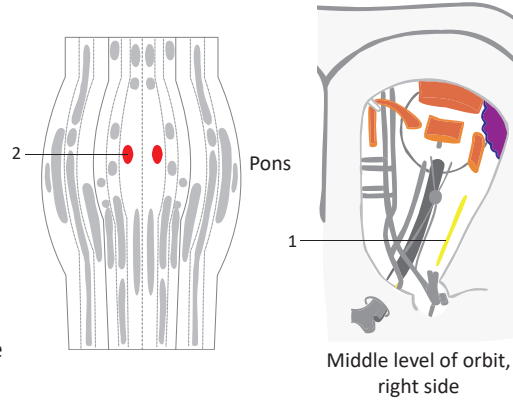
1 Abducens/abducent nerve

Nucleus

- 2 Nucleus of abducens nerve (*nucleus nervi abducentis*)
 - a somatomotor nucleus
 - is located in the pons, inferior to the facial colliculus
 - innervates the lateral rectus

Course

1. emerges from the medullopontine sulcus between the pons and medulla oblongata
2. leaves the brainstem on the ventral side
3. runs on the base of the skull, penetrates the dura mater then runs through Dorello canal to enter the cavernous sinus
4. enters the orbit via the superior orbital fissure and common tendinous ring



Dorello canal / canal of abducent nerve is a passage between the apex of the petrous part of the temporal bone and the petrosphenoidal ligament of Gruber. It contains the abducent nerve and inferior petrosal sinus. The abducent nerve is in an extradural position as it travels in this canal.

The **gustatory nucleus** is located in the rostral part of the solitary tract. It receives taste fibres from the facial nerve and sends impulses to the thalamus and cortex.

Clinical notes

Unilateral paresis of the abducent nerve manifests as convergent strabismus and diplopia (double vision) when looking to the side.

Central paralysis of the facial nerve presents as a drop of the contralateral angle of the mouth. However, innervation of both eyelids remains unaltered. The most common cause is a stroke.

Peripheral paralysis of the facial nerve may arise from a lesion at any point of the nerve. It presents as dysfunction of both the upper and lower branches, so that one whole half of the face is affected. Apart from paresis of the orbicularis oris it has other manifestation, including:

Lagophthalmus – an inability to close the eye caused by dysfunction of the orbicularis oculi.

Xerophthalmia – conjunctival and corneal dryness caused by dysfunction of the lacrimal gland. It increases the risk of corneal scarring and subsequent blindness and eye infections.

Xerostomia – dryness of the mouth due to dysfunction of the sublingual and submandibular glands.

Hyperacusis – pain when hearing loud sounds due to an impaired stapedial reflex.

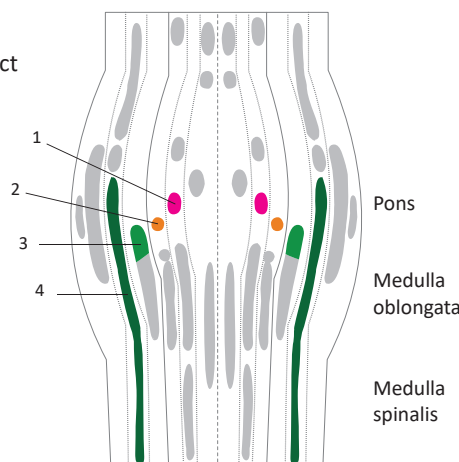
Hypogeusia – decreased perception of taste caused by dysfunction of the chorda tympani.

If the facial nerve is transected (e.g. in parotidectomy), the lesion can be bridged by a graft from either the sural or hypoglossal nerve.

The facial nerve develops from the **2nd pharyngeal arch**. It innervates **all the facial muscles**. It emerges from the pons near the cerebellopontine angle as two roots: the facial motor root and the intermediate nerve. The facial motor root carries somatomotor fibres and the intermediate nerve carries visceromotor and special sensory fibres. The two roots run in the **internal acoustic meatus** and then pass through the **facial nerve canal where they unite**. The facial nerve leaves the skull through the **stylomastoid foramen**. It gives off its terminal branches within the substance of the **parotid gland**.

Nuclei

- 1 **Motor nucleus of facial nerve** (*nucleus nervi facialis*)
 - a somatomotor nucleus for innervation of all facial muscles, platysma, stylohyoid and posterior belly of the digastric
- 2 **Superior salivatory nucleus** (*nucleus salivatorius superior*)
 - a visceromotor nucleus for parasympathetic innervation of the lacrimal gland, nasal glands, palatine glands, nasopharyngeal glands, submandibular gland, sublingual gland and lingual glands
- 3 **Gustatory nucleus** (*nucleus gustatorius*)
 - a special sensory nucleus
 - the rostral part of the nuclei of the solitary tract
 - receives impulses from the chorda tympani
- 4 **Spinal nucleus of trigeminal nerve** (*nucleus spinalis nervi trigemini*)
 - somatosensory fibres from the facial nerve synapse in this nucleus

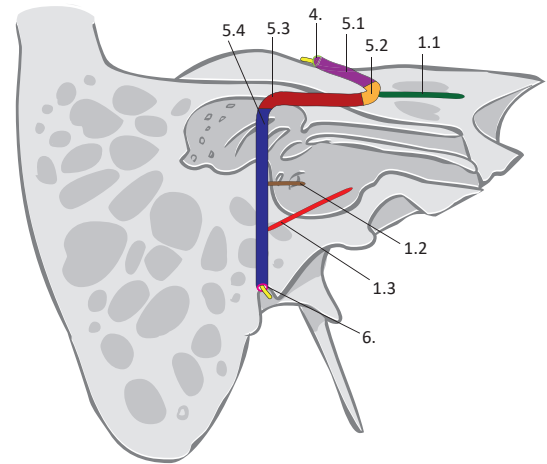


Parts

- 1 **Facial nerve** – somatomotor fibres
- 2 **Nervus intermedius**
 - 2.1 **Visceromotor fibers (for glands)**
 - greater petrosal nerve, chorda tympani
 - 2.3 **Special sensory fibers (taste)**
 - chorda tympani, greater petrosal nerve

Course

1. within the brainstem, it turns around the nucleus of the abducens nerve and thus elevates the facial colliculus on the floor of the fourth ventricle
2. leaves the pons at the **pontocerebellar angle**
3. runs in the posterior cerebral fossa
4. enters the internal acoustic canal and passes through the anterior superior quadrant of the fundus – this is where the **facial canal / Fallopiian canal** (*canalis nervi facialis*) starts
5. travels through the facial canal
 - 5.1 runs ventrolaterally in the **labyrinthine part** (*pars labyrinthica*),
 - 5.2 takes a 90° turn at the **geniculum of the facial canal** (*geniculum canalis nervi facialis*),
 - 5.3 runs dorsolaterally in the **tympanic part** (*pars tympanica*),
 - 5.4 runs caudally in the **mastoid part** (*pars mastoidea*)
6. leaves the skull via the **stylomastoid foramen** (*foramen stylomastoideum*)
7. enters the parotid gland, where it splits into inferior and superior branches, which form the **parotid plexus** (*plexus intraparotideus*), which gives rise to the terminal branches for the facial muscles



Right facial canal scheme, lateral view

Branches and structures innervated

1 Branches arising within the facial canal

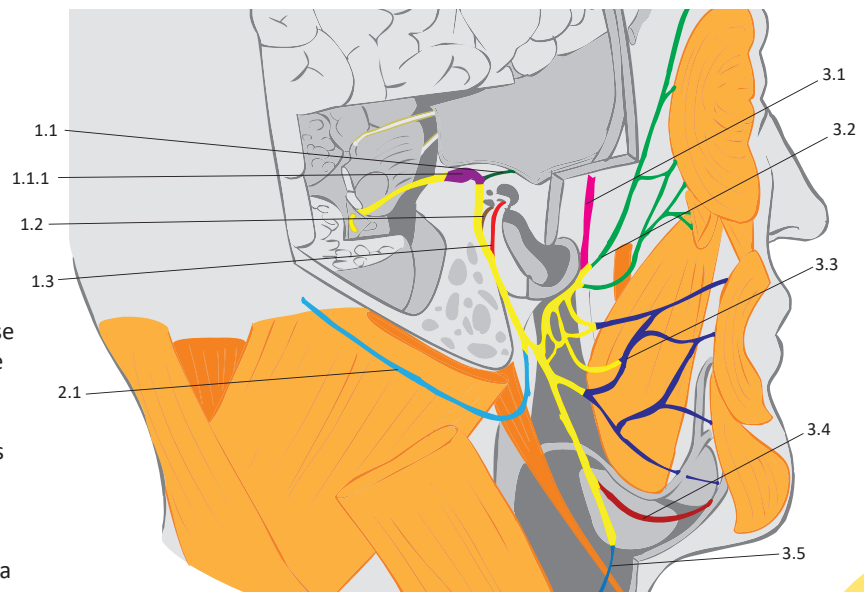
- 1.1 **Greater petrosal nerve** (*nervus petrosus major*)
 - emerges from geniculum of the facial canal and runs in the canal for the greater petrosal nerve
 - travels through the hiatus for the greater petrosal nerve, then lies in the groove for the greater petrosal nerve
 - runs through the sphenopetrosal synchondrosis and the pterygoid canal
 - reaches the pterygopalatine fossa where it terminates in the pterygopalatine ganglion
 - brings parasympathetic fibres from superior salivatory nucleus and some taste fibres to the pterygopalatine ganglion
 - 1.1.1 **Geniculate ganglion** (*ganglion geniculi*) – a special sensory ganglion of the facial nerve
 - located in the first flexure of the facial canal within the petrous part of the temporal bone
 - contains pseudounipolar neurons for the chorda tympani (taste fibres)
- 1.2 **Nerve to stapedius** (*nervus stapedius*) – branches from the facial nerve in the mastoid part of the facial canal
 - innervates the stapedius
- 1.3 **Chorda tympani** – branches from the facial nerve in the mastoid part of the facial canal
 - passes through the canaliculus for the chorda tympani to reach the tympanic cavity
 - runs within a mucosal fold between malleus and stapes and then turns ventrocaudally towards the petrotympanic fissure
 - leaves the tympanic cavity via the petrotympanic fissure to enter the infratemporal fossa
 - joins the lingual nerve to travel to the submandibular ganglion
 - provides parasympathetic innervation for the submandibular and sublingual glands and the lingual glands in the anterior 2/3 of the tongue

2 Branches after the stylomastoid foramen

- 2.1 **Posterior auricular nerve** (*nervus auricularis posterior*) – runs ventrally to the mastoid process on the medial surface of the auricle
 - 2.1.1 **Occipital branch** (*ramus occipitalis*) – innervates the occipital belly of the occipitofrontalis
 - 2.1.2 **Auricular branch** (*ramus auricularis*) – innervates the auricular muscles and the proper muscles of the auricle
- 2.2 **Digastric branch** (*ramus digastricus*)
 - innervates of the posterior belly of the digastric and the stylohyoid

3 Branches forming the parotid plexus

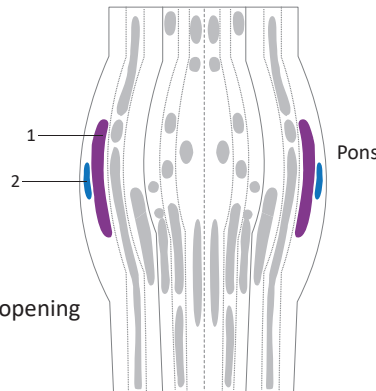
- somatomotor branches
- 3.1 **Temporal branches** (*rami temporales*)
 - innervate the facial muscles of the forehead (frontal region) and temporal region
- 3.2 **Zygomatic branches** (*rami zygomatici*)
 - innervate the orbicularis oculi, zygomatic muscles and muscles of the nose
- 3.3 **Buccal branches** (*rami buccales*) – innervate the muscles of the cheek and upper lip
- 3.4 **Marginal mandibular branch** (*ramus marginalis mandibulae*) – innervates the muscles of the chin and lower lip
- 3.5 **Cervical branch** (*ramus colli*)
 - innervates the platysma and participates in formation of the superficial cervical ansa



The **vestibulocochlear nerve** emerges from the **pons** at the **pontocerebellar angle**. It passes through the **internal acoustic meatus** and bifurcates into the cochlear nerve and the vestibular nerve. The **cochlear nerve** transmits sound information. It consists of axons of the bipolar neurons of the cochlear ganglion, whose cell bodies are located in the modiolus of the bony cochlear. Their dendrites receive signals from the spiral organ of Corti and their axons synapse in the **two cochlear nuclei** (anterior and posterior) in the brainstem. The **vestibular nerve** transmits information about balance, motion and acceleration. It is formed by axons of the **bipolar neurons of the vestibular ganglion** travelling through the internal acoustic meatus. Their dendrites receive information from the **sacculle, utricle and semicircular ducts** and their axons synapse in the **four vestibular nuclei** (superior, inferior, medial and lateral) in the pons and in the cerebellum.

Nuclei

- 1 **Medial, lateral, superior, and inferior vestibular nuclei** (*nucleus vestibularis medialis, lateralis, superior et inferior*)
– special sensory nuclei of the vestibular nerve
- 2 **Anterior and posterior cochlear nuclei** (*nucleus cochlearis anterior et posterior*)
– special sensory nuclei of the cochlear nerve



Course

1. leaves the pons at the pontocerebellar angle
2. runs in the posterior cranial fossa
3. enters the internal acoustic meatus via the internal acoustic opening
– the cochlear nerve penetrates the anterior inferior quadrant (the cochlear area) of the fundus with the tractus spiralis foraminosus
– the vestibular nerve penetrates the superior and inferior posterior quadrants (vestibular areas)

Fundus of the internal acoustic meatus

Anterior inferior quadrant

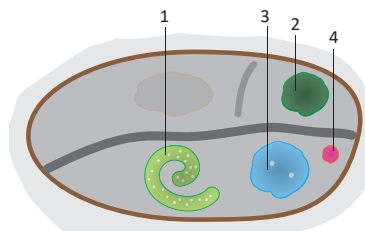
- 1 **Cochlear area** (*area cochlearis*) – cochlear nerve

Superior posterior quadrant

- 2 **Superior vestibular area** (*area vestibularis superior*)
– utriculo-ampullary nerve (*nervus utriculoampullaris*)
from the vestibular nerve (*nervus vestibularis*)

Inferior posterior quadrant

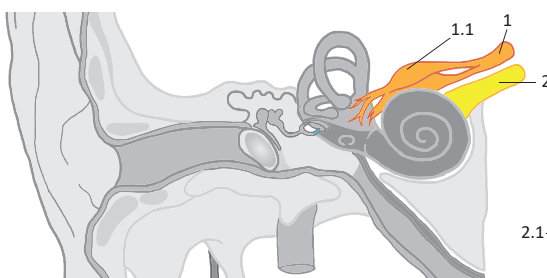
- 3 **Inferior vestibular area** (*area vestibularis inferior*)
– saccular nerve (*nervus saccularis*)
from the vestibular nerve (*nervus vestibularis*)
- 4 **Singular foramen** (*foramen singulare*)
– posterior ampullary nerve (*nervus ampullaris posterior*)



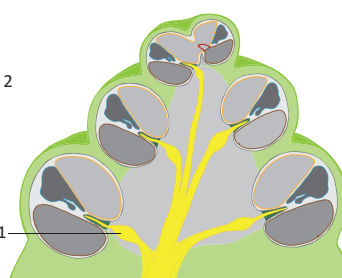
Scheme of the right internal acoustic meatus

Parts

- 1 **Vestibular nerve** (*nervus vestibularis*) – special sensory fibres
– transmit information about position and motion
1.1 **Vestibular ganglion** (*ganglion vestibulare*)
– is located on the fundus of the internal acoustic meatus
- 2 **Cochlear nerve** (*nervus cochlearis*) – special sensory fibers
– transmits auditory impulses from the cochlea
2.1 **Cochlear ganglion** (*ganglion cochleare*)
– is located in the modiolus of the bony cochlea



Frontal section of the external acoustic meatus



Vertical section of the bony cochlea

The **vestibular nuclei** are located in the vestibular area under the floor of the fourth ventricle.

The **cochlear nuclei** are situated under the acoustic tuberculum, which is located in the lateral corner of the fourth ventricle.

The **spiral ganglion of Corti** (*ganglion spirale*) is an obsolete term for the cochlear ganglion

The **stato-acoustic nerve** (*nervus statoacusticus*) is an obsolete for the vestibulocochlear nerve.

The **tensor veli palatini** is the only muscle of the soft palate innervated by the trigeminal nerve and not by the pharyngeal plexus.

Clinical notes

Unilateral lesions of the vestibulocochlear nerve cause hearing loss (hypacusis), tinnitus, vertigo, nystagmus (involuntary repetitive eye movements) and gait disorders (titubation).

Unilateral paresis of the glossopharyngeal nerve presents as dysphagia (difficulty of swallowing), impaired gag reflex, loss of sensitivity and mildly decreased taste perception (hypogeusis).

Glossopharyngeal neuralgia presents as episodic severe pain in the regions supplied by the glossopharyngeal nerve (ear, root of tongue, nasopharynx and tonsils).

The glossopharyngeal nerve emerges from the medulla oblongata at the **retro-olivary groove** and leaves the cranial cavity via the **jugular foramen**. It **forms the lateral mixed system** with the vagus and accessory nerves. It contains **motor, sensory and special sensory fibres**. It innervates the muscles of the pharynx and soft palate, glands and mucosa of the pharynx, posterior third of the tongue, parotid gland and middle ear. It transmits taste impulses from the posterior third of the tongue and the vallate papillae.

Proper nuclei of the glossopharyngeal nerve

- **1 Nucleus ambiguus** – a somatomotor nucleus
 - innervates the stylopharyngeus
- **2 Inferior salivatory nucleus** (*nucleus salivatorius inferior*)
 - a parasympathetic visceromotor nucleus
 - innervates the parotid and buccal glands and the glands of the auditory tube
 - its fibers synapse in the otic ganglion and tympanic ganglion
- **3 Nuclei of solitary tract** (*nuclei tractus solitarii*) – a viscerosensory nucleus
 - 3.1 Gustatory nucleus** (*nucleus gustatorius*) – a special sensory nucleus
 - the rostral part of the nuclei of the solitary tract
 - transmits taste information from the posterior third of the tongue and vallate papillae

Nucleus of the trigeminal nerve, which receives somatosensory fibres from the glossopharyngeal nerve

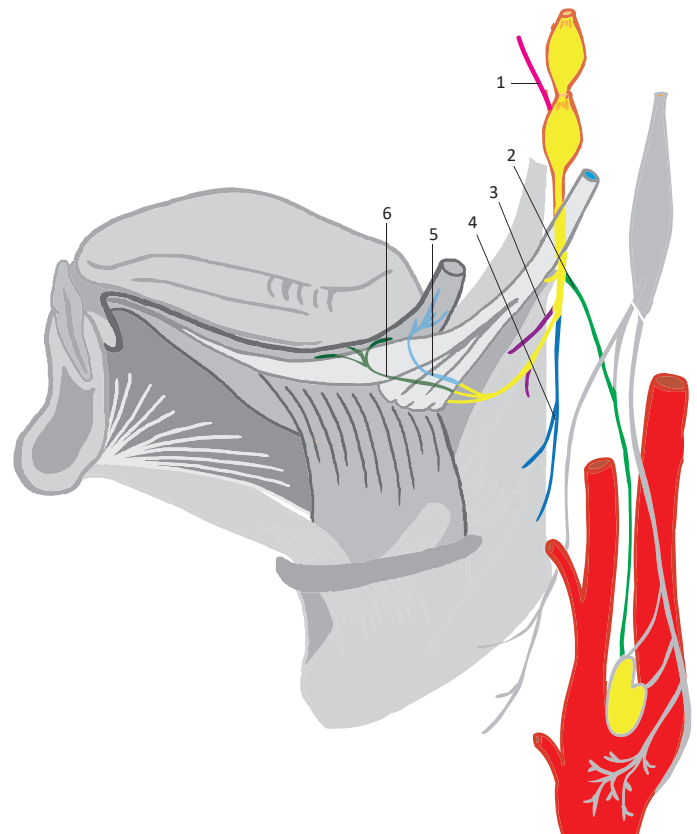
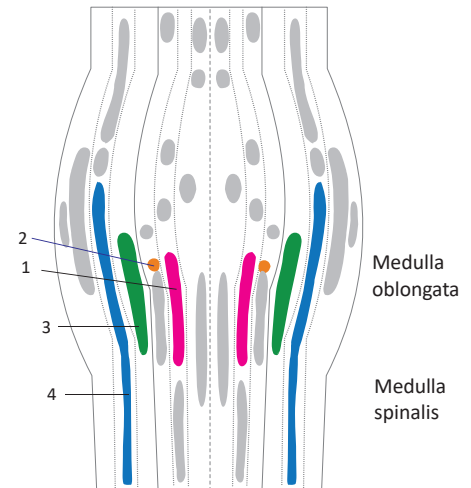
- **4 Spinal nucleus of trigeminal nerve** (*nucleus spinalis nervi trigemini*)
 - a somatosensory nucleus
 - receives somatosensory fibres from the glossopharyngeal nerve

Course

1. emerges from the medulla oblongata at the retro-olivary groove with the vagus and accessory nerves
2. leaves the cranial cavity via the jugular foramen
3. forms the superior ganglion above the jugular foramen and the inferior ganglion just below it
4. after leaving the skull, runs between the internal carotid artery and the internal jugular vein
5. joins the lateral side of the stylopharyngeus, which it innervates, and enters the pharyngeal wall, where it contributes to forming the pharyngeal plexus

Branches and structures innervated

- **1 Tympanic nerve** (*nervus tympanicus*)
 - emerges from the inferior ganglion and enters the tympanic cavity via the tympanic canaliculus
- 1.1 Lesser petrosal nerve** (*nervus petrosus minor*)
 - a branch arising in the tympanic cavity carrying parasympathetic fibres
 - passes through the canal, hiatus and groove for the lesser petrosal nerve
 - penetrates the sphenopetrosal synchondrosis, (often via the petrosal foramen) to reach the infratemporal fossa
 - synapses in the otic ganglion
 - parasympathetic visceromotor innervation of the parotid and buccal glands
- **2 Carotid branch** (*ramus sinus carotici*)
 - descends to the carotid sinus and carotid body
 - transfers viscerosensory information from baroreceptors and chemoreceptors
- **3 Pharyngeal branches** (*rami pharyngei*)
 - form the pharyngeal plexus, which provides motor and sensory innervation of the pharyngeal muscles, pharyngeal mucosa and muscles of the soft palate
- **4 Stylopharyngeal branch** (*ramus stylopharyngeus*)
 - innervates the stylopharyngeus
- **5 Tonsillar branches** (*rami tonsillares*)
 - sensory innervation of the palatine tonsil and the palatoglossal and palatopharyngeal arches
- **6 Lingual branches** (*rami linguales*)
 - carry sensory and taste information from the root of the tongue and the vallate papillae

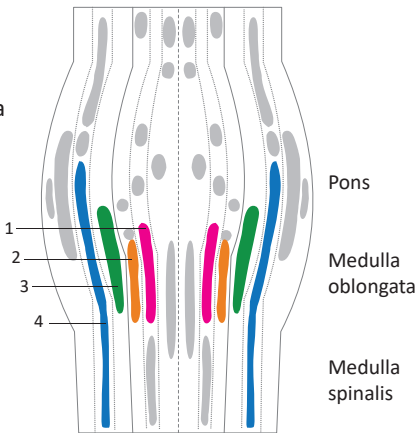


The vagus nerve originates in the **retro-olivary groove** of the medulla oblongata and leaves the cranial cavity via the **jugular foramen**. It forms the **lateral mixed system** with the glossopharyngeal and accessory nerves. It contains **motor, sensory and special sensory** (taste) fibres and is the only cranial nerve to reach the **abdominal cavity**. It gives numerous branches, which provide **parasympathetic and viscerosensory innervation** to the organs of the neck, thorax and abdomen, down to the Cannon-Böhm point at the left end of the transverse colon.

Nuclei

Proper nuclei of the vagus nerve

- **1 Nucleus ambiguus** – a somatomotor nucleus
 - innervates the muscles of the soft palate, pharynx and larynx
 - **2 Posterior nucleus of vagus nerve** (*nucleus posterior nervi vagi*)
 - a parasympathetic visceromotor nucleus
 - innervates the smooth muscle and glands of the internal organs
 - innervates the myocardium and conducting system of the heart
 - **3 Nuclei of solitary tract** (*nuclei tractus solitarii*)
 - a viscerosensory nucleus
 - the axons synapse in the superior and inferior ganglia in the area of the jugular foramen
- 3.1 Gustatory nucleus** (*nucleus gustatorius*)
- a special sensory nucleus
 - the rostral part of the nuclei of the solitary tract
 - receives taste information from the epiglottis, epiglottic valleculae and part of the root of the tongue

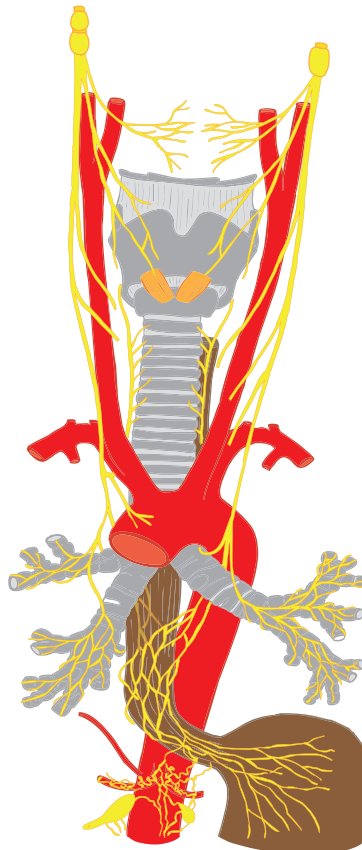


Nucleus of the trigeminal nerve, which receives somatosensory fibres from the vagus nerve

- **4 Spinal nucleus of trigeminal nerve** (*nucleus spinalis nervi trigemini*) – a somatosensory nucleus
 - receives sensory fibres from part of the auricle, external acoustic meatus, dura mater and mucosa of the larynx and pharynx

Course

1. leaves the medulla oblongata in the retro-olivary groove with the glossopharyngeal and accessory nerves
2. leaves the cranial cavity via the jugular foramen
3. its superior ganglion is located above the level of the jugular foramen and inferior ganglion below it
4. runs in the visceral space of the neck with the internal jugular vein and the internal carotid artery (caudally with the common carotid artery) within the carotid sheath
5. descends through the superior thoracic aperture, runs dorsally to the bronchi and caudally joins the oesophagus
 - gives off the right recurrent laryngeal nerve, which passes under the right subclavian artery, and the left recurrent laryngeal nerve, which passes under the aortic arch
6. in the posterior mediastinum, the left vagus nerve moves so it becomes ventral to the oesophagus and the right vagus nerve moves so it becomes dorsal to the oesophagus
7. the left vagus forms the ventral trunk and the right vagus forms the dorsal trunk follows the oesophagus through the diaphragm via the oesophageal hiatus
8. parasympathetic branches in the abdominal cavity give rise to the autonomic plexuses
9. branches reach the individual organs together with vessels



The **anastomosis of Galen** is a term for a connection between the internal branch of the superior laryngeal nerve and the recurrent laryngeal nerve.

The **left vagus nerve** travels ventrally and the right vagus nerve travels dorsally to the oesophagus. This is due to the rotation of the primitive gut of 90° clockwise during embryogenesis.

The **vagus nerve** innervates the pharyngeal muscles (except for the stylopharyngeus), muscles of the soft palate (except for the tensor veli palatini), laryngeal muscles and muscles of the upper half of the esophagus.

The **function of the vagus nerve** may be assessed by the following reflexes:

The **pharyngeal (gag) reflex** is triggered by touching the root of the tongue, pharyngeal wall or palatine arches. It evokes contraction of the pharyngeal muscles and causes nausea.

The **oculocardial reflex** (Aschner-Dagnini reflex) is induced by stimulating the vagus nerve by applying pressure on the eyeballs. This causes a decrease in pulse rate. The reflex can also be stimulated by pressing the carotid sinus or hitting the epigastrium (to irritate the coeliac plexus) or the testes.

The **auricular branch of the vagus nerve** (Arnold nerve) innervates a small area in the concha of the auricle and external acoustic meatus called the Ramsay Hunt zone.

The **principal gastric branch of Latarjet** (from the anterior gastric branch) supplies the lesser curvature of the stomach. Selective vagotomy used to be performed to treat peptic ulcer by surgically interrupting this nerve.

Clinical notes

Unilateral paresis of the vagus nerve causes dysphagia, hoarseness and divergence of the uvula.

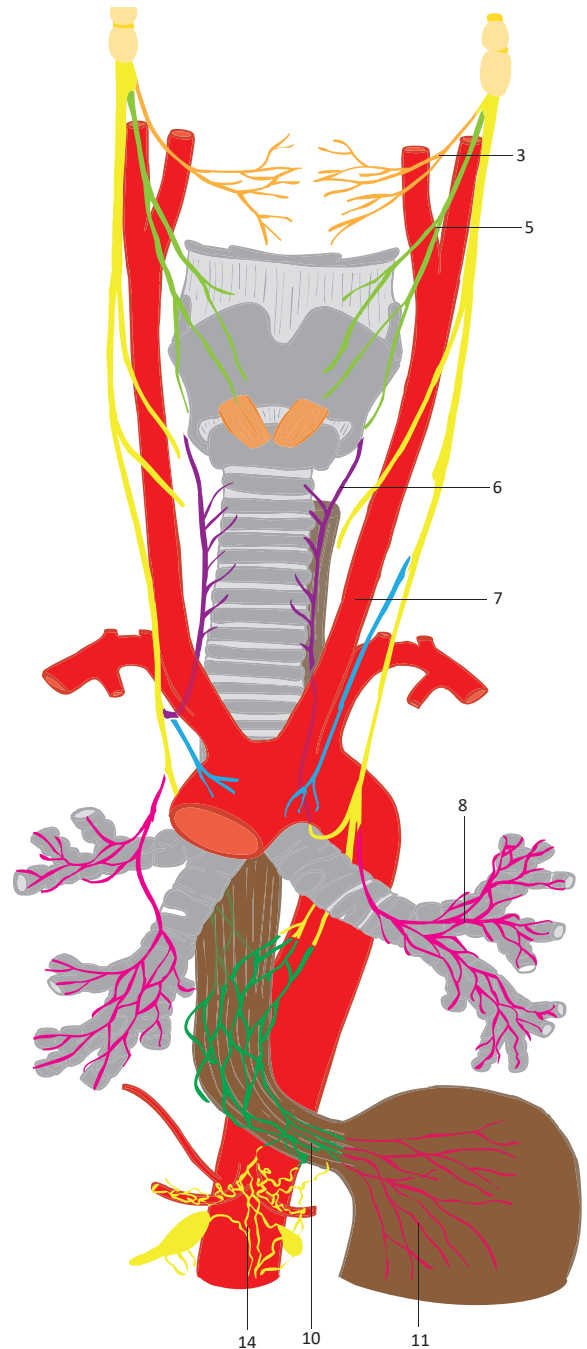
Bilateral paresis of the vagus nerve causes rhinolalia, speech defects (dysarthria) and paresis of both vocal cords.

Irritation of the vagus nerve may cause bradycardia, smooth muscle spasms (laryngospasm, bronchospasm) and disturbances in gastrointestinal passage.

In **pharmacoresistant depression** or epilepsy, vagal nerve stimulation (VNS) is a therapeutic option.

Branches and structures innervated

- 1 **Meningeal branch** (*ramus meningeus*) – somatosensory innervation of the dura mater in the posterior cranial fossa
- 2 **Auricular branch** (*ramus auricularis*) – somatosensory innervation of the external acoustic meatus and concha of the auricle
- 3 **Pharyngeal branches** (*rami pharyngei*) – somatomotor and sensory innervation of the pharynx
 - 3.1 **Pharyngeal plexus** (*plexus pharyngeus*) – a mixed plexus for the mucosa, glands and muscles of the pharynx
- 4 **Branch for carotid body** (*ramus glomi carotici*) – viscerosensory fibers entering the branches of the glossopharyngeal nerve for the carotid body and carotid sinus
 - carries baroreceptor and chemoreceptor information
- 5 **Superior laryngeal nerve** (*nervus laryngeus superior*) – a mixed branch originating from the inferior ganglion
 - 5.1 **Internal branch** (*ramus internus*) – a sensory branch for the upper larynx, root of the tongue and epiglottis
 - enters the larynx through the thyrohyoid membrane
 - 5.2 **External branch** (*ramus externus*) – a mixed branch, which travels with the superior thyroid artery and innervates the cricothyroid and certain pharyngeal muscles
 - carries parasympathetic and viscerosensory branches for the thyroid and parathyroid glands
- 6 **Recurrent laryngeal nerve** (*nervus laryngeus recurrens*)
 - innervates the majority of the laryngeal muscles and the laryngeal mucosa below the vocal cords
 - on the right side, it passes under the right subclavian artery, on the left side under the aortic arch
 - runs cranially between the trachea and oesophagus in the tracheo-oesophageal groove
 - 6.1 **Tracheal branches** (*rami tracheales*) – visceromotor and viscerosensory fibers for the tracheal mucosa
 - 6.2 **Oesophageal branches** (*rami oesophagei*) – visceromotor and viscerosensory branches for the cervical part of the oesophagus
 - 6.3 **Pharyngeal branches** (*rami pharyngei*) – visceromotor and viscerosensory branches for the pharyngeal mucosa
 - 6.4 **Terminal branch / inferior laryngeal nerve** (*nervus laryngeus inferior*) – innervates the laryngeal muscles (except for the cricothyroid) and laryngeal mucosa below the vocal folds
- 7 **Cardiac branches** (*rami cardiaci*) – visceromotor and viscerosensory fibers for the heart
 - 7.1 **Superior cervical, inferior cervical and thoracic cardiac branches** (*rami cardiaci cervicales superiores, cervicales inferiores et thoracici*) – emerge from the vagus at different levels
 - 7.2 **Cardiac plexus** (*plexus cardiacus*) – a mixed plexus for the heart
 - contain the parasympathetic cardiac branches from the vagus nerve and sympathetic cardiac nerves from the sympathetic trunk
- 8 **Bronchial branches** (*rami bronchiales*) – viscerosensory and visceromotor branches for the mucosa, glands and smooth muscles of the bronchi
- 9 **Pulmonary branches** (*rami pulmonales*) – viscerosensory and visceromotor fibres
 - the continuation of the bronchial branches into the lungs
 - 9.1 **Pulmonary plexus** (*plexus pulmonalis*) – a mixed plexus in the lungs consisting of both parasympathetic and sympathetic fibres
- 10 **Oesophageal plexus** (*plexus oesophageus*) – visceromotor and viscerosensory fibres for the oesophageal mucosa, glands and muscles
- 11 **Anterior and posterior gastric branches** (*rami gastrici anteriores et posteriores*) – visceromotor and viscerosensory fibers for the stomach
- 12 **Hepatic branches** (*rami hepatici*) – visceromotor and viscerosensory fibers for the liver
 - run along the common hepatic artery and hepatic artery proper
- 13 **Renal branches** (*rami renales*) – viscerosensory and visceromotor branches for the kidneys, upper part of the urinary tract, adrenals and (partially) testes/ovaries
- 14 **Coeliac branches** (*rami coeliaci*) – visceromotor and viscerosensory branches, which along with the sympathetic nerves form the coeliac plexus
 - plexuses for the adrenals, kidneys, pancreas, ovaries, testes, small intestine, caecum, ascending colon and 2/3 of the transverse colon (to the Cannon–Böhm point, where they meet with the parasympathetic branches from the sacral plexus)
 - 14.1 **Coeliac plexus** (*plexus coeliacus*) – a mixed plexus for the abdominal organs containing the parasympathetic and viscerosensory fibers from the n. X and sympathetic fibers from the splanchnic nerves



The accessory nerve emerges from the medulla oblongata at the **retro-olivary groove** and leaves the cranial cavity via the **jugular foramen**. It forms the **lateral mixed system** with the vagus and glossopharyngeal nerves. It contains **somatomotor fibres** that innervate the sternocleidomastoid and trapezius. The accessory nerve is the only cranial nerve, which has a **nucleus outside the brainstem** (in the cervical spinal cord). Its fibres merge inside the skull at the level of the foramen magnum to form the trunk of the accessory nerve.

1 Accessory nerve

Nuclei

2 Nucleus ambiguus

- the cranial root (*radix cranialis*)
- is located in the medulla oblongata

3 Spinal nucleus of accessory nerve

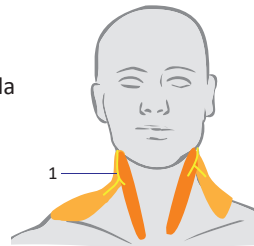
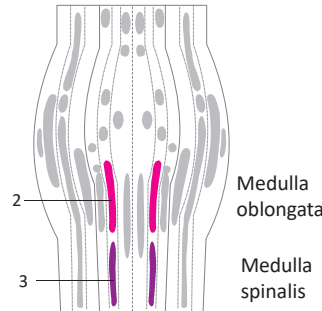
- (*nucleus spinalis nervi accessorii*)
- spinal roots (*radix spinalis*) from spinal segments C1–C6
- ascend laterally to the spinal cord and reach the posterior cranial fossa via the foramen magnum

Course

1. leaves the cranial cavity via the jugular foramen
2. passes behind the internal jugular vein and the transverse process of the atlas
3. continues through the sternocleidomastoid towards the trapezius
4. runs within the trapezius parallel to the medial border of the scapula

Branches and structures innervated

- 1 **Internal branch** (*ramus internus*) – joins the vagus nerve and participates in the innervation of the larynx
- 2 **External branch** (*ramus externus*) – innervates the sternocleidomastoid and trapezius



The name of the accessory nerve originates from its older name *nervus accessorius ad vagum* (accessory to vagus), because some of its fibres join the vagus nerve.

The **sternocleidomastoid and trapezius** receive somatomotor fibers from the accessory nerve and spinal segments C2–C4. The spinal root of the accessory innervates muscles derived from somites and was secondarily encompassed into the cranial cavity and therefore separated from direct muscular branches.

The **hypoglossal nerve** does not participate in the innervation of the thyrohyoid. Fibres from the cervical plexus (C1) run with the hypoglossal to innervate the thyrohyoid.

ANS is an abbreviation for the **Autonomic Nervous System**.

The **vegetative nervous system** is an obsolete term for the ANS.

The **sympathetic nervous system** is called the thoraco-lumbar part of the ANS and the parasympathetic nervous system the cranio-sacral part of the ANS.

The hypoglossal nerve emerges from the medulla oblongata at the **pre-olivary groove** and leaves the cranial cavity via the **hypoglossal canal**. It forms an arch in its course in the neck. **Somatomotor fibres** from the cervical plexus (the superior root of the deep cervical ansa) join the nerve, but leave it after a short distance. The hypoglossal nerve innervates the **muscles of the tongue**.

1 Hypoglossal nerve

Nucleus

- 2 **Nucleus of hypoglossal nerve** (*nucleus nervi hypoglossi*) – a somatomotor nucleus

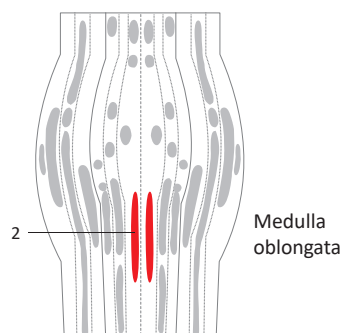
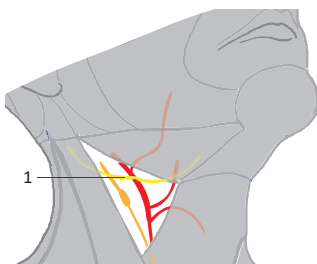
Course

1. leaves the cranial cavity via the hypoglossal canal
2. passes ventrally between the internal jugular vein and the internal carotid artery
3. forms an arch in the carotid triangle and continues into the submandibular triangle
4. enters the root of the tongue where it gives off its terminal branches

Proper branches and structures innervated

Lingual branches (*rami linguales*)

- innervate the muscles of the tongue, except for the palatoglossus (n. X)



Clinical notes

The **external branch of the accessory nerve** can be easily harmed at the spine of the scapula. This can occur from falls on to the back. Lesions in this location impair the ascending part of the trapezius, which is then not able to rotate the scapula, preventing the arm from being abducted above the level of the shoulder. The anterior serratus has a synergistic function, but it is not able to perform this movement alone due to the excessive mass of the limb. The superior angle of the scapula protrudes outwards in what is called a winged scapula (*scapula alata*).

Lesions of the nucleus of the hypoglossal nerve cause fine trembling of muscle fibres (fasciculation).

Unilateral paresis of the hypoglossal nerve cause deviation of the tongue to the side of the lesion (*hemiglossoplegia*) and atrophy of the muscles and mucosa of the tongue.

The autonomic nervous system (ANS) is a relatively independent part of the nervous system. It innervates smooth muscle in the internal organs, skin and vessels and regulates the activity of the heart and glands. The effector component of the ANS consists of preganglionic and postganglionic visceromotor fibres. Preganglionic fibres enter autonomic ganglia, where they branch extensively and form synapses with postganglionic neurons. The postganglionic fibres travel to the target organs. The hypothalamus and reticular formation are the highest centres of the ANS. The ANS is functionally divided into the sympathetic part (*pars sympathica*) and the parasympathetic parts (*pars parasympathica*). The enteric nervous system is located within the wall of the gastrointestinal tract (myenteric and submucous plexus). It regulates the function of the gastrointestinal tract and is relatively independent from the rest of the ANS.

Centres and nuclei

1 Central part of the ANS

- 1.1 Hypothalamus
- 1.2 Reticular formation (*formatio reticularis*)
- 1.3 Limbic system

2 Parasympathetic part

- 2.1 Nuclei of the brainstem – parasympathetic nuclei of n. III, n. VII, n. IX and n. X
- 2.2 Intermediolateral nuclei (*nuclei intermediolaterales*) S2–S4
 - parasympathetic nuclei in the sacral segments of the spinal cord

3 Sympathetic part

- 3.1 Intermediolateral nuclei (*nuclei intermediolaterales*) C8–L3
 - sympathetic nuclei in the thoracic and lumbar segments of the spinal cord

Nerves, ganglia, plexuses

1 Parasympathetic part

- purely parasympathetic ganglia are only found in the head
- ganglia are situated closely to the innervated organs

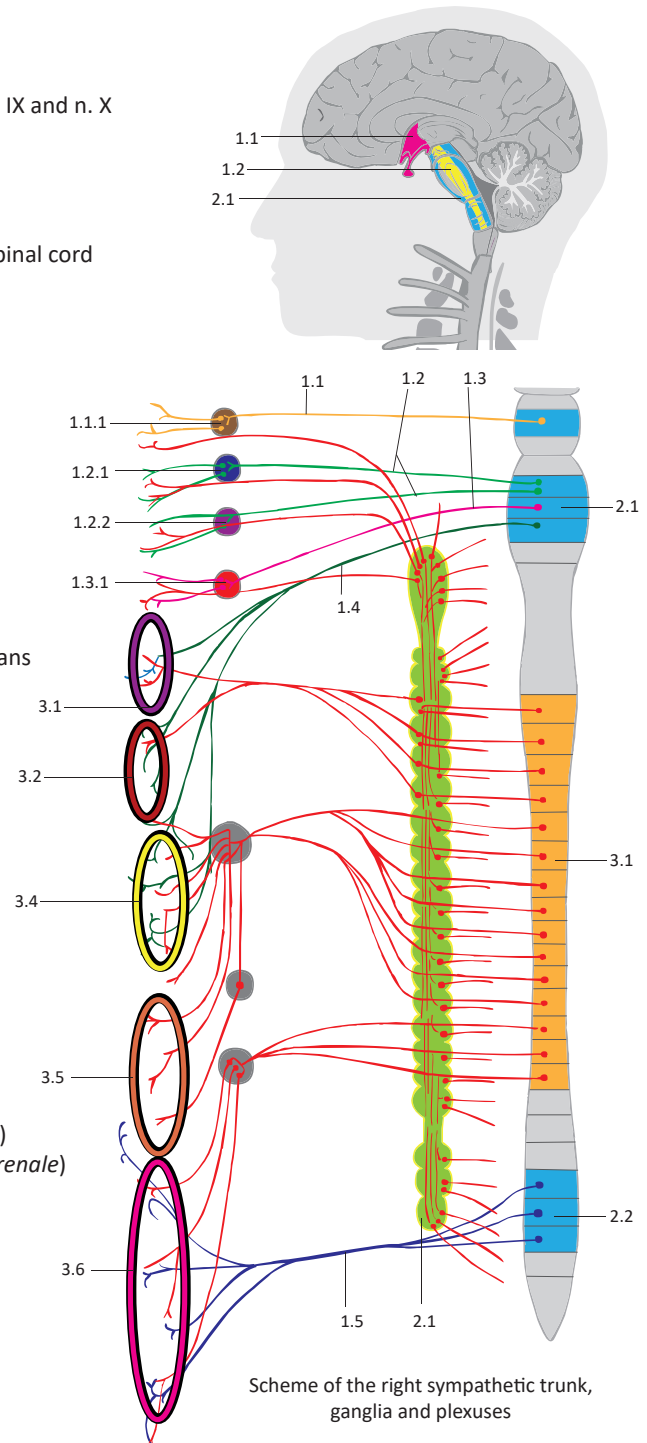
- 1.1 Oculomotor nerve (*nervus oculomotorius*)
 - 1.1.1 Ciliary ganglion (*ganglion ciliare*)
- 1.2 Facial nerve (*nervus facialis*)
 - 1.2.1 Pterygopalatine ganglion (*ganglion pterygopalatinum*)
 - 1.2.2 Submandibular ganglion (*ganglion submandibulare*)
- 1.3 Glossopharyngeal nerve (*nervus glossopharyngeus*)
 - 1.3.1 Otic ganglion (*ganglion oticum*)
- 1.4 Vagus nerve (*nervus vagus*) – forms ganglia within the walls of organs
- 1.5 Sacral splanchnic nerves (*nervi splanchnici sacrales*)

2 Sympathetic part

- purely sympathetic paravertebral ganglia are located in the sympathetic trunk
- 2.1 Sympathetic trunk (*truncus sympathicus*)
 - consists of the paravertebral ganglia interconnected by interganglionic branches (*rami interganglionares*)

3 Mixed plexuses

- are composed of both sympathetic and parasympathetic fibres
- 3.1 Cardiac plexus (*plexus cardiacus*)
- 3.2 Pulmonary plexus (*plexus pulmonalis*)
- 3.3 Oesophageal plexus (*plexus oesophageus*)
- 3.4 Abdominal aortic plexus (*plexus aorticus abdominalis*)
 - 3.4.1 Coeliac plexus (*plexus coeliacus*)
 - contains the paired coeliac ganglion (*ganglion coeliacum*)
 - 3.4.2 Renal and suprarenal plexuses (*plexus renalis et suprarenalis*)
 - contain the paired aorticorenal ganglion (*ganglion aorticorenale*)
 - 3.4.3 Testicular and ovarian plexus (*plexus testicularis et ovaricus*)
 - 3.4.4 Ureteric plexus (*plexus uretericus*)
 - 3.4.5 Superior mesenteric plexus (*plexus mesentericus superior*)
 - contains the unpaired superior mesenteric ganglion (*ganglion mesentericum superius*)
 - 3.4.6 Inferior mesenteric plexus (*plexus mesentericus inferior*)
 - contains the unpaired inferior mesenteric ganglion (*ganglion mesentericum inferius*)
- 3.5 Superior hypogastric plexus (*plexus hypogastricus superior*)
- 3.6 Inferior hypogastric plexus (*plexus hypogastricus inferior*)



Scheme of the right sympathetic trunk, ganglia and plexuses

The **sympathetic nervous system** consists of both central and peripheral parts. The central part consists of the **intermediolateral nuclei** in the grey mater of spinal cord segments C8–L3. The **paired sympathetic trunks** make up the peripheral part. They lie lateral to the vertebral bodies and extend all the way from the base of the skull to the coccyx. They penetrate the diaphragm as they pass from the thorax to the abdomen, and then lie in the retroperitoneal space. The sympathetic trunks consist of a series of ganglia, interconnected by interganglionic branches. The sympathetic nervous system **increases energy expenditure and catabolism. It increases heart rate and myocardial contractility, dilates bronchi, increases blood pressure and indirectly increases blood glucose and lipid levels.**

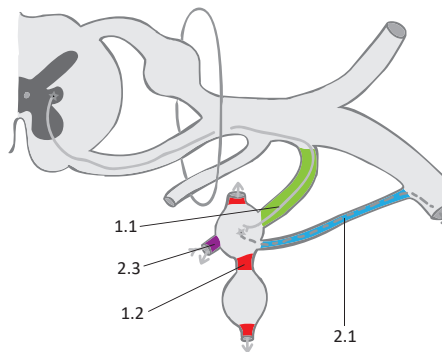
Connection of fibres in the sympathetic nervous system

1 Preganglionic fibers

- neurones of the intermediolateral nuclei (C8–L3)
 - **1.1 White rami communicantes** (*rami communicantes albi*)
 - heavily myelinated axons of neurones from the intermediolateral nucleus
 - enter the ganglia of the sympathetic trunk
 - the preganglionic fibres enter ganglia from the level of the cervicothoracic ganglion (*ganglion cervicothoracicum*) to the lumbar ganglion (*ganglion lumbale*) L3
 - **1.2 Interganglionic branches** (*rami interganglionares*)
 - preganglionic fibers reach the more remote ganglia via interganglionic branches

2 Postganglionic fibers

- **2.1 Grey rami communicantes** (*rami communicantes grisei*)
 - emerge from all ganglia of the sympathetic trunk
 - join the spinal nerves and innervate cutaneous glands and smooth muscle in the skin (the erector pili muscles)
- **2.2 Vascular branches** (*rami vasculares*)
 - travel with arteries
 - form plexuses in the adventitia
 - provide sympathetic innervation of the organs and limbs
- **2.3 Visceral branches, splanchnic nerves** (*rami viscerales, nervi splanchnici*)
 - travel towards the organs



Sympathetic trunk

1 Cervical ganglia (*ganglia cervicalia*) – 3 pairs

- are located lateral to the cervical vertebral column, within the prevertebral layer of the cervical fascia

● 1.1 Superior cervical ganglion (*ganglion cervicale superius*)

● 1.2 Middle cervical ganglion (*ganglion cervicale medium*)

● 1.3 Inferior cervical ganglion (*ganglion cervicale inferius*)

1.3.1 Cervicothoracic, stellate, ganglion

(*ganglion cervicothoracicum, stellatum*)

- is formed by fusion of the inferior cervical ganglion with first thoracic ganglion (*ganglion thoracicum primum*)
- is present in the majority of cases

● 2 Thoracic ganglia (*ganglia thoracica*) – 10–11 pairs

- are located lateral to the thoracic vertebral column

● 3 Lumbar ganglia (*ganglia lumbalia*) – 4–5 pairs

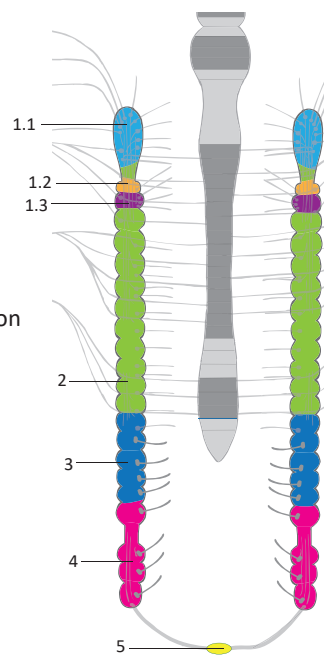
- are located lateral to the lumbar vertebral column

● 4 Sacral ganglia (*ganglia sacralia*) – 4 pairs

- are located on the pelvic surface of the sacrum, medial to the anterior sacral foramina (*foramina sacralia anteriora*)

● 5 Ganglion impar – the caudal ganglion

- is unpaired
- is located in front of the coccyx
- connects the left and right sympathetic trunks



Scheme of the right and left sympathetic trunks

The part of the internal carotid plexus within the carotid sinus is sometimes called the cavernous plexus (*plexus cavernosus*).

The **subclavian ansa** (*ansa subclavia*) is a connection between the middle cervical ganglion and cervicothoracic ganglion, which forms a loop around the subclavian artery.

The **splanchnic nerves** (*nervi splanchnici*) consist mainly of preganglionic fibres, which synapse in either the prevertebral or intramural ganglia.

The **sacral ansa** (*ansa sacralis*) is the terminal part of both sympathetic trunks. It connects the 4th sacral ganglia with the ganglion impar.

Clinical notes

Impairment of the sympathetic internal carotid plexus (*plexus caroticus internus*) may be caused by dissection of the internal carotid artery (usually traumatic or postoperative separation of the intima from the media) or by damage to the cervicothoracic ganglion (by tumors in the apex of the lungs). The clinical manifestation is called **Claude Bernard-Horner syndrome**. It presents as ptosis (drooping of the upper eyelid), miosis (constricted pupil), enophthalmos (posterior displacement of the eyeball) and anhidrosis in the face (absence of sweating).

Cervical ganglia

1 Superior cervical ganglion (*ganglion cervicale superius*)

- is located on the surface of the longus capitis at the level of C2–C4
- its preganglionic fibers originate from spinal segments C8–T1

Postganglionic fibers:

1.1 **Grey rami communicantes** (*rami communicantes grisei*) – join the first 4 cervical nerves

1.2 **Jugular nerve** (*nervus jugularis*) – travels towards the jugular foramen

- connects the inferior ganglion of the glossopharyngeal nerve with the inferior ganglion of the vagus nerve
- functionally it is a grey ramus communicans composed of postganglionic fibers

1.3 **Internal carotid nerve** (*nervus caroticus internus*) – forms the internal carotid plexus (*plexus caroticus internus*) within the adventitia of the carotid artery, travels the cavernous sinus and then splits along the branches of the internal carotid artery

1.3.1 **Caroticotympanic nerves** (*nervi caroticotympanici*) – pass through the caroticotympanic canaliculi into the tympanic cavity

● 1.3.2 **Deep petrosal nerve** (*nervus petrosus profundus*) – carries sympathetic fibers into the pterygopalatine ganglion

- enters the base of the skull at the sphenopetrosal synchondrosis

- fuses with the greater petrosal nerve to form the nerve of the pterygoid canal, then travels to the pterygopalatine fossa

● 1.3.3 **Sympathetic root of ciliary ganglion** (*radix sympathica ganglii ciliaris*)

- arises from the ophthalmic plexus (*plexus ophthalmicus*)

- passes through the ciliary ganglion without synapsing and forms the short ciliary nerves

- synapse with unmyelinated postganglionic fibers takes place in the superior cervical ganglion

1.3.4 **Orbital branches** (*rami orbitales*) – from the ophthalmic plexus

- innervates superior and inferior tarsal muscle and orbitalis

1.4 **External carotid nerves** (*nervi carotici externi*)

- approach the external carotid artery, form the external carotid plexus (*plexus caroticus externus*) within adventitia of the external carotid artery and then continues downward to the common carotid artery

● 1.4.1 **Sympathetic root of submandibular ganglion**

(*radix sympathica ganglii submandibularis*)

- arises from the plexus of the facial artery (*plexus arteriae facialis*)

● 1.4.2 **Sympathetic root of otic ganglion**

(*radix sympathica ganglii otici*)

- arises from the plexus of the middle meningeal artery (*plexus arteriae meningae mediae*)

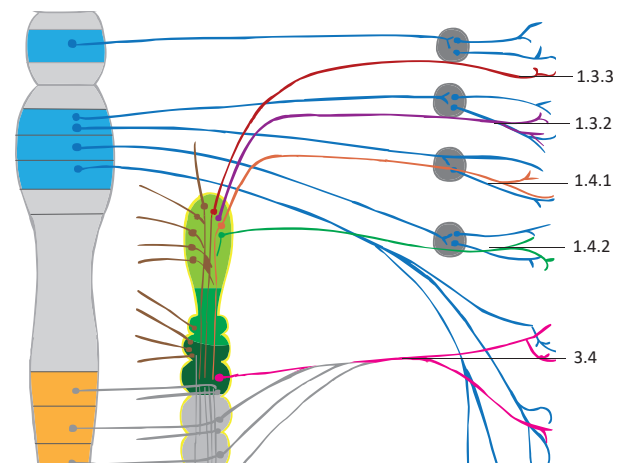
1.4.3 **Branches for carotid body** (*rami glomi carotici*)

1.5 **Laryngopharyngeal branches** (*rami laryngopharyngei*)

- join the pharyngeal branches from the glossopharyngeal and vagus nerves and form the pharyngeal plexus

1.6 **Superior cervical cardiac nerve** (*nervus cardiacus cervicalis superior*)

- on the left, it runs behind the common carotid artery, on the right it, runs behind the brachiocephalic trunk
- enters the mediastinum and terminates in the cardiac plexus (*plexus cardiacus*)



Scheme of the left sympathetic trunk, ganglia and plexuses

2 Middle cervical ganglion (*ganglion cervicale medium*) – is located on the surface of the longus colli at the level of C6

- in this area, the sympathetic trunk splits into several bundles that surround and form a loop around the subclavian artery
- the preganglionic fibres are derived from spinal segments C8–T1

Postganglionic fibers:

2.1 **Grey rami communicantes** (*rami communicantes grisei*) – join the 5th and 6th cervical nerves

2.2 **Vascular branches** (*rami vasculares*) – head towards the common carotid artery,

- form the common carotid plexus (*plexus caroticus communis*)

2.3 **Visceral branches** (*rami viscerales*) – innervate the thyroid gland and parathyroid glands

2.4 **Middle cervical cardiac nerve** (*nervus cardiacus cervicalis medius*)

- terminates in the cardiac plexus with the superior cervical cardiac nerve

3 Inferior cervical ganglion (*ganglion cervicale inferius*) – in the majority of cases, merges with the 1st thoracic ganglion forming the cervicothoracic/stellate ganglion (*ganglion cervicothoracicum/stellatum*)

- is located anterior to the transverse process (*processus transversus*) of C7 and posterior to the vertebral artery
- its preganglionic fibers originate from spinal segments C8–T1

Postganglionic fibers:

3.1 **Grey rami communicantes** (*rami communicantes grisei*) – join the 6th and 7th cervical nerves and the 1st thoracic nerve

3.2 **Vascular branches** (*rami vasculares*) – travel towards the subclavian artery and form the subclavian plexus for the upper limb

3.3 **Visceral branches** (*rami viscerales*) – innervate the thyroid gland and parathyroid glands

● 3.4 **Inferior cervical cardiac nerve** (*nervus cardiacus cervicalis inferior*) – terminates in the cardiac plexus

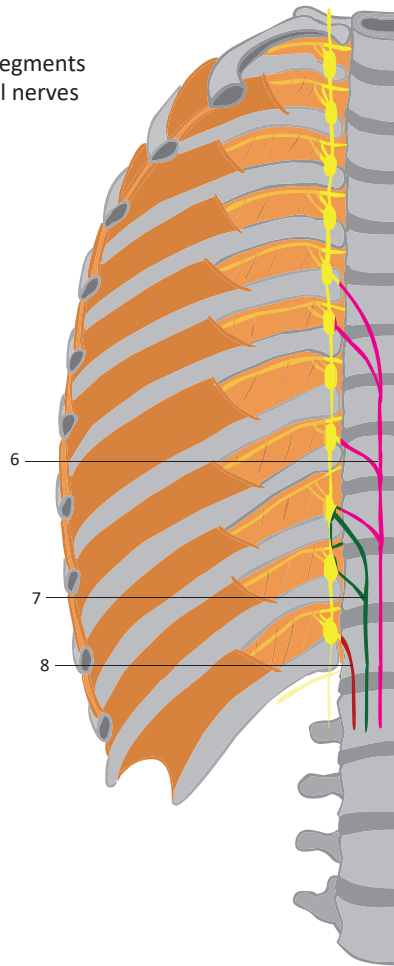
3.5 **Vertebral nerve** (*nervus vertebralis*) – travels towards the vertebral artery and forms a plexus around it

Thoracic ganglia

- 10–11 ganglia
- the trunk passes through the lumbar part of the diaphragm
- the preganglionic fibres originate from the thoracic spinal segments
- the white rami communicantes emerge from the intercostal nerves

Postganglionic fibers:

- 1 Grey rami communicantes** (*rami communicantes grisei*)
 - join the intercostal nerves
- 2 Vascular branches** (*rami vasculares*)
 - follow the individual branches of the thoracic aorta
- 3 Thoracic cardiac nerves** (*nervi cardiaci thoracici*)
 - arise from the 2nd – 4th thoracic ganglia
 - terminate in the cardiac plexus (*plexus cardiacus*)
- 4 Pulmonary branches** (*rami pulmonales*)
 - arise from the 2nd – 5th thoracic ganglia
 - form the pulmonary plexus (*plexus pulmonalis*) with parasympathetic fibres from the vagus nerve
- 5 Oesophageal branches** (*rami oesophagei*)
 - form the oesophageal plexus (*plexus oesophageus*) with parasympathetic branches from the left and right vagus nerves
- 6 Greater splanchnic nerve** (*nervus splanchnicus major*)
 - arises from the 5th (6th) – 9th thoracic ganglia
 - enters the abdominal cavity through the lumbar part of the diaphragm (through the innominate foramen)
 - terminates in the coeliac plexus (*plexus coeliacus*)
- 7 Lesser splanchnic nerve** (*nervus splanchnicus minor*)
 - originates from the 10th and 11th thoracic ganglia
 - enters the abdomen through the lumbar part of the diaphragm
 - terminates in the coeliac, renal and adrenal plexuses (*plexus coeliacus, plexus renalis et suprarenalis*)
- 8 Least splanchnic nerve** (*nervus splanchnicus imus*)
 - arises from the 12th thoracic ganglion
 - terminates in the renal and adrenal plexus (*plexus renalis et suprarenalis*)



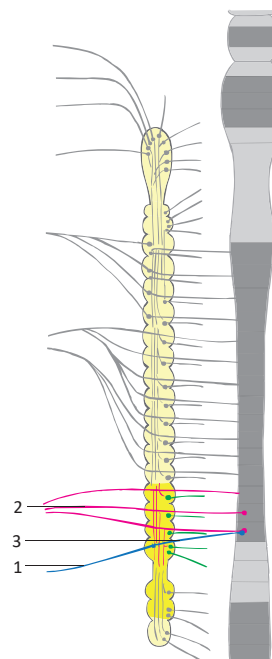
Right sympathetic trunk and its splanchnic nerves

Abdominal ganglia

- 4–5 ganglia in the lumbar region and 4–5 ganglia in the sacral region
- the preganglionic fibers originate from spinal segments L1–L3
- the right sympathetic trunk (*truncus sympathicus dexter*) lies medially to the psoas major and posteriorly to the inferior vena cava
- the left sympathetic trunk (*truncus sympathicus sinister*) lies medially to the psoas major and laterally to the abdominal aorta
- both trunks enter the lesser pelvis, travel caudally and merge together at the ganglion impar

Postganglionic fibers:

- 1 Grey rami communicantes** (*rami communicantes grisei*)
 - join the lumbar and sacral spinal nerves
- 2 Lumbar splanchnic nerves** (*nervi splanchnici lumbales*)
 - terminate in the abdominal aortic plexus and the plexuses around the branches of the abdominal aorta
- 3 Sacral splanchnic nerves** (*nervi splanchnici sacrales*)
 - terminate in the inferior hypogastric plexus



Scheme of the abdominal and splanchnic nerves

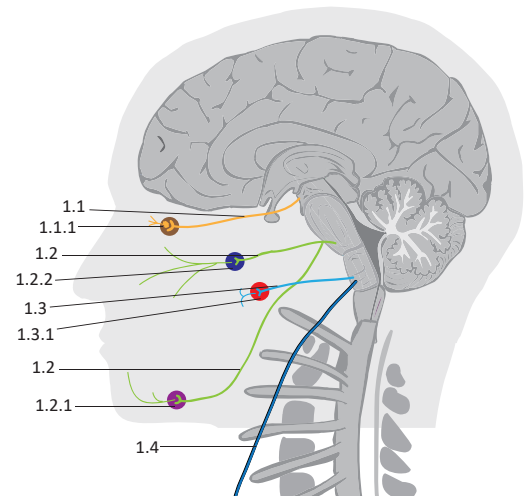
The **anastomosis of Jacobson** is formed by the tympanic nerve, tympanic plexus and minor petrosal nerve. It contains autonomous fibres for the otic ganglion.

The **Cannon-Böhm point** corresponds to a developmental border between the midgut (innervated by the vagus nerve) and hindgut (innervated by the sacral splanchnic nerves).

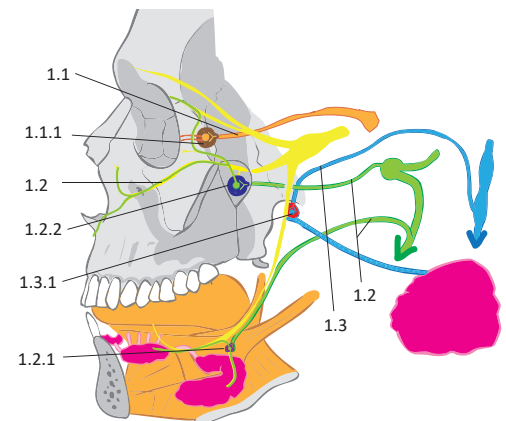
The parasympathetic nervous system is **divided into a cranial part (*pars cranialis*) and a sacral part (*pars sacralis*)**. The cranial part consists of the cranial nerves that carry parasympathetic fibres and their associated ganglia. The border between parasympathetic innervation from the cranial and sacral parts lies at the left colic flexure and is called the **Cannon-Böhm point**. The **sacral part consists of the 2nd to 4th sacral nerves**. The preganglionic fibres originate in the intermediolateral nucleus of the spinal segments **S2–S4** and travel as the **sacral splanchnic nerves** to the **inferior hypogastric plexus**. They synapse with neurones in ganglia within the plexus or pass through the plexus to synapse with neurones in the walls of the target organs. The sacral part innervates the large intestines aboral to the left colic flexure, the urinary bladder, erectile bodies and the internal genital organs (except for the gonads). The parasympathetic nervous system is **responsible for activities associated with restoring energy stores and anabolic reactions. It slows down the heart rate, constricts bronchi, increases intestinal peristalsis and gland secretion and contracts of the detrusor muscle**.

Cranial part

- 1.1 **Oculomotor nerve (*nervus oculomotorius*)**
 - innervates the ciliary body and sphincter pupillae
 - 1.1.1 **Ciliary ganglion (*ganglion ciliare*)** – is located in the orbit
- 1.2 **Facial nerve (*nervus facialis*)**
 - 1.2.1 **Submandibular ganglion (*ganglion submandibulare*)**
 - is located in the submandibular triangle
 - innervates the submandibular, sublingual and lingual glands
 - fibres from the chorda tympani reach the submandibular ganglion via the lingual nerve
 - 1.2.2 **Pterygopalatine ganglion (*ganglion pterygopalatinum*)**
 - is located in the pterygopalatine fossa
 - innervates the lacrimal, palatine, nasal and nasopharyngeal glands
- 1.3 **Glossopharyngeal nerve (*nervus glossopharyngeus*)**
 - innervates the parotid gland
 - preganglionic fibres reach the otic ganglion (*ganglion oticum*) via the tympanic nerve, tympanic plexus (*plexus tympanicus*) and lesser petrosal nerve
 - the postganglionic fibres travel to the parotid gland via the auriculotemporal nerve and to the buccal glands via the buccal nerve
 - 1.3.1 **Otic ganglion (*ganglion oticum*)**
 - is located in the infratemporal fossa
- 1.4 **Vagus nerve (*nervus vagus*)**
 - innervates of smooth muscle, glands, internal organs and myocardium



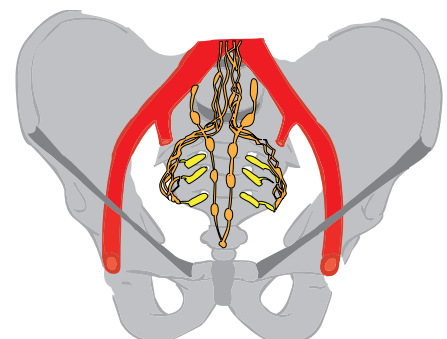
Scheme of the cranial parasympathetic system



Scheme of the cranial parasympathetic system and secretory innervation of the salivary glands

Sacral part

- Intermediolateral nucleus (*nucleus intermediolateralis*) S2–S4**
 – sacral parasympathetic nuclei in the lateral horns of the spinal cord
- Sacral splanchnic nerves (*nervi splanchnici sacrales*)**
 – preganglionic fibers
 – parasympathetic fibres into the inferior hypogastric plexus (*plexus hypogastricus inferior*)
 – preganglionic fibres enter the mixed pelvic plexuses around the organs of the lesser pelvis and synapse with postganglionic fibres in the purely parasympathetic ganglia

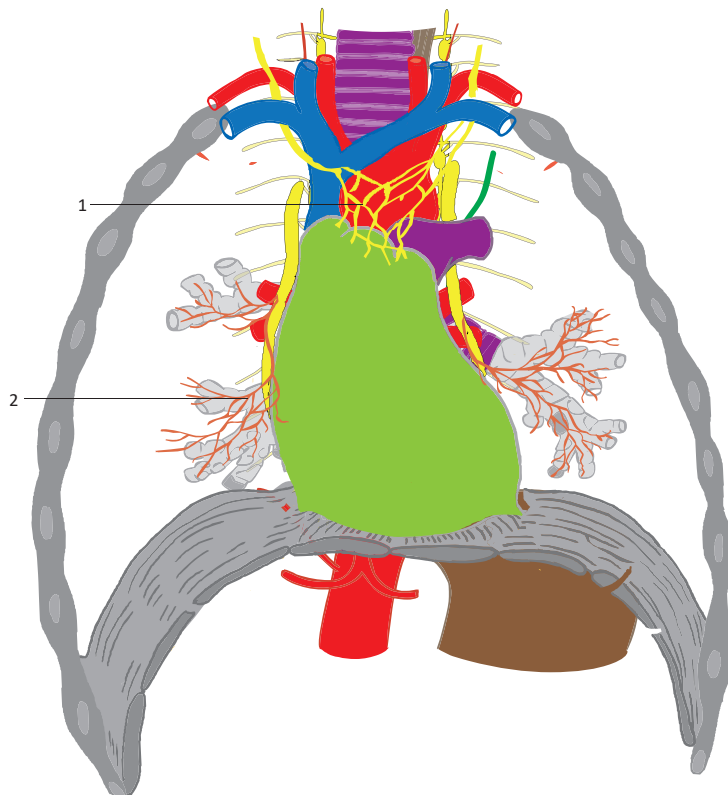


Scheme of the sacral parasympathetic system

The mixed plexuses of the thorax are the **cardiac plexus** (*plexus cardiacus*), **pulmonary plexus** (*plexus pulmonalis*), and **thoracic aortic plexus** (*plexus aorticus thoracicus*). The thoracic aortic plexus passes along the aorta through the aortic hiatus of the diaphragm into the retroperitoneal space, where it follows the branches of the aorta to reach the abdominal organs.

Thorax

- 1 **Cardiac plexus** (*plexus cardiacus*)
 - spreads around the aortic arch and
 - is subdivided according to its relation to the aortic arch
 - **Superficial cardiac plexus** (*plexus cardiacus superficialis*)
 - lies between the aortic arch and the pulmonary trunk
 - **Deep cardiac plexus** (*plexus cardiacus profundus*)
 - lies between the aortic arch and the trachea
- 1.1 **Sympathetic fibres**
 - increases heart rate, conduction speed, contractility and excitability
 - 1.1.1 **Superior cervical cardiac nerve** (*nervus cardiacus cervicalis superior*)
 - 1.1.2 **Middle cervical cardiac nerve** (*nervus cardiacus cervicalis medius*)
 - 1.1.3 **Inferior cervical cardiac nerve** (*nervus cardiacus cervicalis inferior*)
 - 1.1.4 **Thoracic cardiac nerves** (*nervi cardiaci thoracici*)
- 1.2 **Parasympathetic fibres**
 - decreases heart rate, conduction speed, contractility and excitability
 - 1.2.1 **Superior cardiac branch** (*ramus cardiacus superior*)
 - 1.2.2 **Inferior cardiac branch** (*ramus cardiacus inferior*)
 - 1.2.3 **Thoracic cardiac branches** (*rami cardiaci thoracici*)
- 2 **Pulmonary plexus** (*plexus pulmonalis*) – a mixed plexus supplying the lungs
 - 2.1 **Sympathetic fibres**
 - dilate bronchial smooth muscle, widening the bronchial lumen
 - inhibit secretion of mucosal glands
 - 2.1.1 **Thoracic pulmonary branches** (*rami pulmonales thoracici*)
 - 2.2 **Parasympathetic fibres**
 - 2.2.1 **Bronchial branches** (*rami bronchiales*)
 - increase tension of bronchial smooth muscle, narrowing the bronchial lumen
 - stimulate secretion of mucosal glands
- 3 **Oesophageal plexus** (*plexus oesophageus*)



Difference in nomenclature:

- **nerves** (*nervi*) are formed by **sympathetic** fibres
- **branches** (*rami*) are formed by **parasympathetic** fibres

Accelerating nerves (*nervi accelerantes*) is an obsolete name for the sympathetic cardiac nerves, which accelerate heart rate. Retarding nerves (*nervi retardantes*) is an obsolete name for the parasympathetic cardiac branches, which slow the heart rate.

The **cardiac plexus** continues along the left and right coronary arteries as the left and right coronary plexuses and participates in adjusting the diameter of their lumen. It reaches the conducting system of the heart along the branches of the coronary arteries, where it regulates heart rate. Sympathetic fibres dilate the coronary arteries increasing perfusion of the heart. An increase in sympathetic activation results in tachycardia, higher contraction force, and a higher demand for oxygen and nutrients. Parasympathetic fibres constrict the coronary arteries. Increased parasympathetic activity causes bradycardia and lowers the metabolic demands of the myocardium.

The **pulmonary sympathetic fibres** dilate the bronchi. This occurs in response to increased oxygen demand. The dilated bronchi increase respiratory gas exchange. The work needed for active inspiration is indirectly decreased by bronchodilation.

The **activity of smooth muscle, myocardium, and gland secretion** is controlled by signal molecules and hormones. Epinephrine (adrenaline) causes tachycardia and increases heart contraction force. Nitric oxide (NO) causes vasodilation. Angiotensin II is the most potent vasoconstrictor and serotonin potentiates intestinal peristalsis.

Clinical notes

Renal denervation is an intravascular procedure that uses radiofrequency pulses to ablate the sympathetic fibres of the renal plexus. It is performed by entering a catheter into the femoral artery and passing it up the renal artery. It is used to treat intractable hypertension.

Abdomen

- 1 **Aortic abdominal plexus** (*plexus aorticus abdominalis*)
 - surrounds the abdominal aorta
 - splits into two iliac plexuses at the aortic bifurcation
 - ramifies into several smaller plexuses, which innervate individual organs

1.1 Sympathetic fibres

- come from the sympathetic trunk (*truncus sympathicus*)
- 1.1.1 **Greater splanchnic nerve** (*nervus splanchnicus major*)
- 1.1.2 **Lesser splanchnic nerve** (*nervus splanchnicus minor*)
- 1.1.3 **Least splanchnic nerve** (*nervus splanchnicus imus*)
- 1.1.4 **Lumbar splanchnic nerves** (*nervi splanchnici lumbales*)

1.2 Parasympathetic fibres

- come from the vagus nerve (*nervus vagus*)
- 1.2.1 **Coeliac branches** (*rami coeliaci*)
- 1.2.2 **Gastric branches** (*rami gastrici*)
- 1.2.3 **Hepatic branches** (*rami hepatici*)
- 1.2.4 **Renal branches** (*rami renales*)

Subplexuses of the abdominal aortic plexus

2 Coeliac plexus (*plexus coeliacus*)

- is located around the coeliac trunk (*truncus coeliacus*)
- 2.1 **Right and left coeliac ganglia** (*ganglion coeliacum dextrum et sinistrum*)
 - are located on both sides of the coeliac trunk
 - follow the branches of the coeliac trunk

Subplexuses:

- 2.2 **Hepatic plexus** (*plexus hepaticus*)
- 2.3 **Gastric plexus** (*plexus gastricus*)
- 2.4 **Splenic plexus** (*plexus splenicus*)
- 2.4 **Pancreatic plexus** (*plexus pancreaticus*)
- 2.5 **Renal and suprarenal plexuses** (*plexus renalis et suprarenalis*)
 - paired plexuses
 - innervate the kidneys and adrenal glands
 - 2.5.1 **Aorticorenal ganglion** (*ganglion aorticorenale*)
 - a paired ganglion located at the origin of the renal artery

3 Testicular/ovarian plexus (*plexus testicularis/ovaricus*)

- innervates the gonads
- follows the gonadal vessels to the testes/ovaries

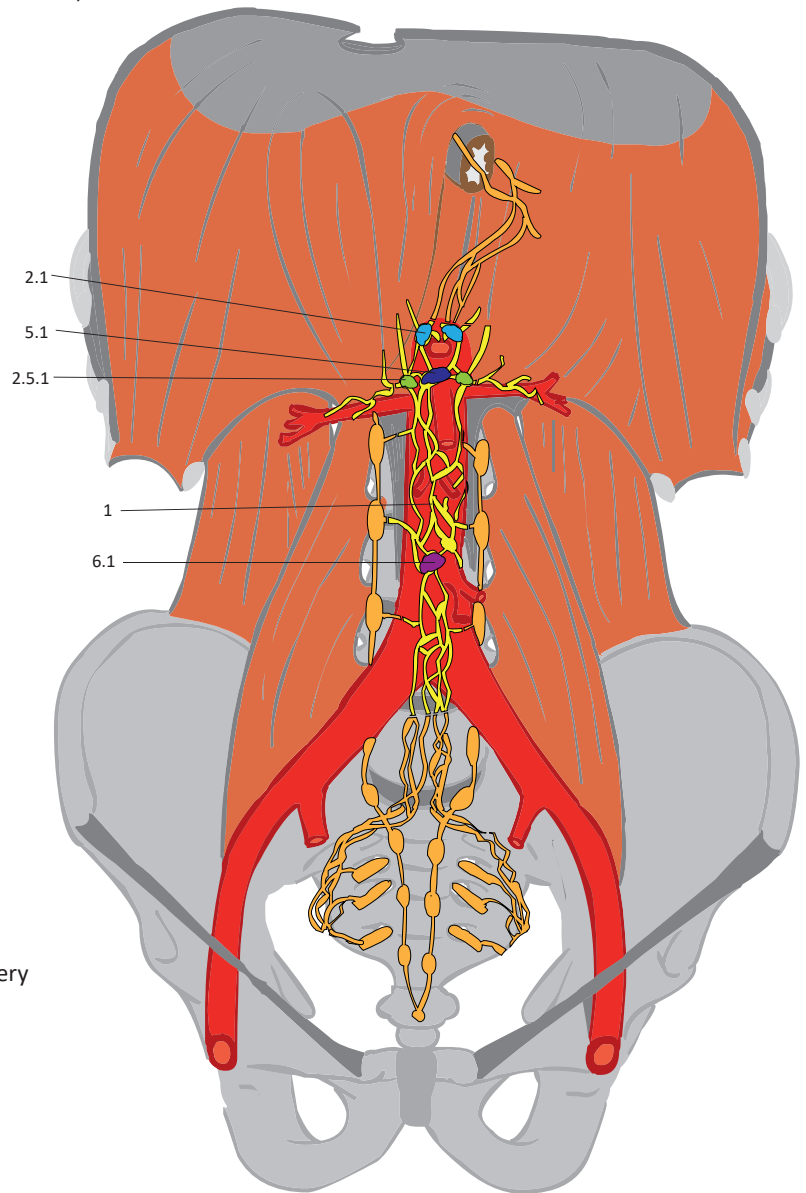
4 Ureteric plexus (*plexus uretericus*) – innervates the ureters

5 Superior mesenteric plexus (*plexus mesentericus superior*)

- is located at the point where the superior mesenteric artery emerges from the aorta
- participates in the innervation of the intestines from the major duodenal papilla to the left colic flexure
- 5.1 **Superior mesenteric ganglion** (*ganglion mesentericum superius*)
 - an unpaired ganglion
 - is located at the point where the superior mesenteric artery emerges from the aorta

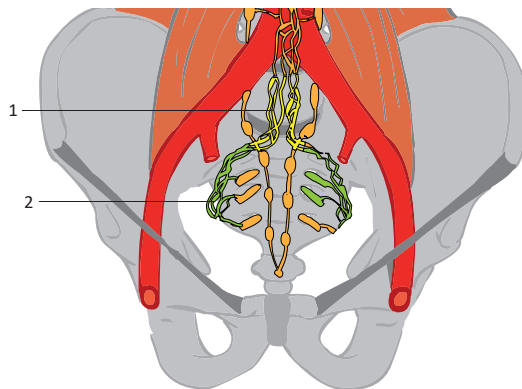
6 Inferior mesenteric plexus (*plexus mesentericus inferior*)

- is located at the level of the inferior mesenteric artery
- parasympathetic fibres come from the sacral part of the parasympathetic nervous system
- innervates the descending colon, sigmoid colon and rectal ampulla
- 6.1 **Inferior mesenteric ganglion** (*ganglion mesentericum inferius*)
 - unpaired ganglion
 - there may be multiple inferior mesenteric ganglia
 - located at the origin of the inferior mesenteric artery



Pelvis

- 1 **Superior hypogastric plexus / presacral nerve**
(*plexus hypogastricus superior / nervus presacralis*)
 - the continuation of the abdominal aortic plexus (*plexus aorticus abdominalis*)
 - splits into the right and left hypogastric nerves (*nervus hypogastricus dexter et sinister*) at the sacral promontory
 - 1.1 **Lumbar and sacral splanchnic nerves (L4, L5 and S1)**
(*nervi splanchnici lumbales et sacrales*)
 - 1.2 **Aortic abdominal plexus** (*plexus aorticus abdominalis*)
- 2 **Inferior hypogastric plexus / pelvic plexus** (*plexus hypogastricus inferior / plexus pelvici*)
 - a mixed plexus forming a continuation of the hypogastric nerves
 - passes laterally to the rectum into the lesser pelvis
 - the parasympathetic fibres come from the white communicating branches of S2–4 via the sacral splanchnic nerves
 - parasympathetic fibres synapse in ganglia close to the pelvic organs
 - branches into smaller plexuses which innervate the individual organs of the lesser pelvis and follow the branches of the internal iliac artery
 - 2.1 **Rectal plexus** (*plexus rectalis*) – innervates the rectum
 - 2.2 **Vesical and ureteric plexus** (*plexus vesicalis et uretericus*)
 - innervates the urinary bladder and the pelvic part of the ureter
 - 2.3 **Prostatic plexus** (*plexus prostaticus*)
 - innervates the smooth muscle of the prostate, prostatic part of urethra, prostatic and urethral glands and seminal glands
 - 2.4 **Deferential plexus** (*plexus deferentialis*) – innervates the ductus deferens
 - 2.5 **Uterovaginal plexus** (*plexus uterovaginalis*) – innervates the ovaries, uterus and vagina
 - 2.6 **Cavernous nerves of clitoris/penis** (*nervi cavernosi clitoridis/penis*)
 - innervates the cavernous bodies of the penis and clitoris

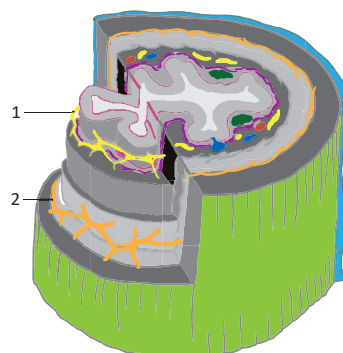


4.4

Enteric nervous system

The **enteric nervous system** is a separate part of the ANS. It is located in the **wall of the digestive tract**. It **controls gastrointestinal functions** more or less independently from the rest of the ANS.

- 1 **Submucous plexus / Meissner's plexus**
(*plexus submucosus*)
 - an autonomic plexus in the submucosa of the digestive tract
 - plays an important role in the regulation of secretion of substances into the intestinal lumen and blood supply of the intestinal wall
 - operates predominantly on a local level
- 2 **Myenteric plexus / Auerbach's plexus** (*plexus myentericus*)
 - a plexus in the smooth muscle of the digestive tract
 - controls and coordinates of peristalsis
 - operates on larger segments of the digestive tract

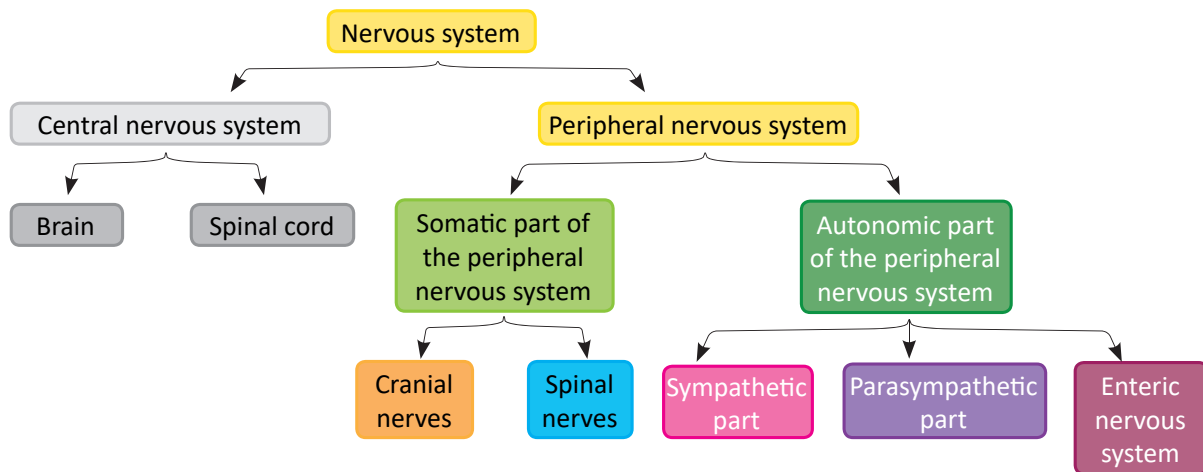


Scheme of the autonomic plexuses in the digestive tube (submucous and myenteric plexus)

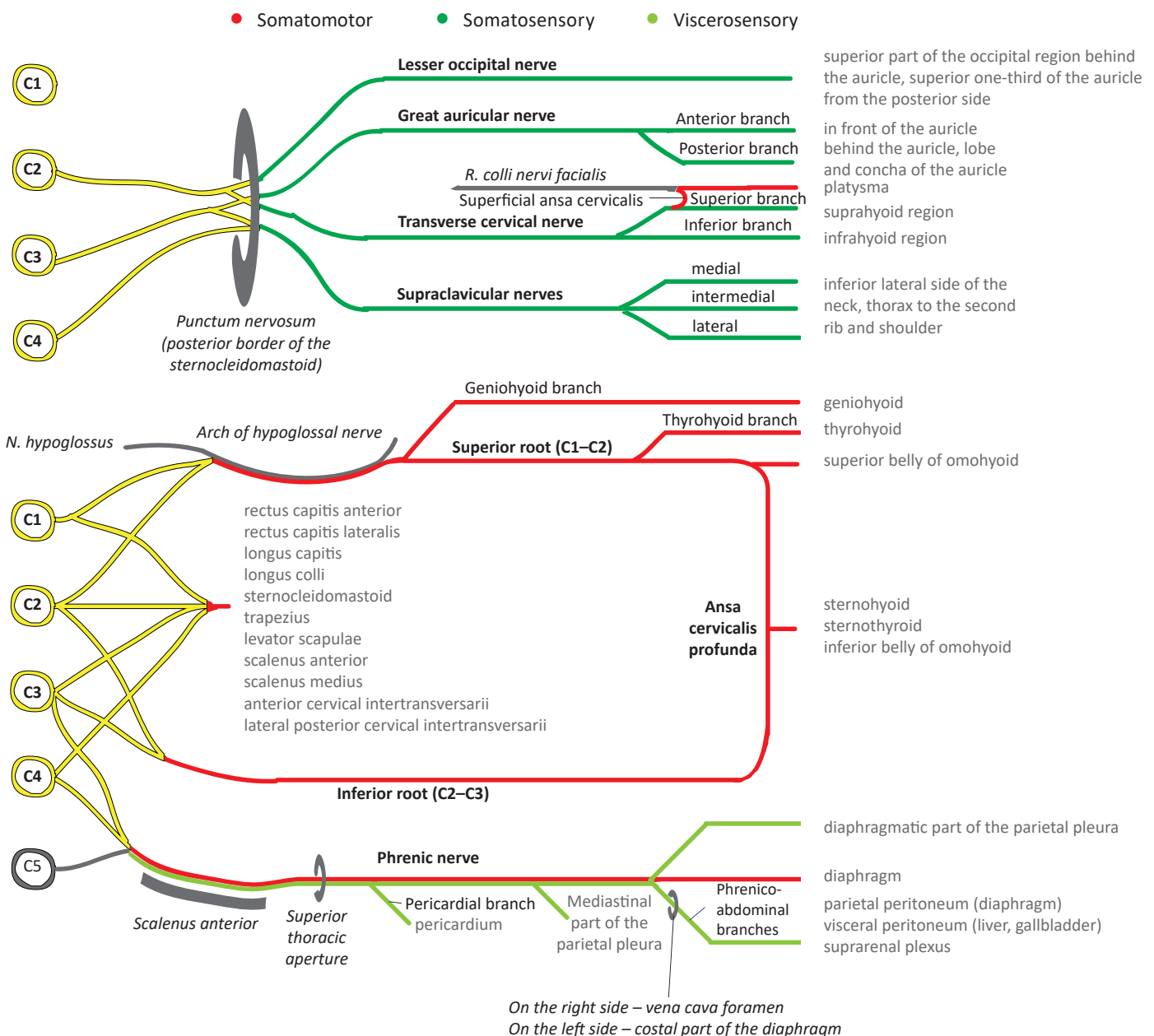
Clinical notes

Hirschsprung disease (congenital megacolon / intestinal aganglioneosis) is a congenital absence of the ganglia of the enteric nervous system in the aboral part of the colon. It leads to permanent contraction of the aganglionic segment causing accumulation of stool above the contracted segment dilating the colon.

Cajal cells are found in the digestive tract. They are believed to be the pacemaker cells of intestinal peristalsis. GIST tumors (gastrointestinal stromal tumors) arise from these cells.

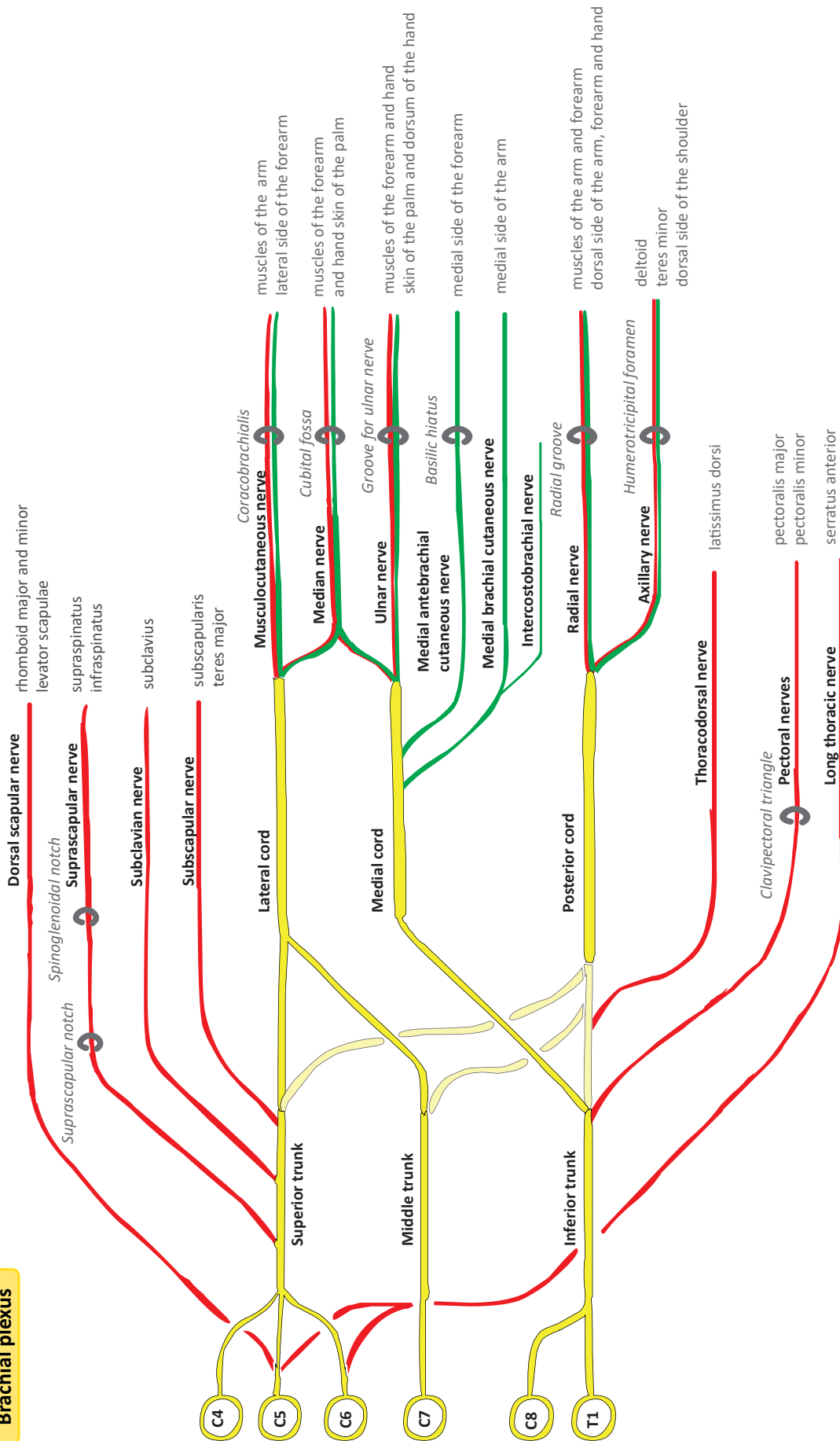


Cervical plexus



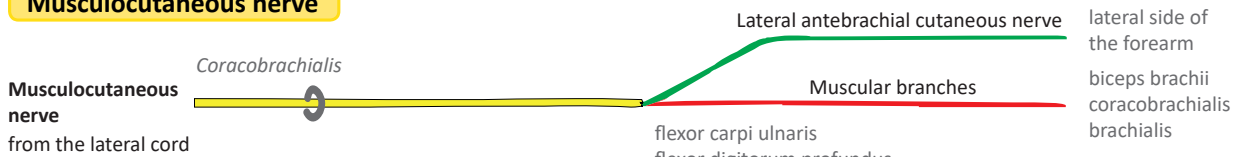
● Somatomotor ● Somatosensory

Brachial plexus

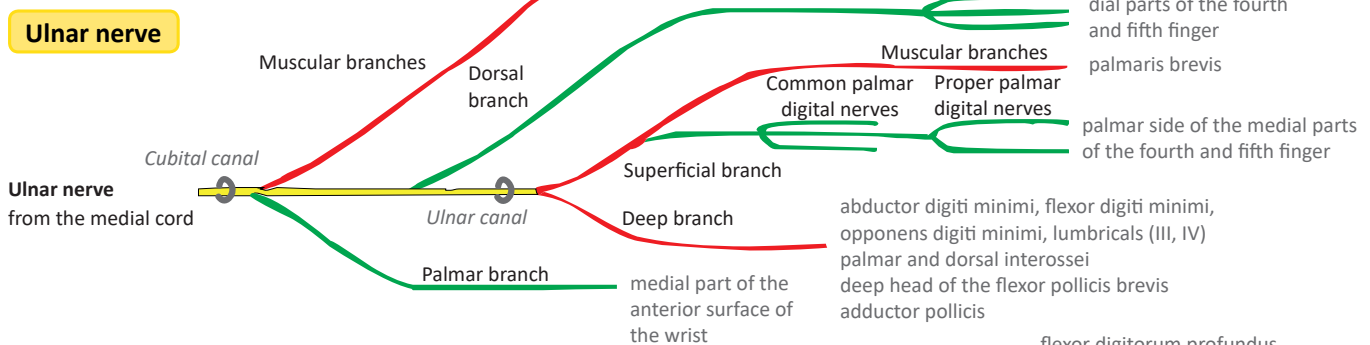


● Somatomotor ● Somatosensory

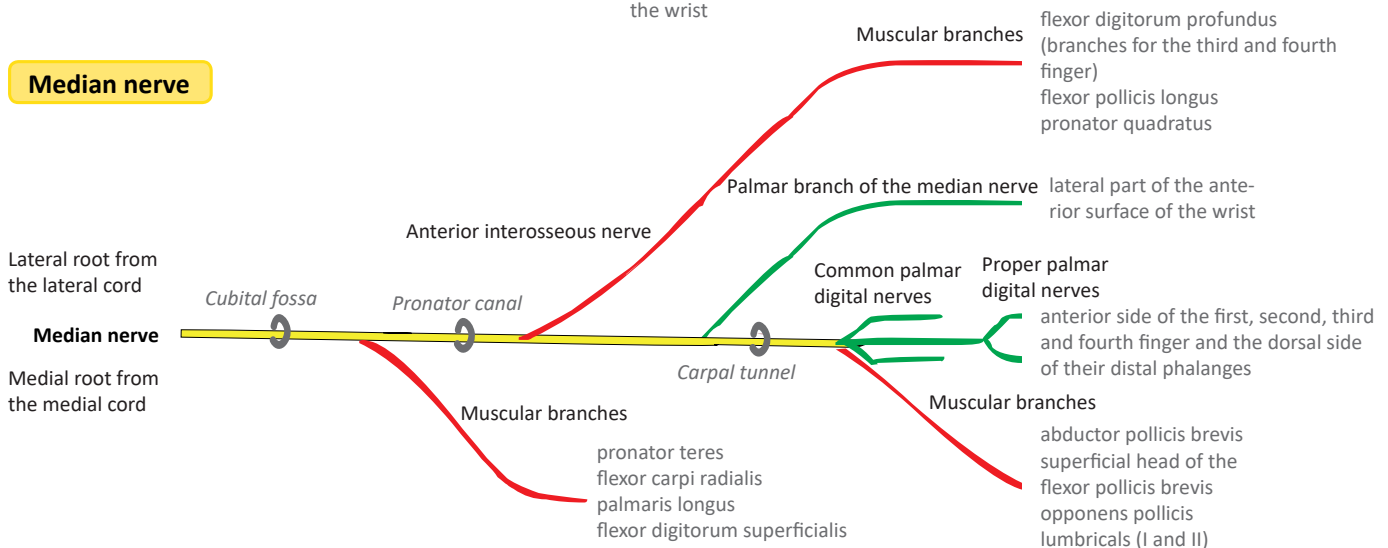
Musculocutaneous nerve



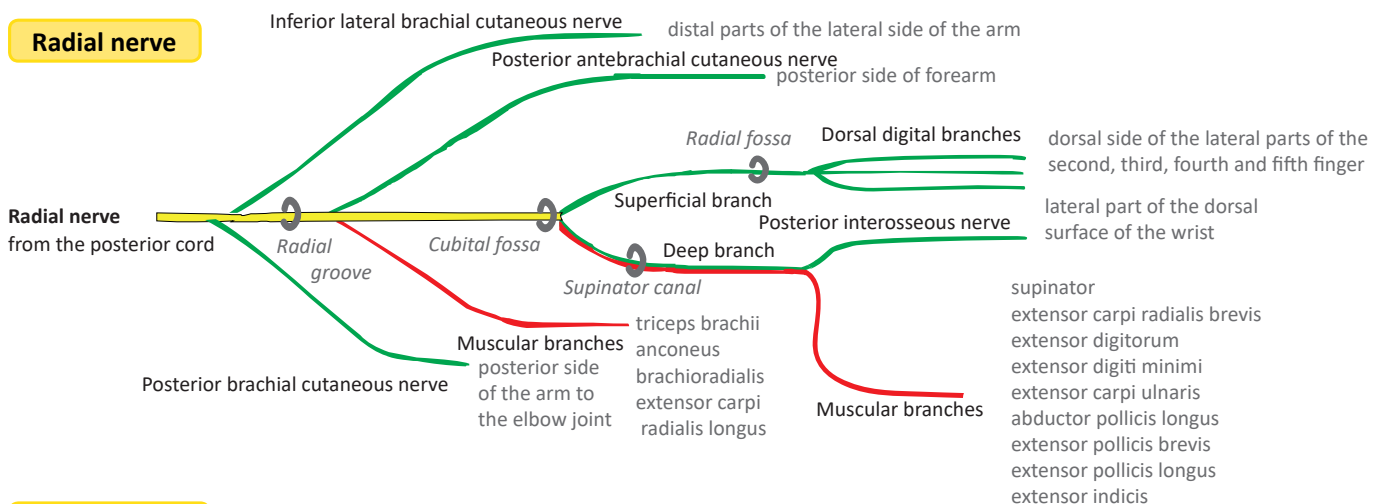
Ulnar nerve



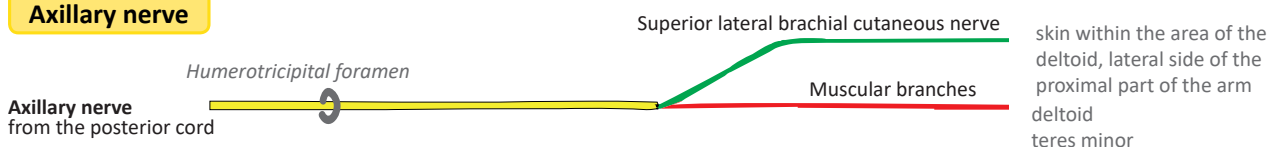
Median nerve



Radial nerve

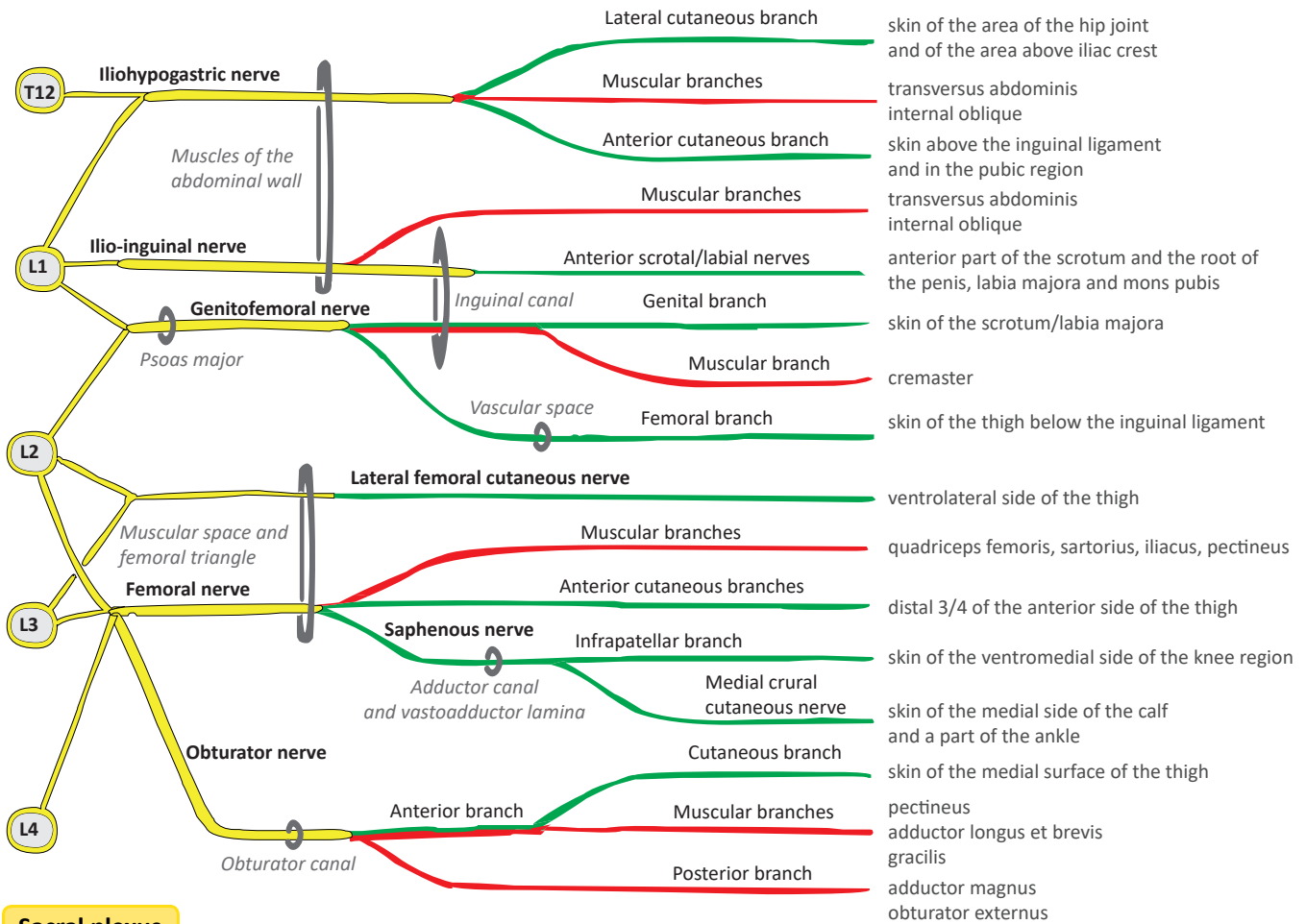


Axillary nerve

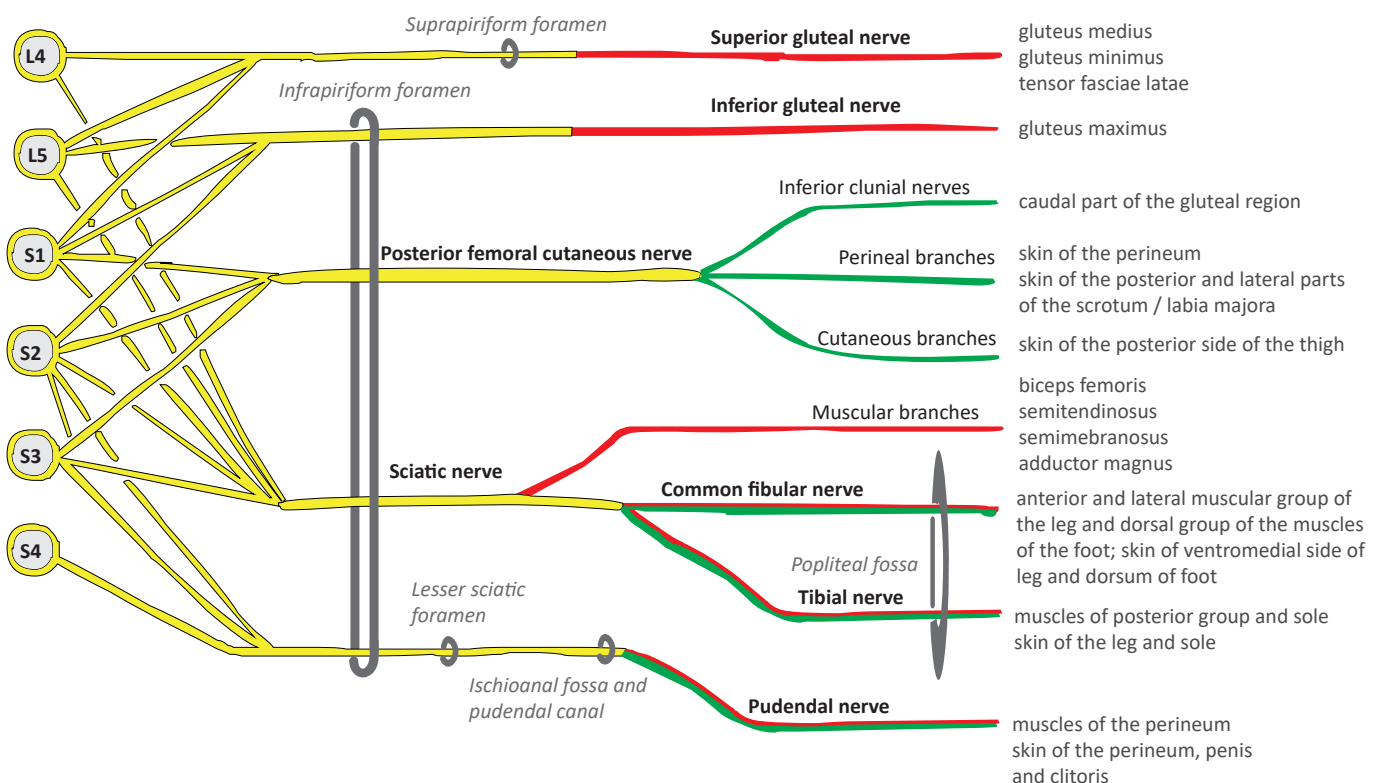


● Somatomotor ● Somatosensory

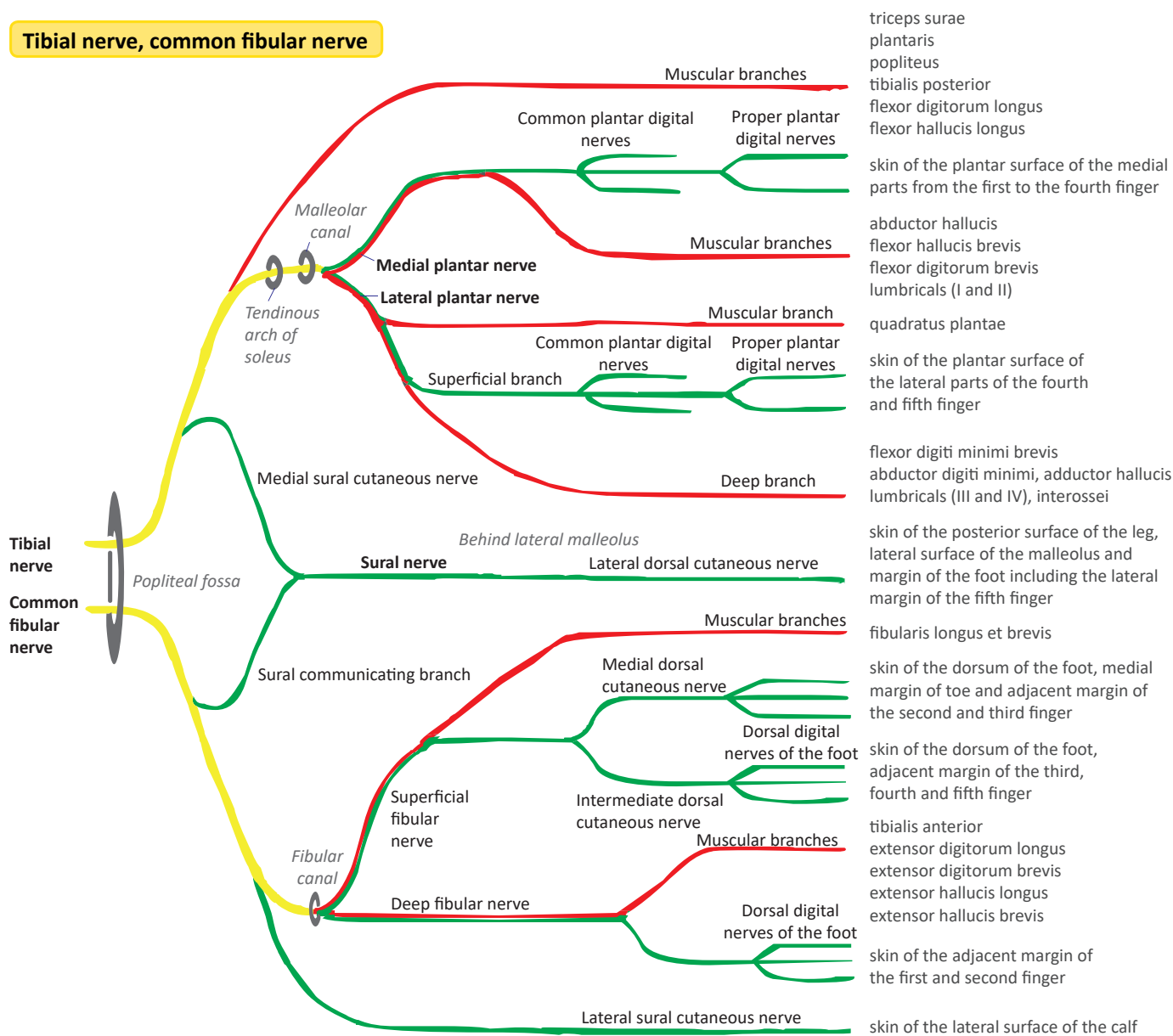
Lumbar plexus



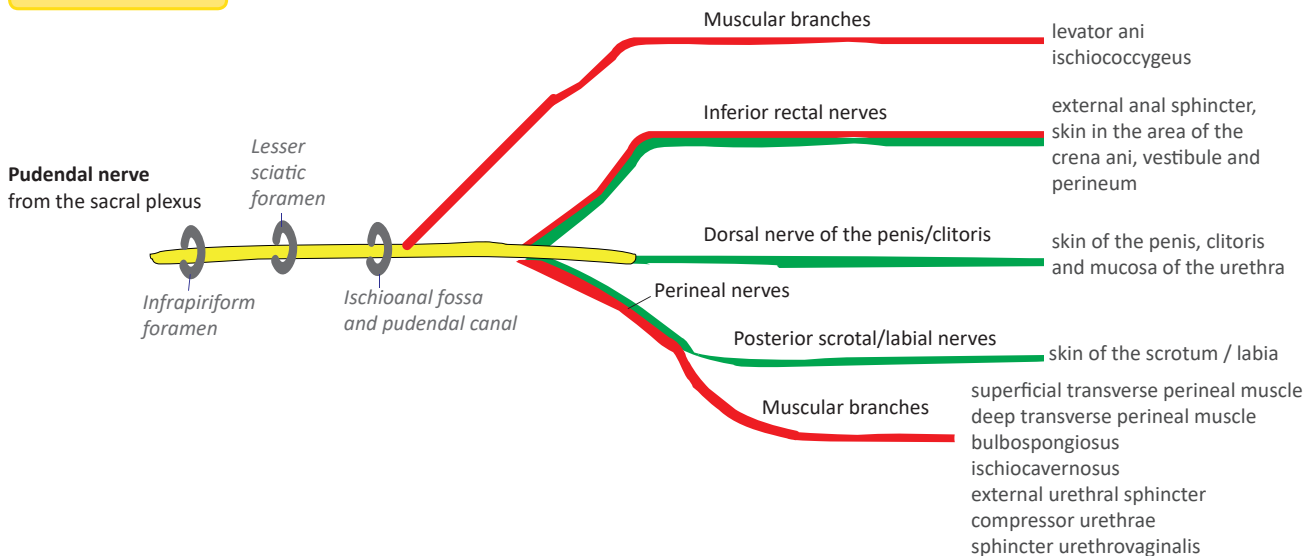
Sacral plexus



Tibial nerve, common fibular nerve



Pudendal nerve

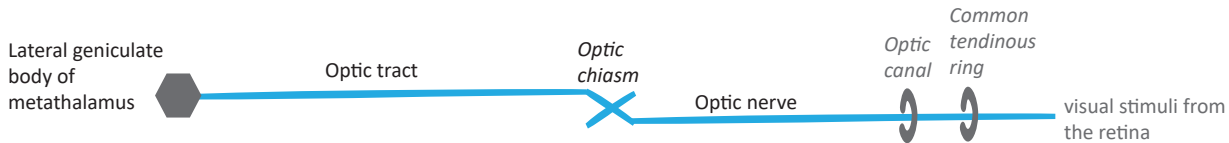


- Somatomotor ● Parasympathetic part ● Sensory

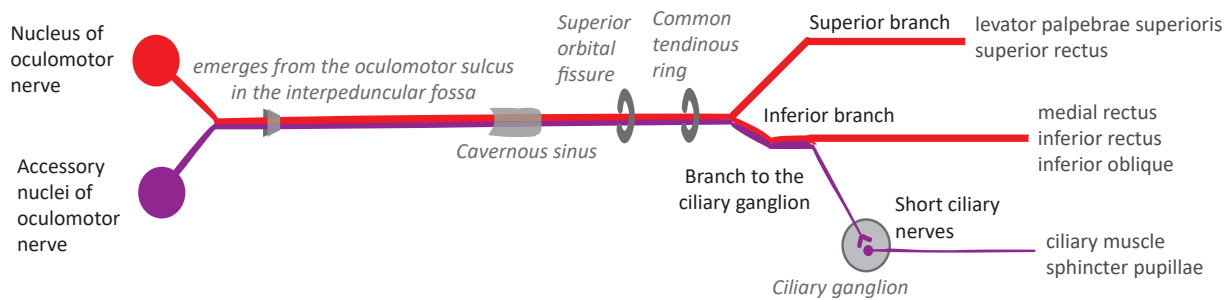
Olfactory nerve



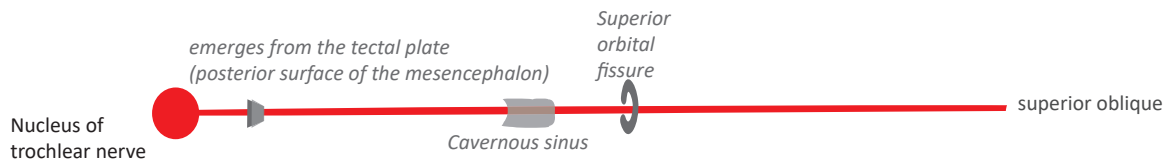
Optic nerve



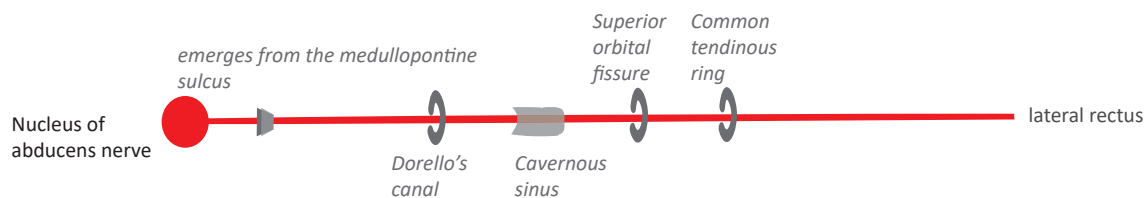
Oculomotor nerve

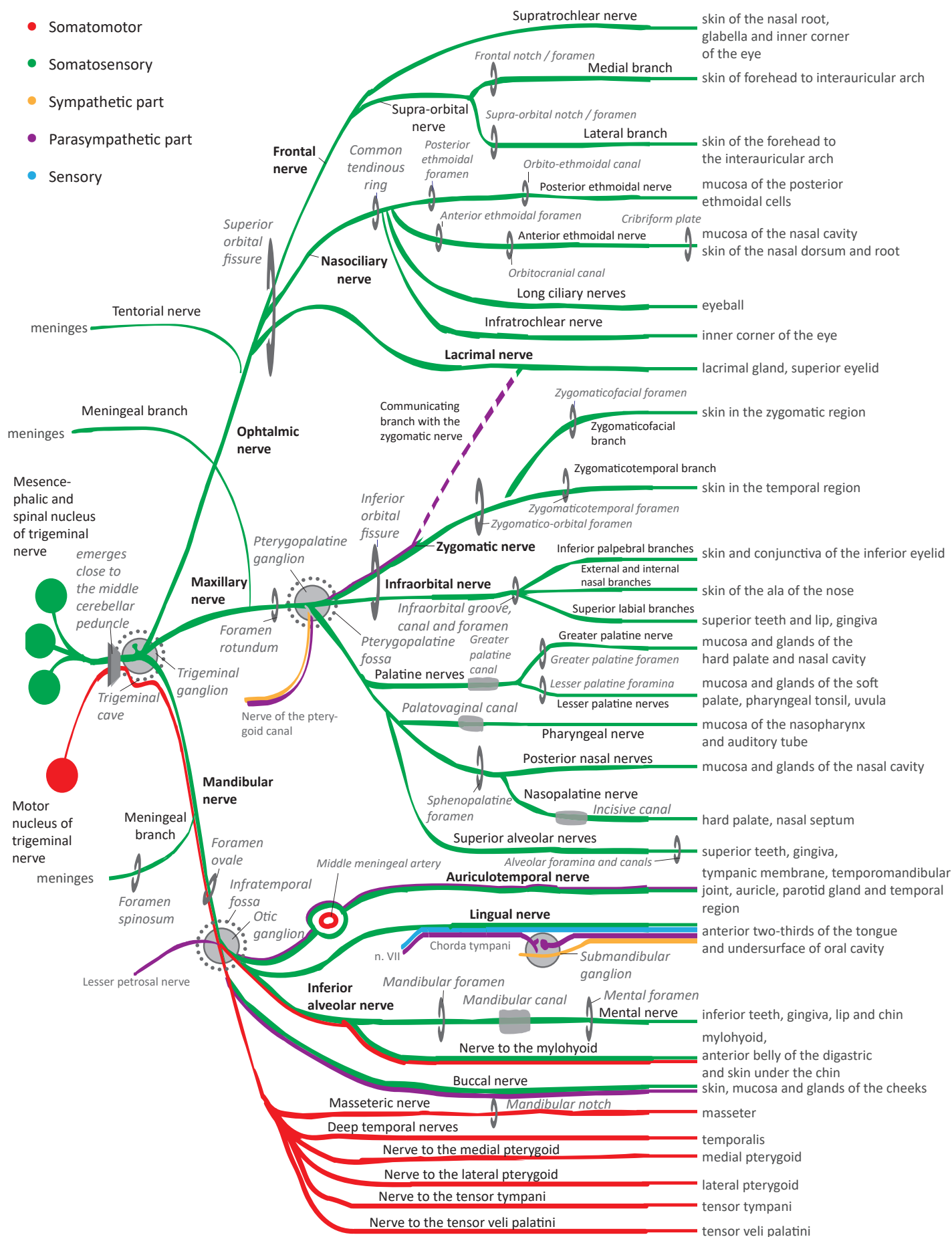


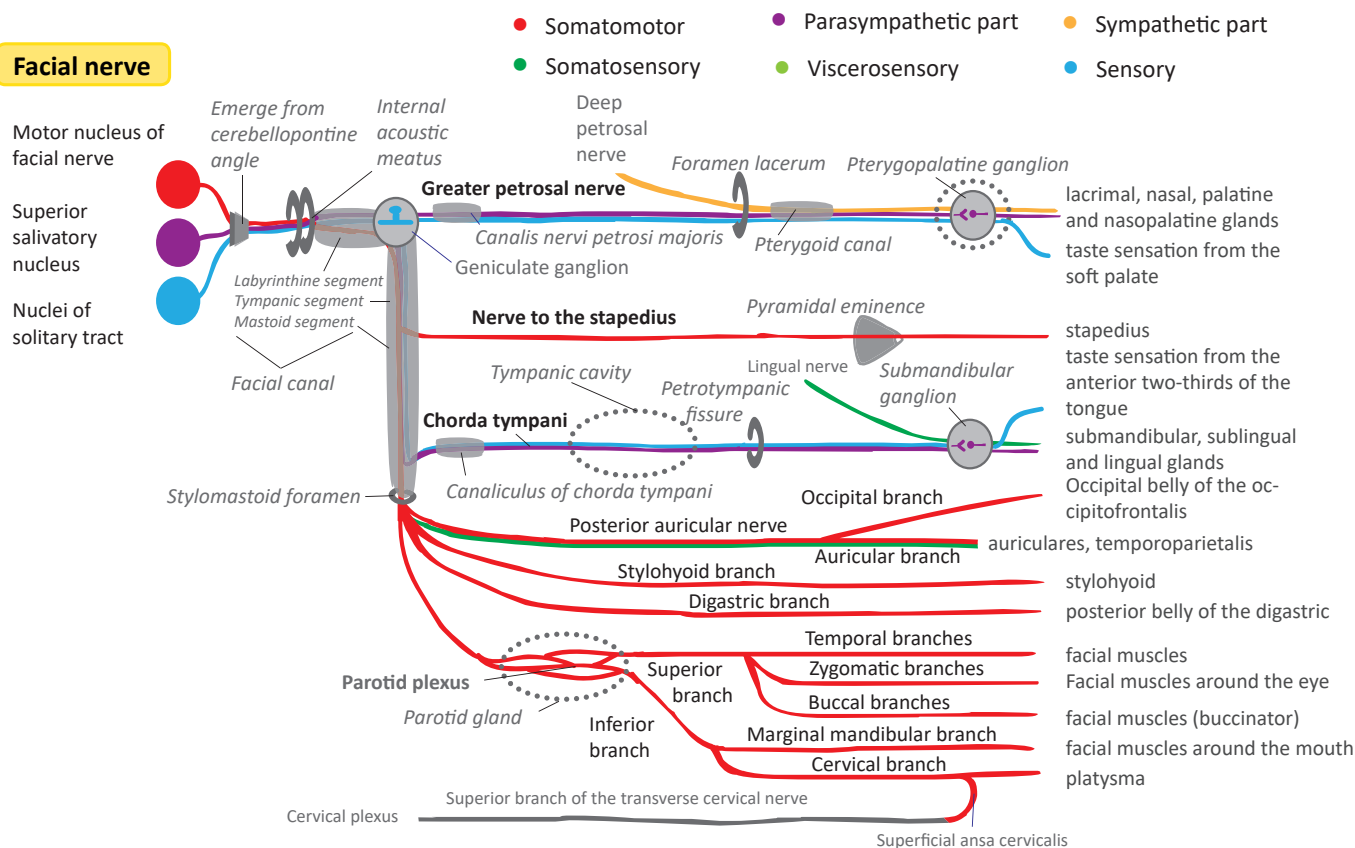
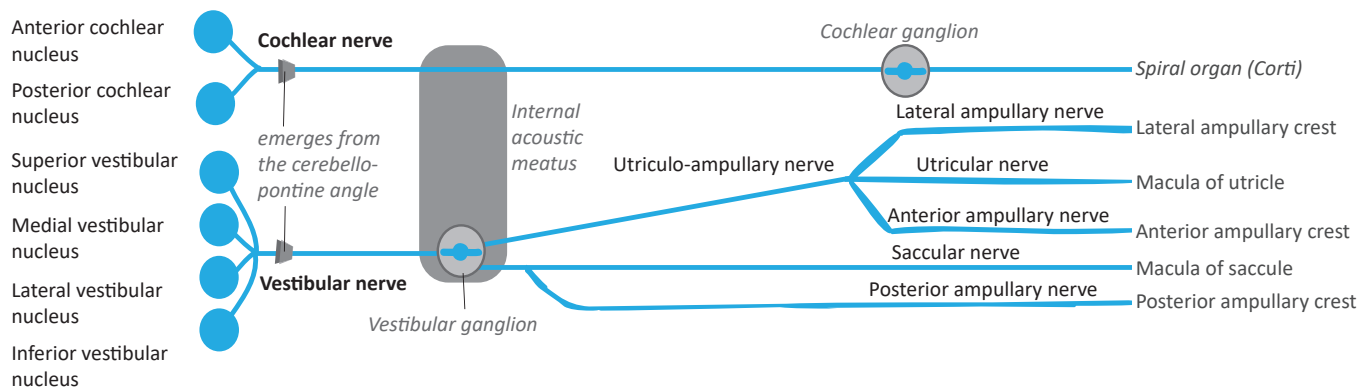
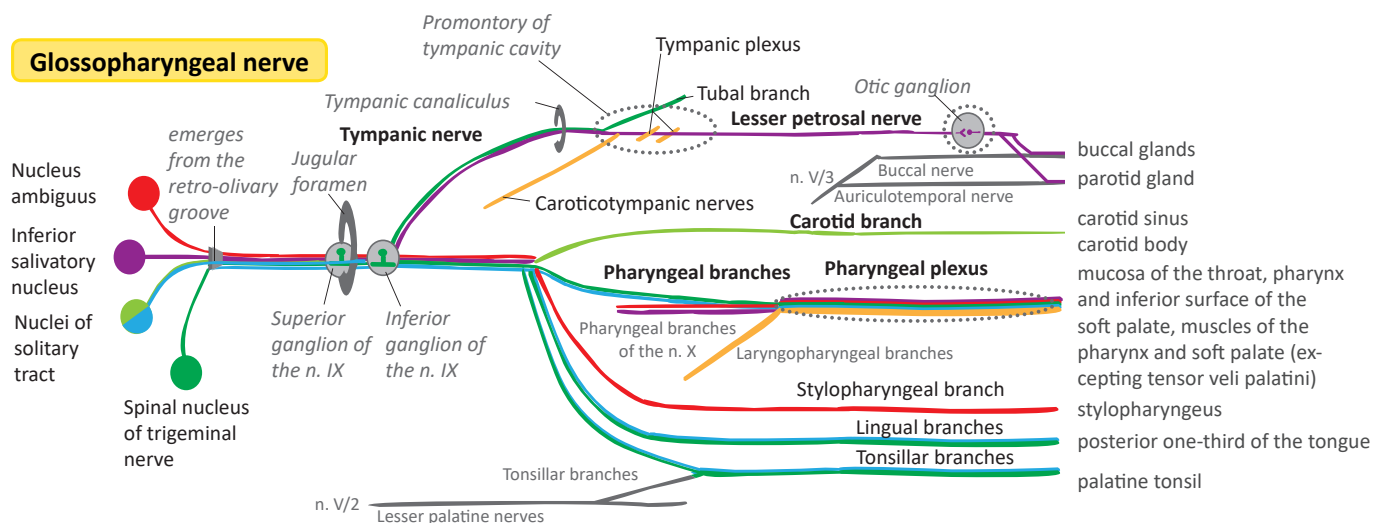
Trochlear nerve



Abducens nerve

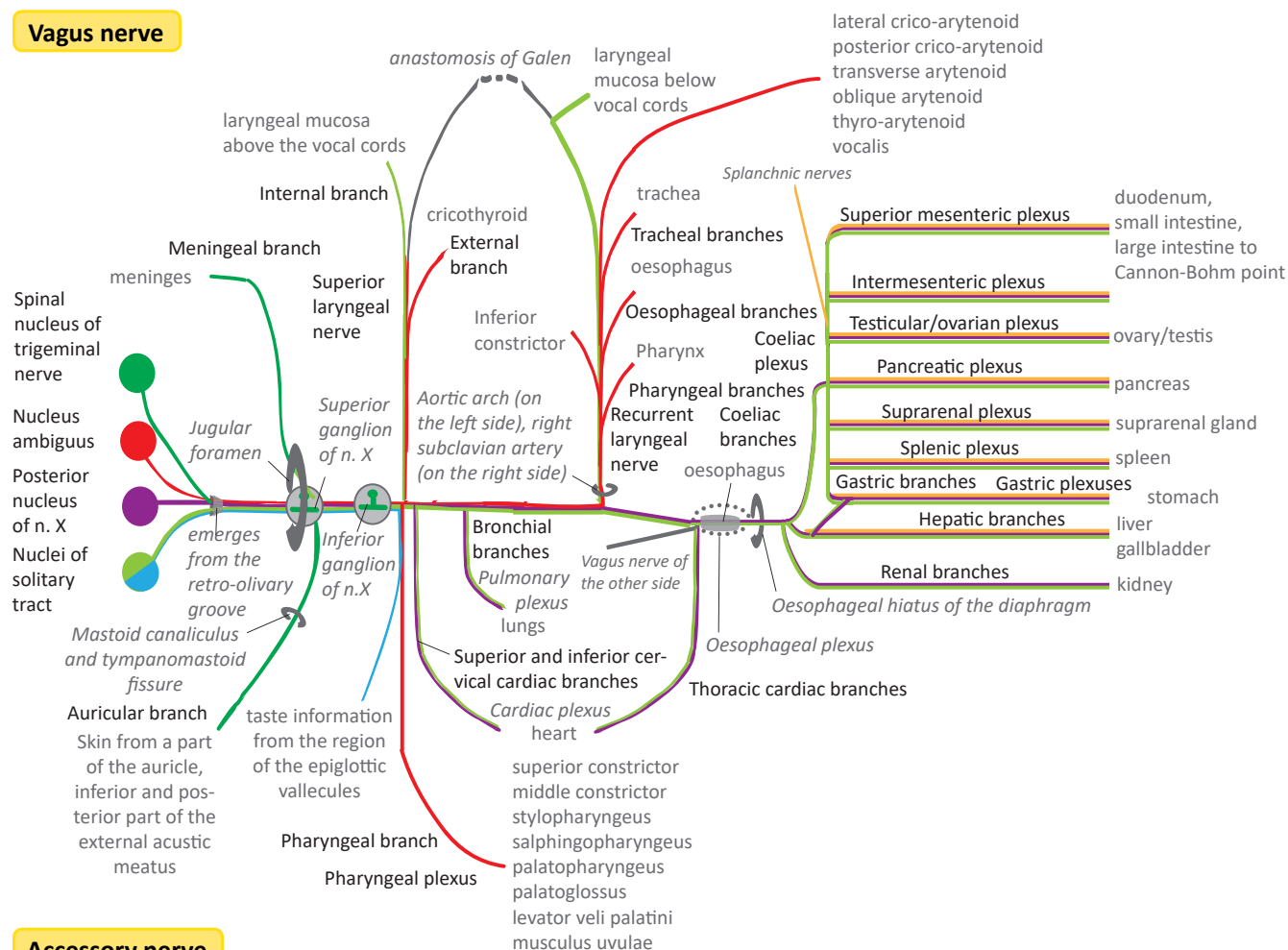




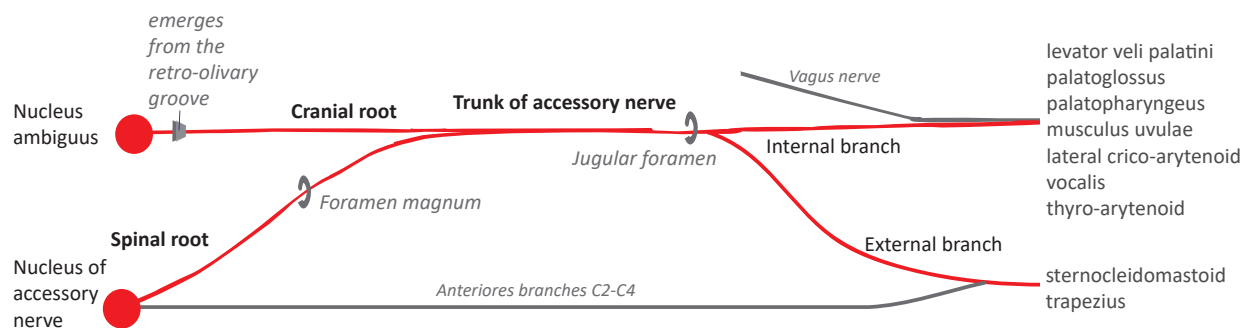
Facial nerve**Vestibulocochlear nerve****Glossopharyngeal nerve**

● Somatomotor ● Parasympathetic part ● Sympathetic part ● Somatosensory ● Viscerosensory ● Sensory

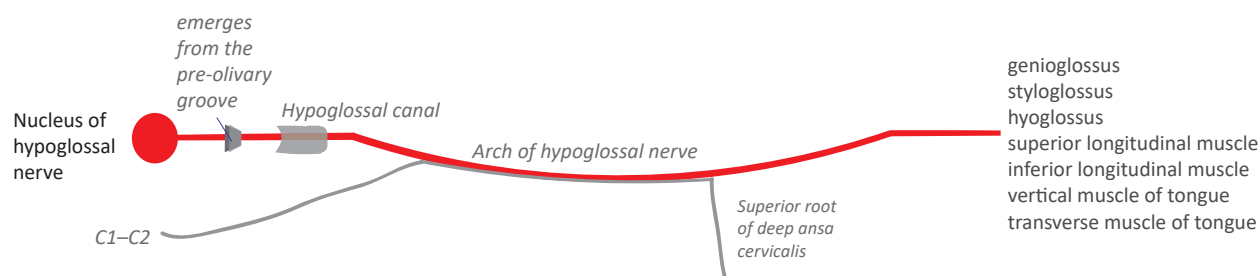
Vagus nerve

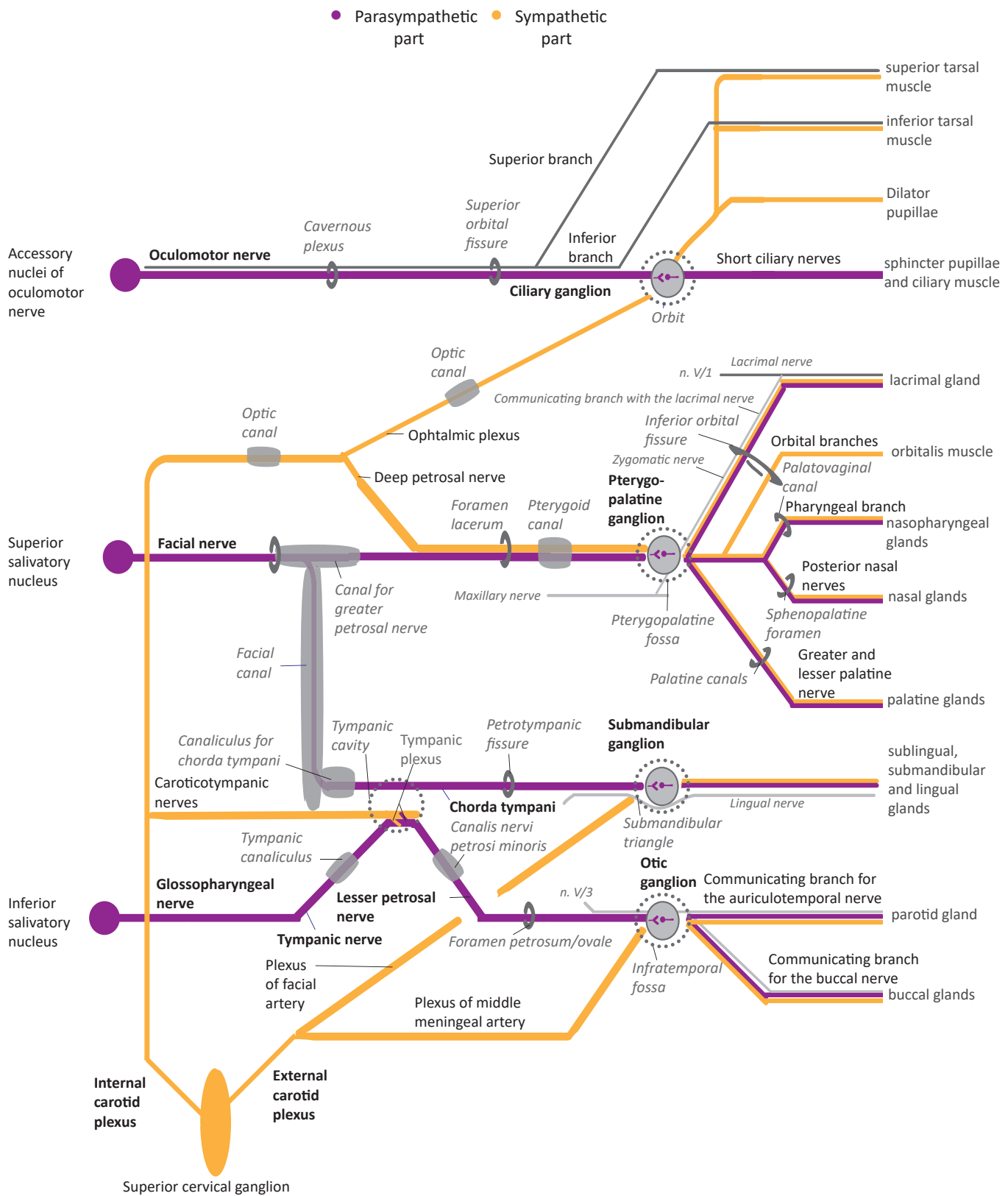


Accessory nerve



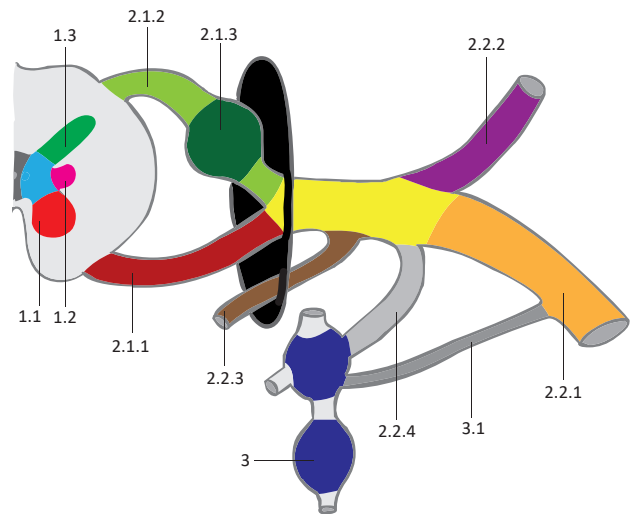
Hypoglossal nerve





I. General structure of the peripheral nervous system

1. Describe the classification of the peripheral nervous system. (p. 336)
2. Describe the structure of a peripheral nerve. (p. 336)
3. Describe the general structure of a spinal nerve. (p. 337)
4. State the types of fibres found within a peripheral nerve. (p. 337)
5. Describe the afferent and efferent arms of the reflex arch. (p. 338)
6. Describe a monosynaptic reflex. (p. 338)
7. List the differences between the anterior and posterior branches of spinal nerves. (p. 338)
8. Describe the reflex arc in the figure. (p. 338)



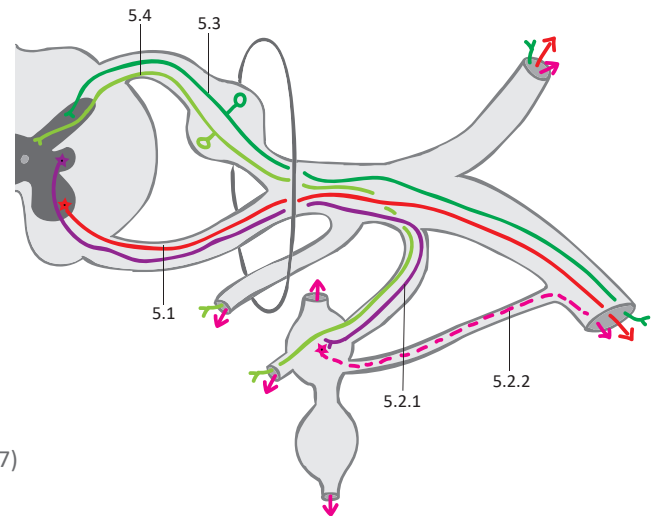
Describe the elementary structure of the peripheral nerve

II. Cervical plexus

9. List all the sensory branches of the cervical plexus. (p. 340)
10. Describe the superficial cervical loop. (p. 340)
11. Describe the deep cervical loop and list the structures it innervates. (p. 341)
12. Describe the course of the both phrenic nerves. (p. 341)

III. Brachial plexus

13. Describe the basic morphology of the brachial plexus. (p. 342)
14. Name the seven principal branches emerging from the supraclavicular part of the brachial plexus (p. 342)
15. List the eight principal branches emerging from the infraclavicular portion of the brachial plexus. (p. 343)
16. Describe the course of the median nerve. (p. 343)
17. List the muscles innervated by the ulnar nerve. (p. 344)
18. Describe the course of the radial nerve. (p. 345)
19. Name the four structures innervated by the axillary nerve. (p. 345)



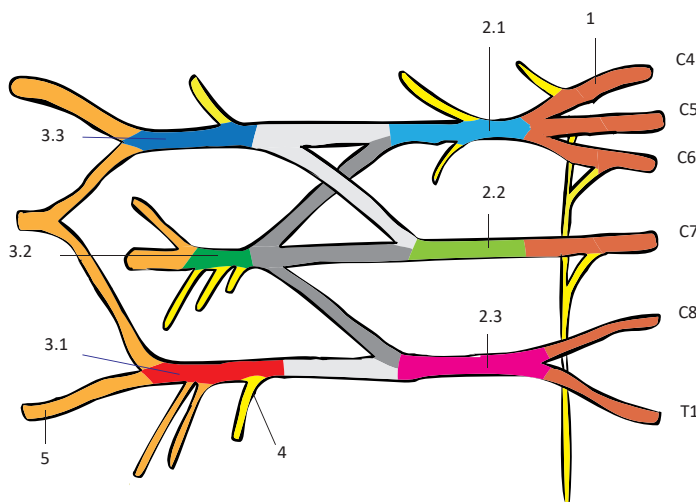
Describe particular nerve fibers in the peripheral nerve

IV. Thoracic nerves

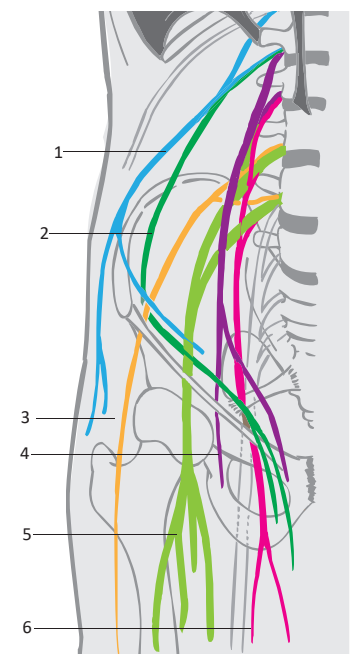
20. Describe the course of the intercostal nerve. (p. 346)
21. List the 13 muscles innervated by the thoracic nerves. (p. 346)

V. Lumbar plexus

22. Name the six principal branches of the lumbar plexus. (p. 346)
23. Describe the pathway of the ilioinguinal nerve. (p. 347)
24. Name the four structures innervated by the genitofemoral nerve. (p. 347)
25. Describe the course of the femoral nerve. (p. 347)
26. List the five structures innervated by the obturator nerve. (p. 347)



Describe the structure of the brachial plexus



Describe the structure of the lumbar plexus

VI. Sacral plexus

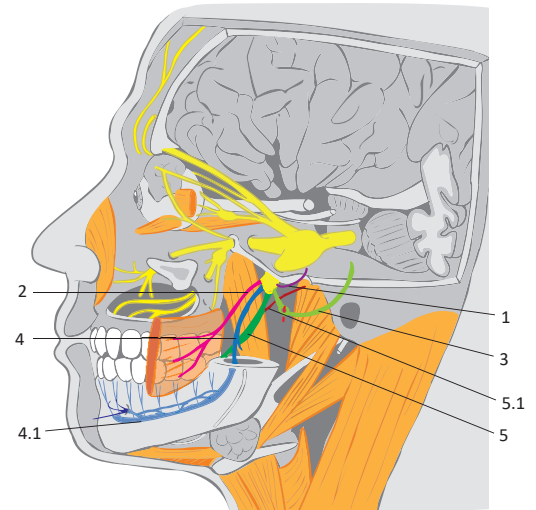
27. Name the five main branches of the sacral plexus. (p. 348)
28. List the four nerves that pass through the infrapiriform foramen. (p. 348)
29. Describe the course of the pudendal nerve. (p. 348)
30. Name the five structures innervated by the pudendal nerve. (p. 348)
31. Describe the course of the sciatic nerve. (p. 349)
32. Explain the point of origin of the sural nerve. (p. 349)
33. Describe the course of the common fibular nerve and state the structures it innervates. (p. 349)
34. Describe the course of the tibial nerve and state the structures it innervates. (p. 349)

VII. Dermatomes and peripheral nerve fields

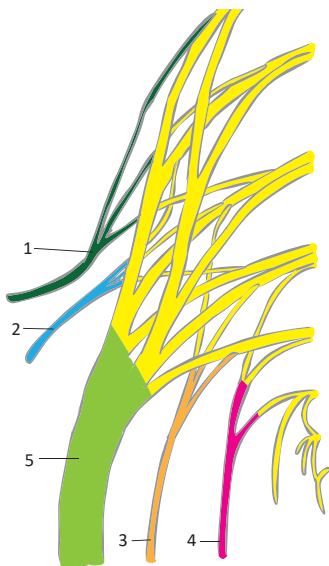
35. Define what dermatomes and myotomes are. (p. 350)
36. Explain what Head zones are and describe their significance. (p. 350)
37. Explain the difference between motor and sensory peripheral nerve fields. (p. 351)

VIII. Cranial nerves

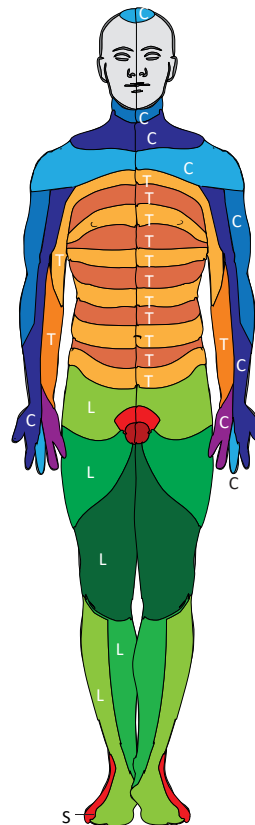
38. Name the twelve cranial nerves. (p. 352)
39. Classify the cranial nerves according to the nature of their fibres. (p. 352)
40. State the four cranial nerves, which contain visceromotor fibers. (p. 352)
41. State the three cranial nerves, which contain somatosensory fibers. (p. 352)
42. Explain what the mixed lateral system is. (p. 352)
43. List the two cranial nerves, which originate from the mesencephalon. (p. 353)
44. List the three cranial nerves, which originate from the medulla oblongata. (p. 353)
45. State the foramina, through which the 12 cranial nerves leave the skull. (p. 353)
46. Describe the course of the olfactory nerve. (p. 354)
47. Explain what the optic nerve consists of and describe its course. (p. 354)
48. State the two branches of the oculomotor nerve and describe its course. (p. 355)
49. Give the eponymous term for the parasympathetic nucleus of the oculomotor nerve. (p. 355)
50. Describe the consequences of a unilateral lesion of the oculomotor nerve. (p. 354)
51. Describe the course of the trochlear nerve and state the muscle it innervates. (p. 355)
52. Describe the course of the trigeminal nerve and its origin in the brainstem. (p. 356)



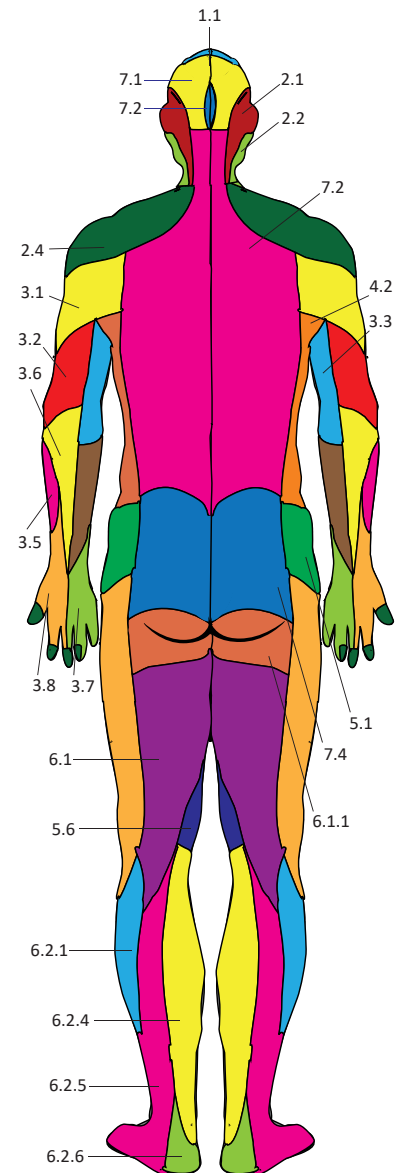
Describe the branches of the mandibular nerve



Describe the branches of the sacral plexus



Describe the dermatomes



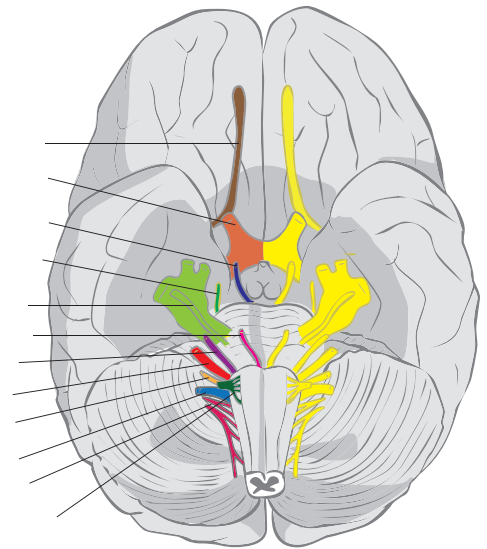
Describe the sensory nerve areas

VIII. Cranial nerves

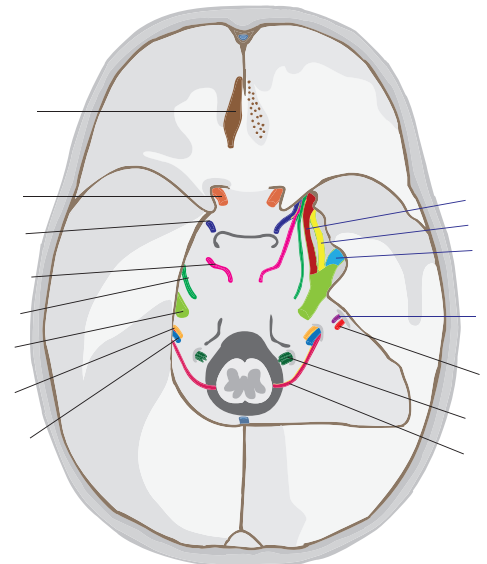
53. State the four nuclei, which contribute to the trigeminal nerve. (p. 356)
54. Describe the function of the mesencephalic nucleus of the trigeminal nerve. (p. 356)
55. State the three principal branches of the trigeminal nerve and describe their course. (p. 356)
56. State the three branches of the ophthalmic nerve and list the structures they innervate. (p. 357)
57. State the five branches of the maxillary nerve and list the structures they innervate. (p. 358)
58. State the five branches of the mandibular nerve and list the structures they innervate. (p. 359)
59. Describe the course of the abducent nerve and state the muscle it innervates. (p. 360)
60. Describe the course of the facial nerve and list its three nuclei. (p. 360)
61. List the ten principal branches of the facial nerve and state the structures they innervate. (p. 361)
62. Describe the clinical presentation of peripheral paralysis of the facial nerve. (p. 360)
63. Explain the function of the gustatory nucleus. (p. 360)
64. Describe the course of the vestibulocochlear nerve and state its two nuclei. (p. 362)
65. State the two principal branches of the vestibulocochlear nerve and list the structures they innervate. (p. 362)
66. Describe the course of the vestibulocochlear nerve through the internal acoustic meatus. (p. 362)
67. Describe the course of the glossopharyngeal nerve and state its four nuclei. (p. 363)
68. State the six branches of the glossopharyngeal nerve. (p. 363)
69. Describe the course of the vagus nerve through the body and state its four nuclei. (p. 364)
70. Describe the course of the accessory nerve and list the three structures it innervates. (p. 365)
71. Describe the course of the hypoglossus nerve and describe its relation to the deep ansa cervicalis. (p. 366)

III. Autonomic nervous system

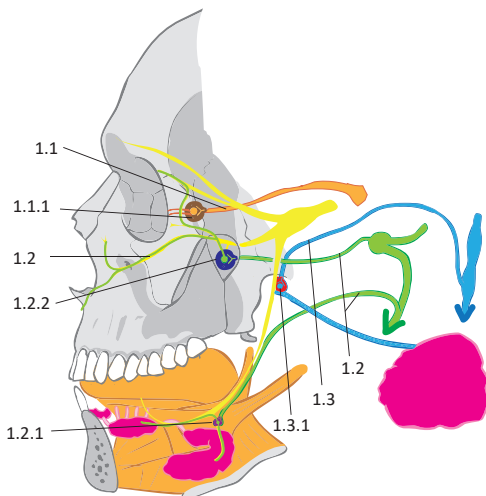
72. Describe interconnections of the sympathetic fibres. (p. 368)
73. List all functions of the sympathetic system. (p. 368)
74. Describe the interconnections of the parasympathetic fibers. (p. 371)
75. State the functions of parasympathetic nerves. (p. 371)
76. List the three cervical sympathetic ganglia. (p. 370)
77. Describe the classification of the parasympathetic portion of the ANS. (p. 371)
78. Explain what the Cannon-Böhm point is. (p. 371)
79. State the nine parts of the mixed nervous plexuses of the thorax and abdomen. (p. 372–373)
80. Explain the function of the enteric nervous system. (p. 374)



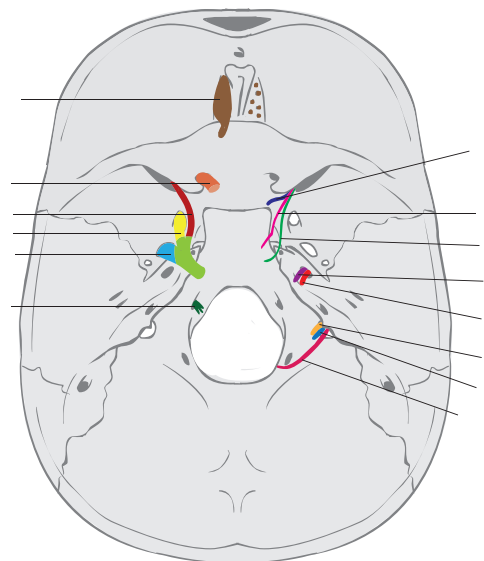
Describe how the cranial nerves leave the skull



Describe how the cranial nerves pass through the dura mater



Describe the nerves and ganglia of the cranial part of the parasympathetic system



Describe how the cranial nerves pass through the skull

We would like to **thank the following anatomists, clinicians and medical students** for their invaluable help, devotion and feedback in the preparation of this chapter.

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